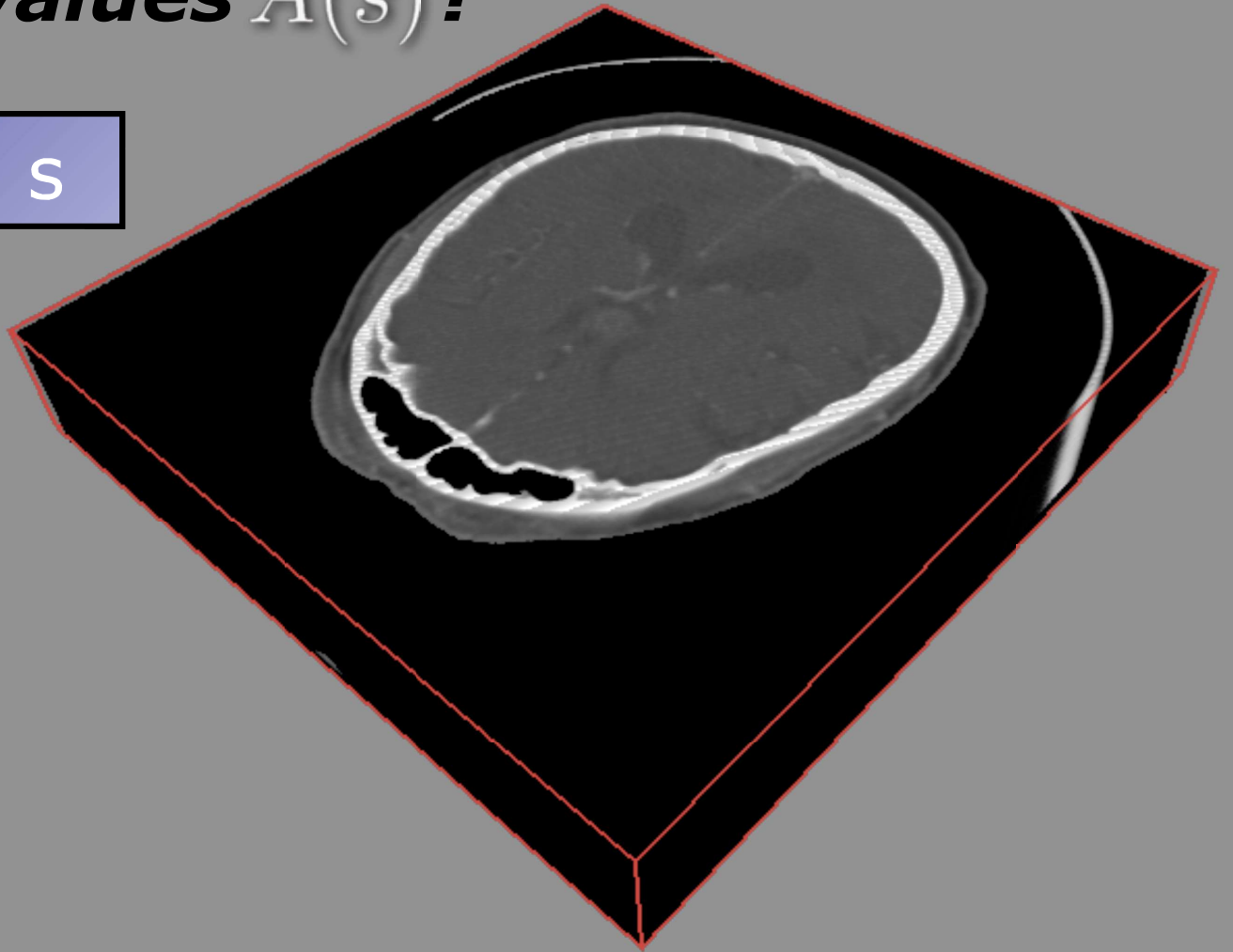


Transfer Functions

Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s



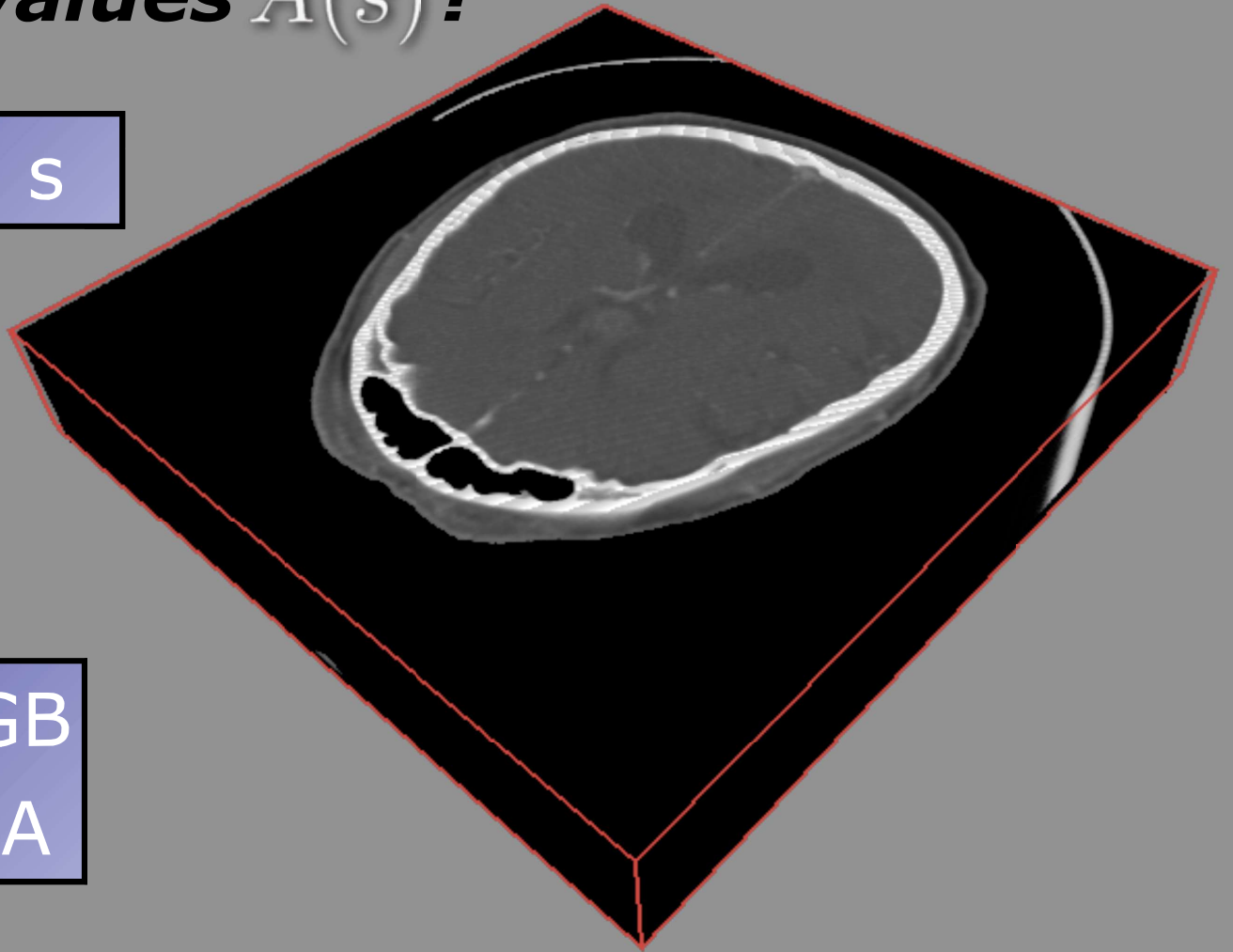
Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s

$T(s)$

emission RGB
absorption A



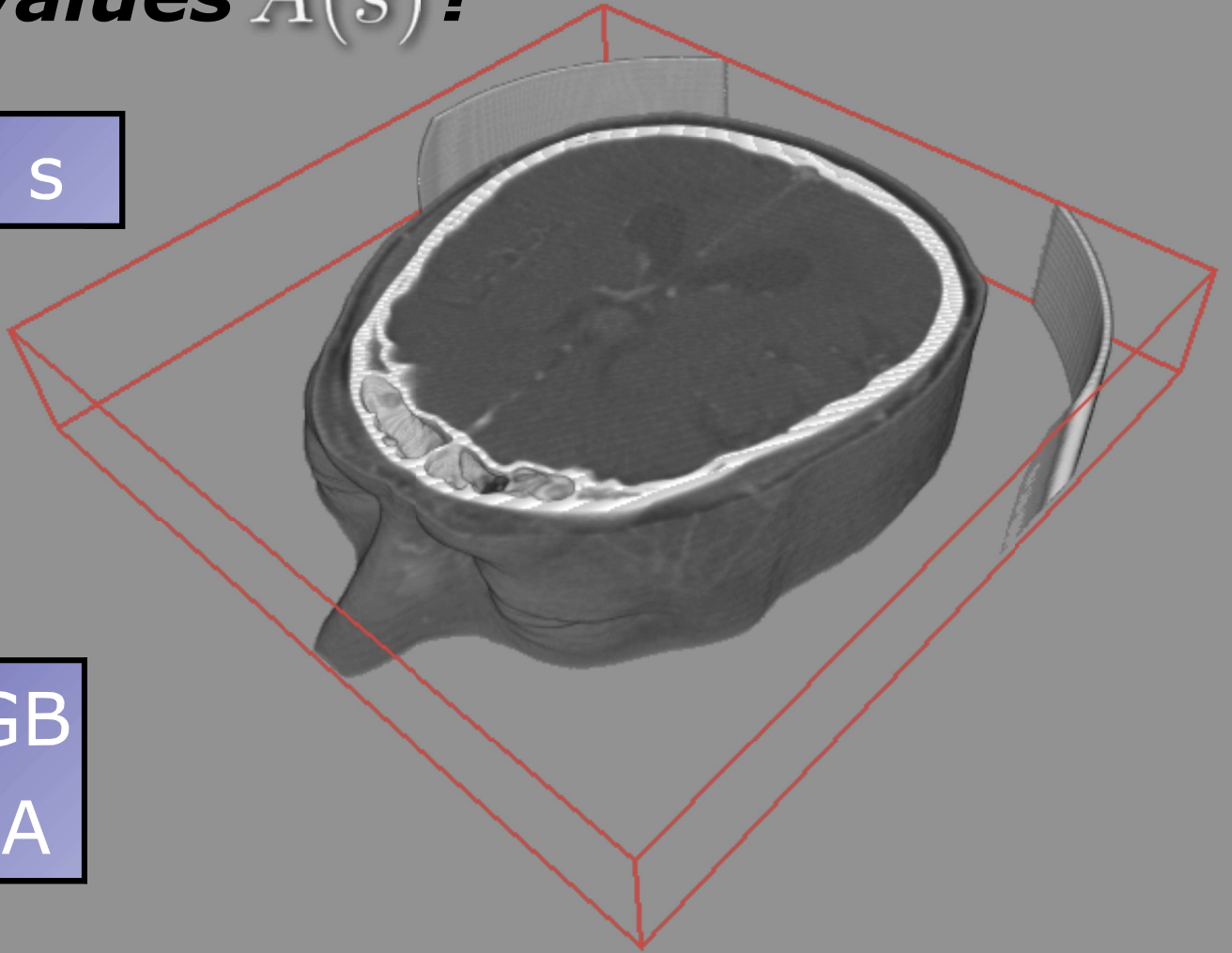
Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s

$T(s)$

emission RGB
absorption A



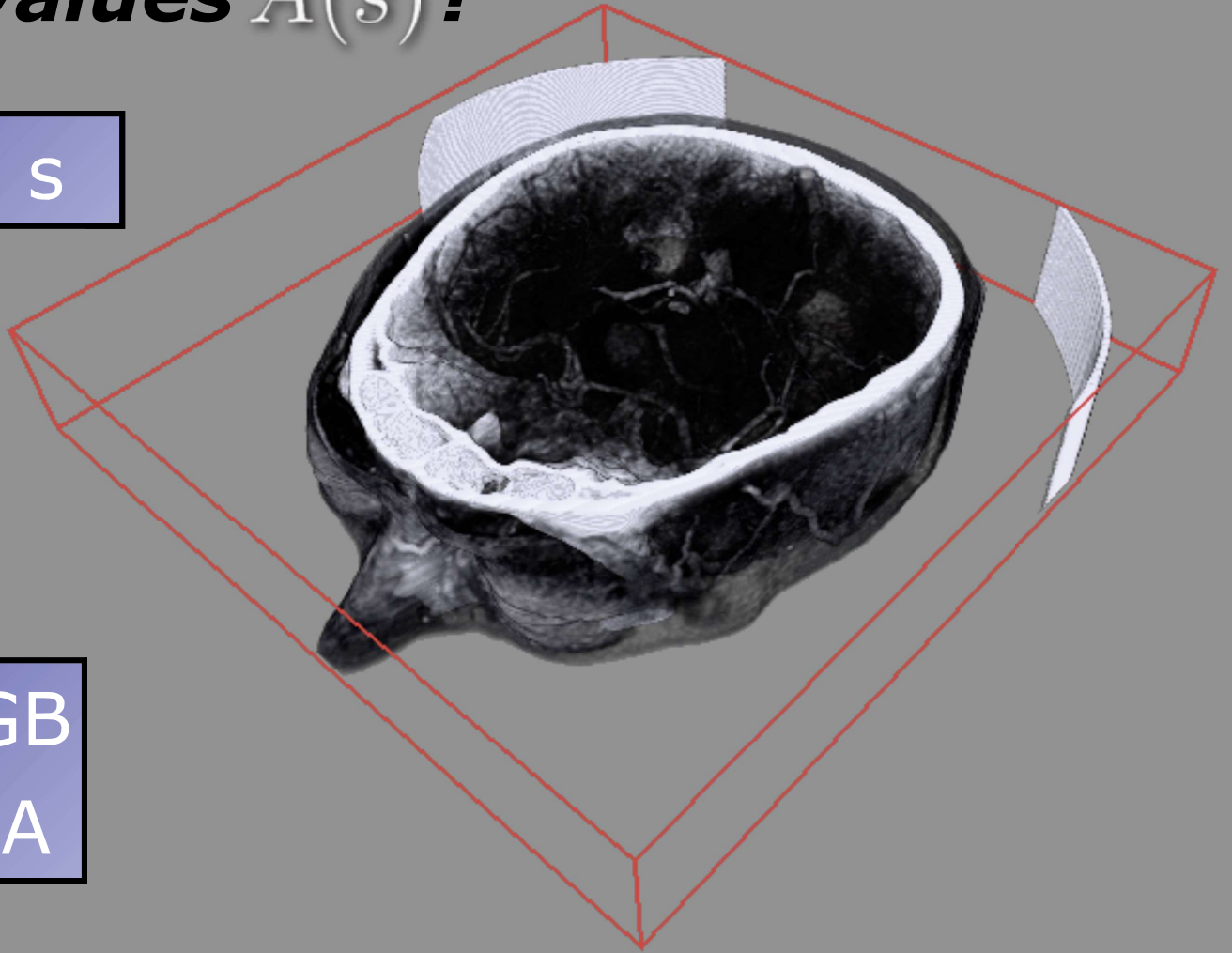
Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s

$T(s)$

emission RGB
absorption A



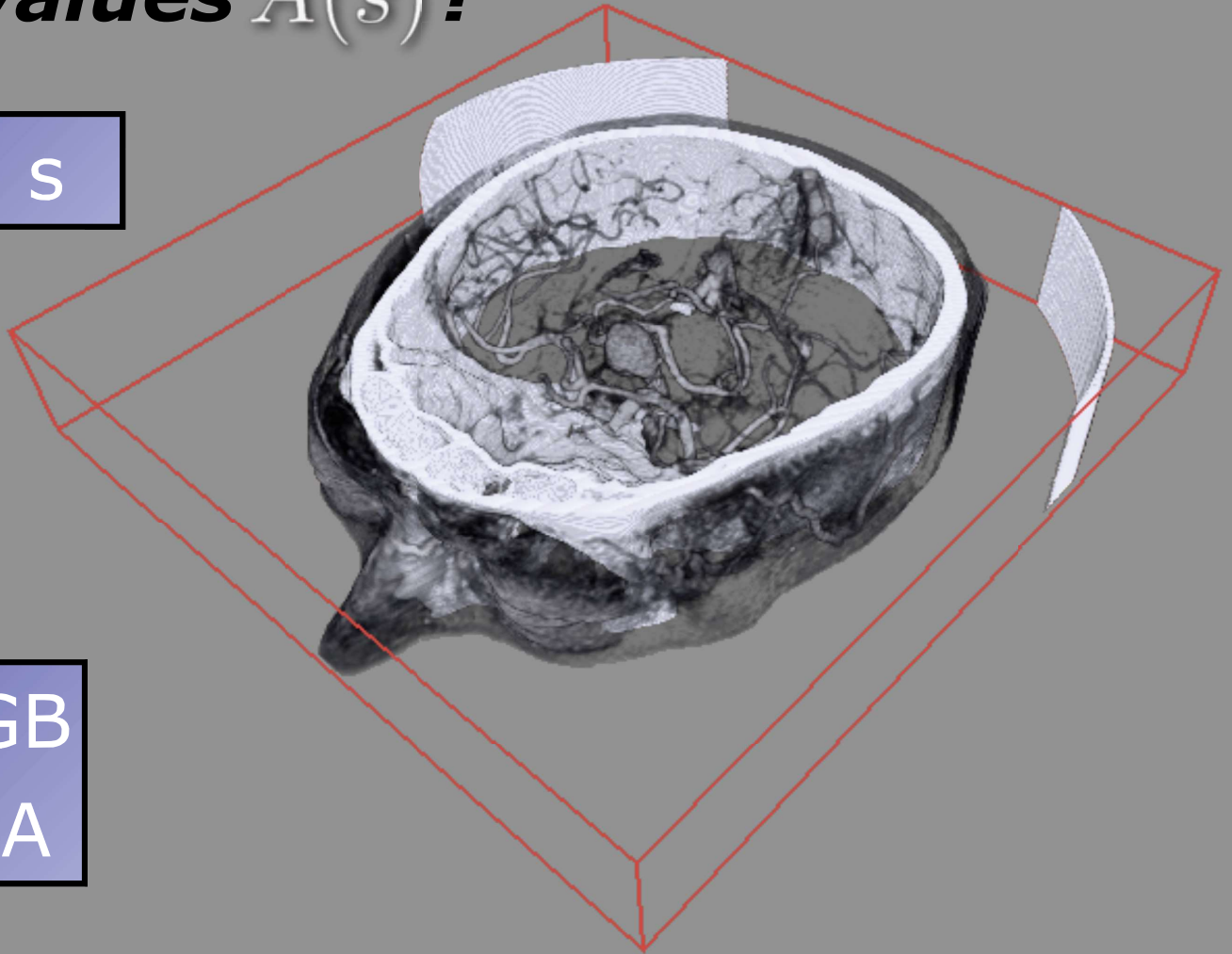
Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s

$T(s)$

emission RGB
absorption A



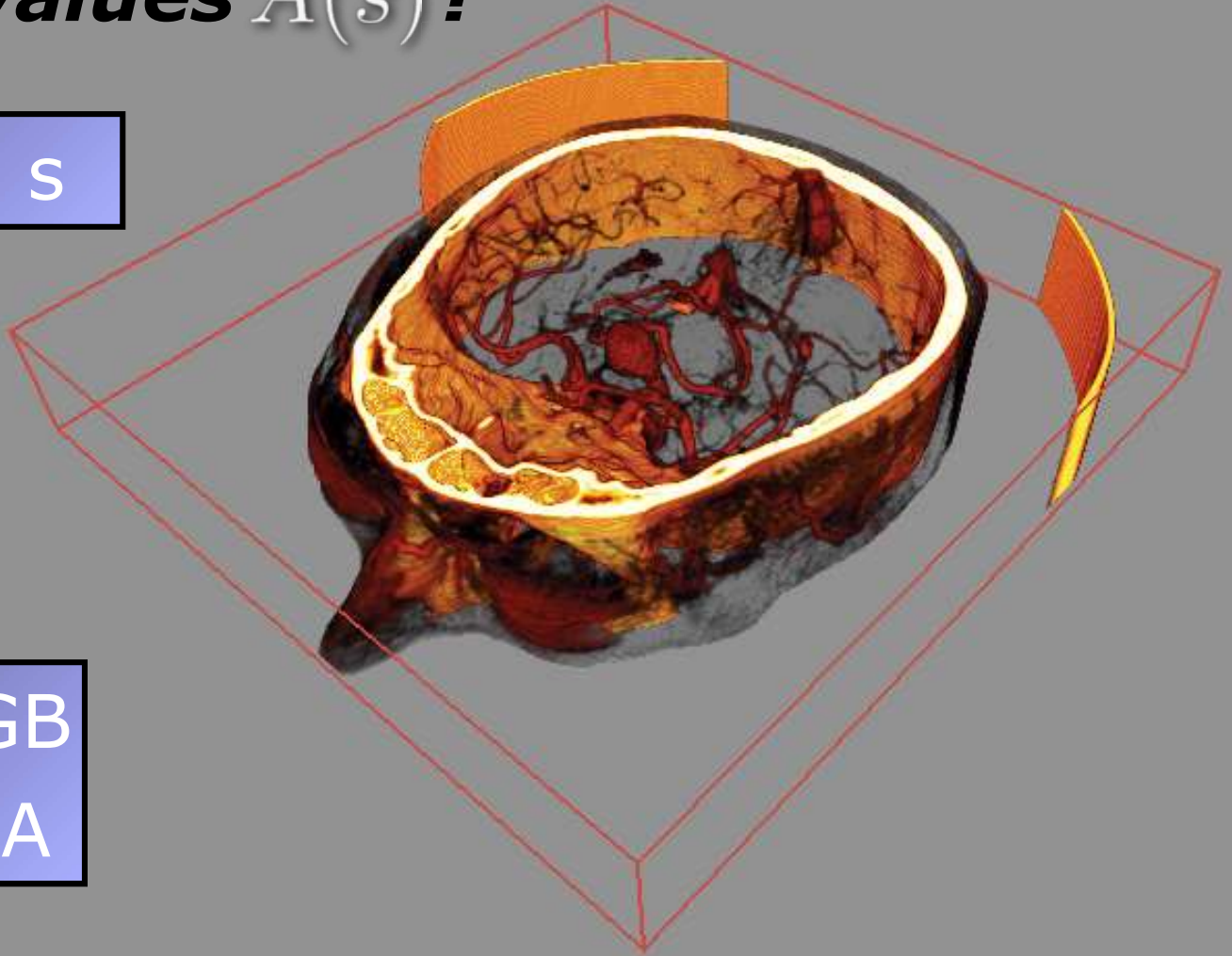
Classification

How do I obtain the emission values $q(s)$ and Absorption values $A(s)$?

scalar value s

$T(s)$

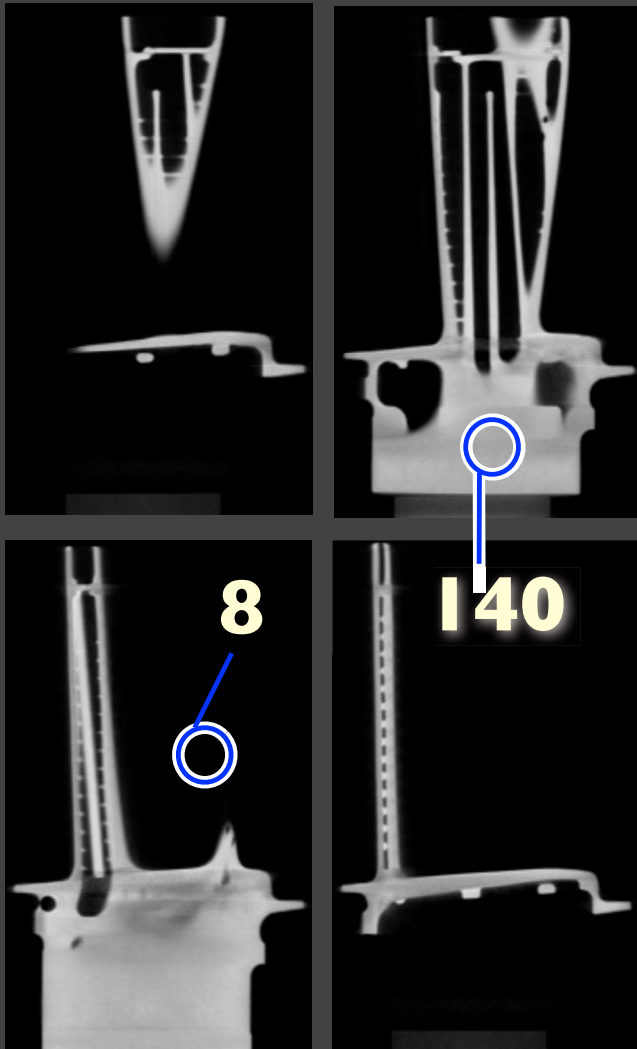
emission RGB
absorption A



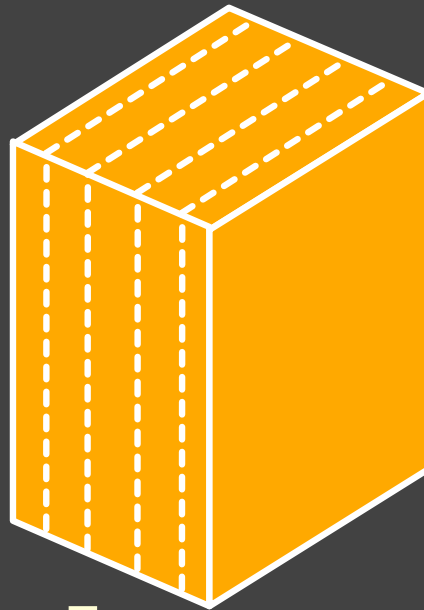
Introduction

Transfer functions make volume data visible by mapping data values to optical properties

slices:



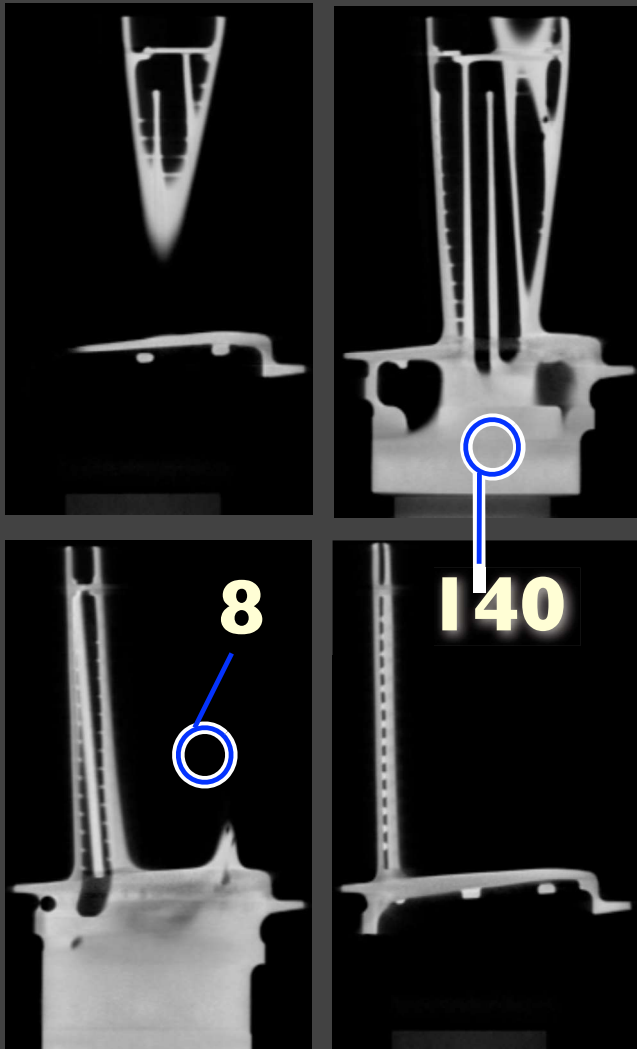
volume data:



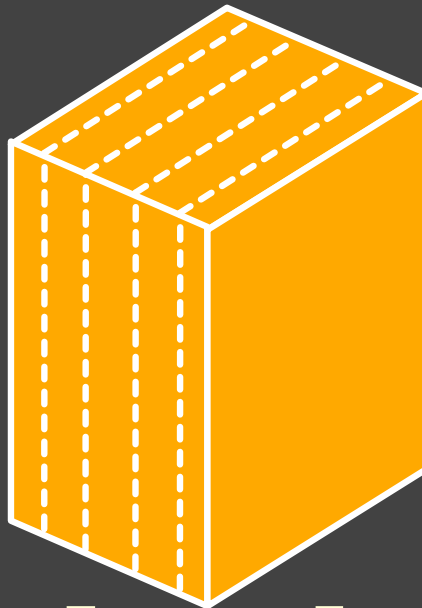
Introduction

Transfer functions make volume data visible by mapping data values to optical properties

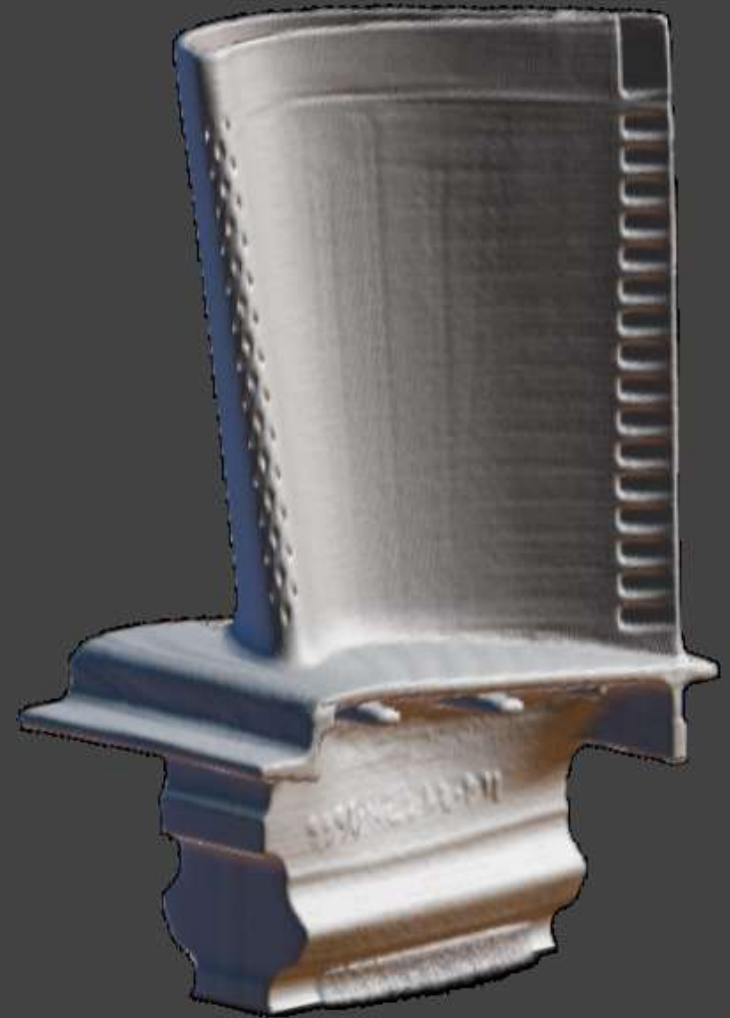
slices:



volume data:

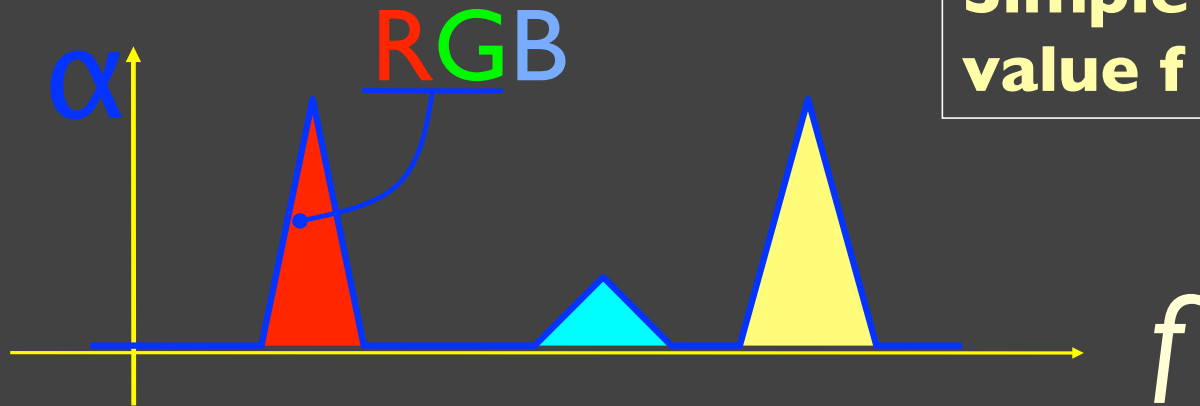


volume rendering:



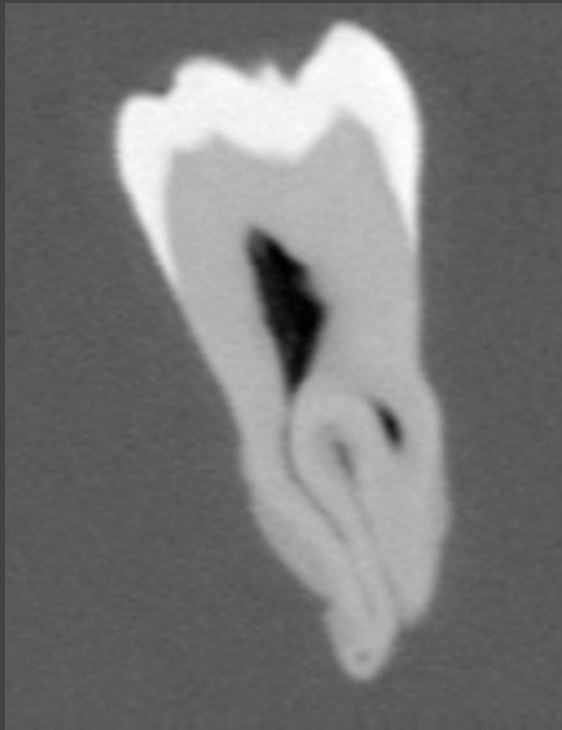
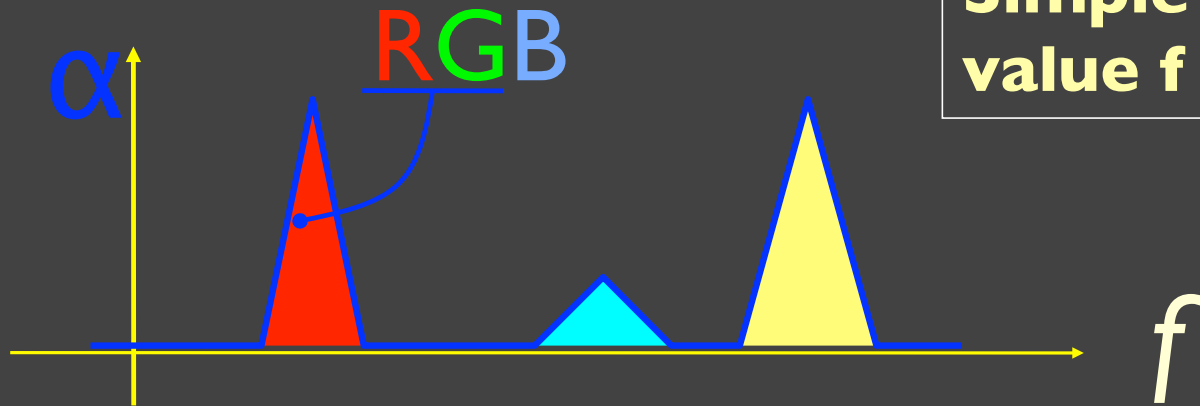
Transfer Functions (TFs)

Simple (usual) case: Map data value f to color and opacity



Transfer Functions (TFs)

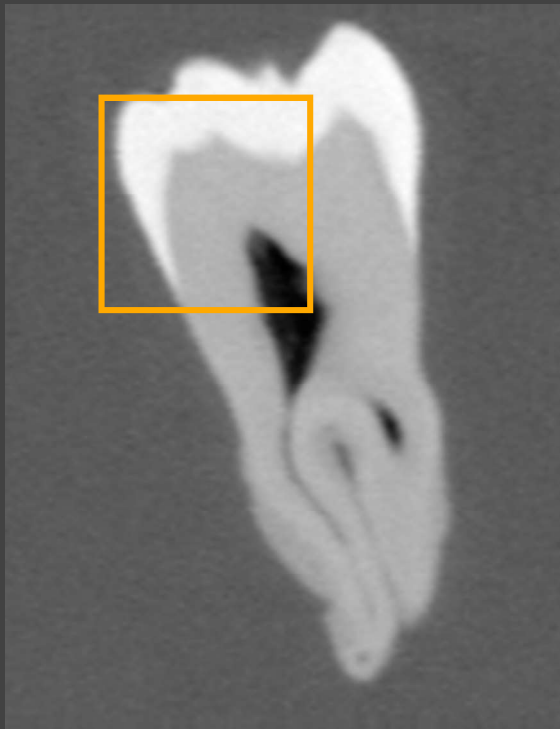
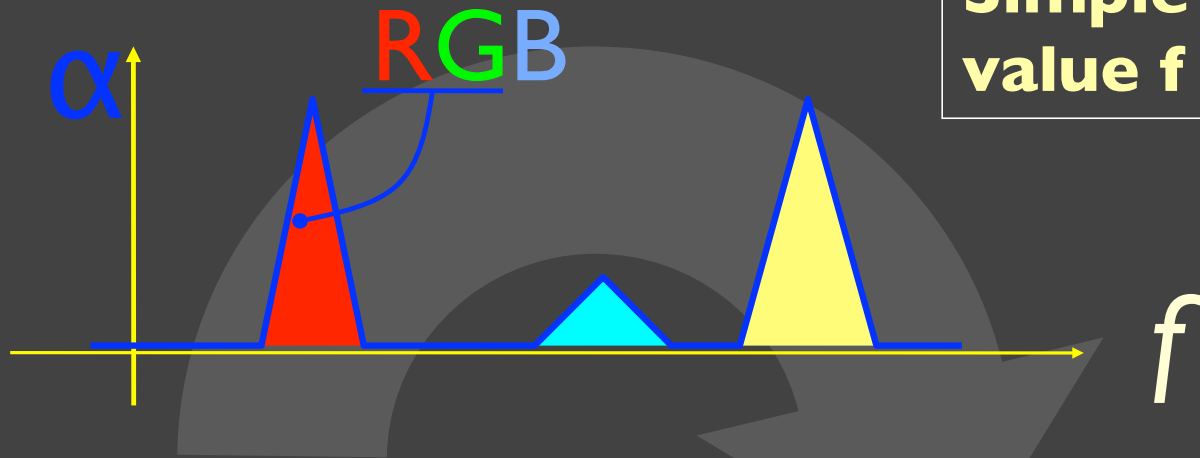
Simple (usual) case: Map data value f to color and opacity



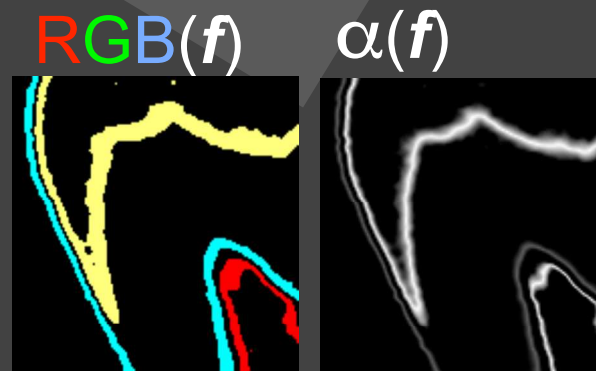
Human Tooth CT

Transfer Functions (TFs)

Simple (usual) case: Map data value f to color and opacity

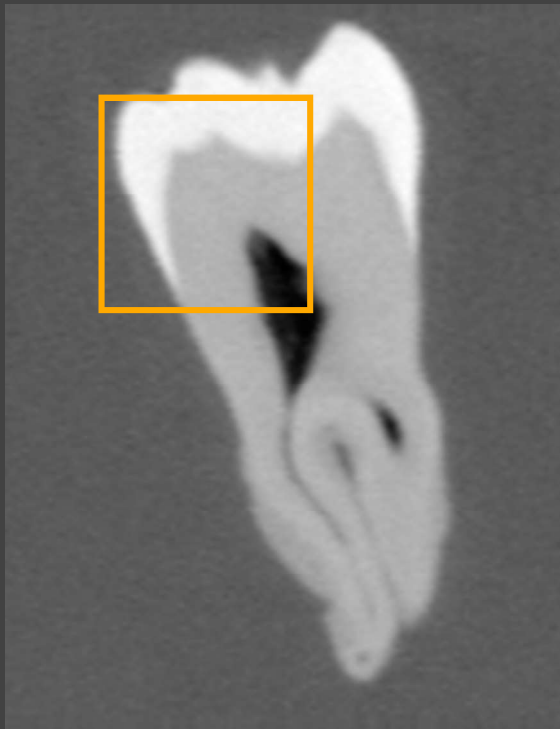
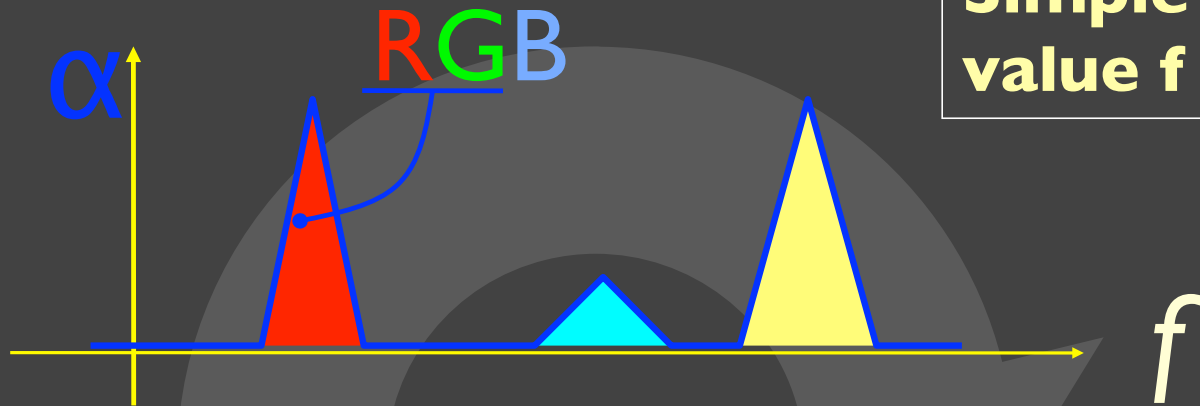


Human Tooth CT

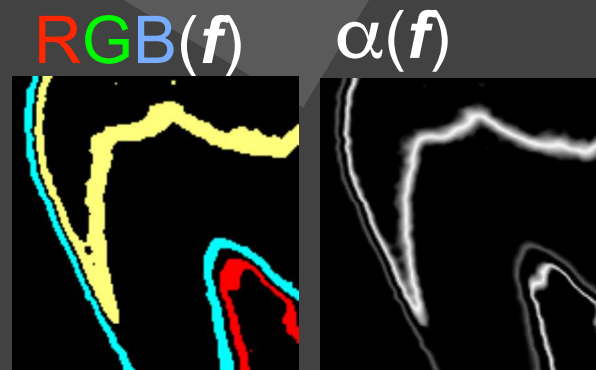


Transfer Functions (TFs)

Simple (usual) case: Map data value f to color and opacity



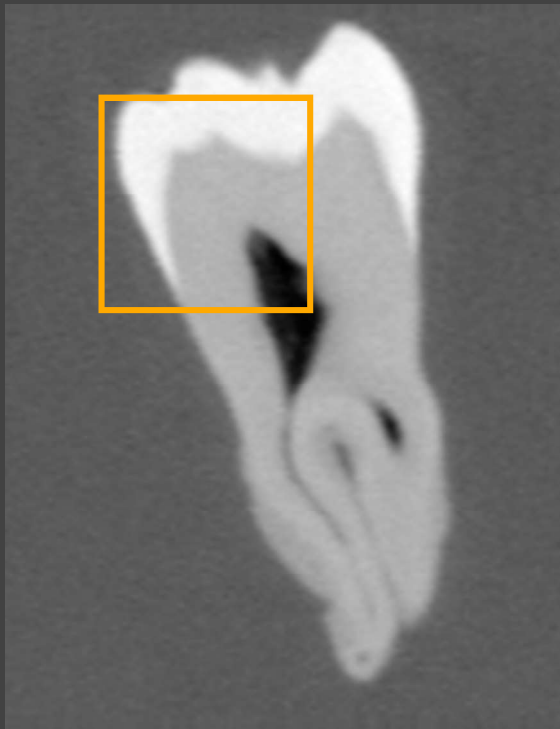
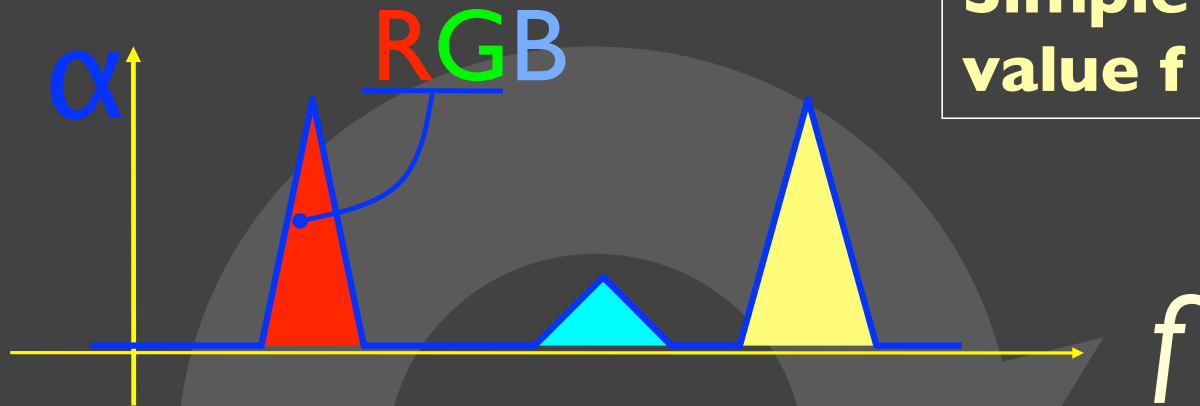
Human Tooth CT



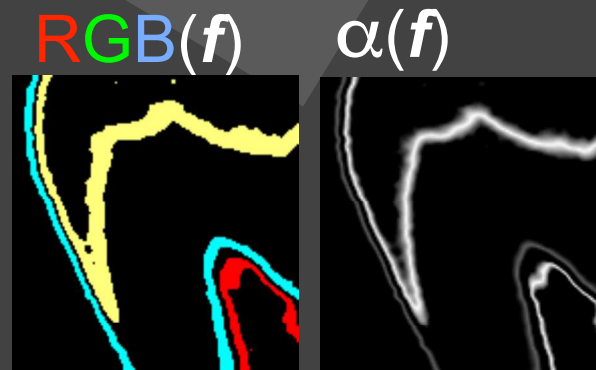
Shading,
Compositing...

Transfer Functions (TFs)

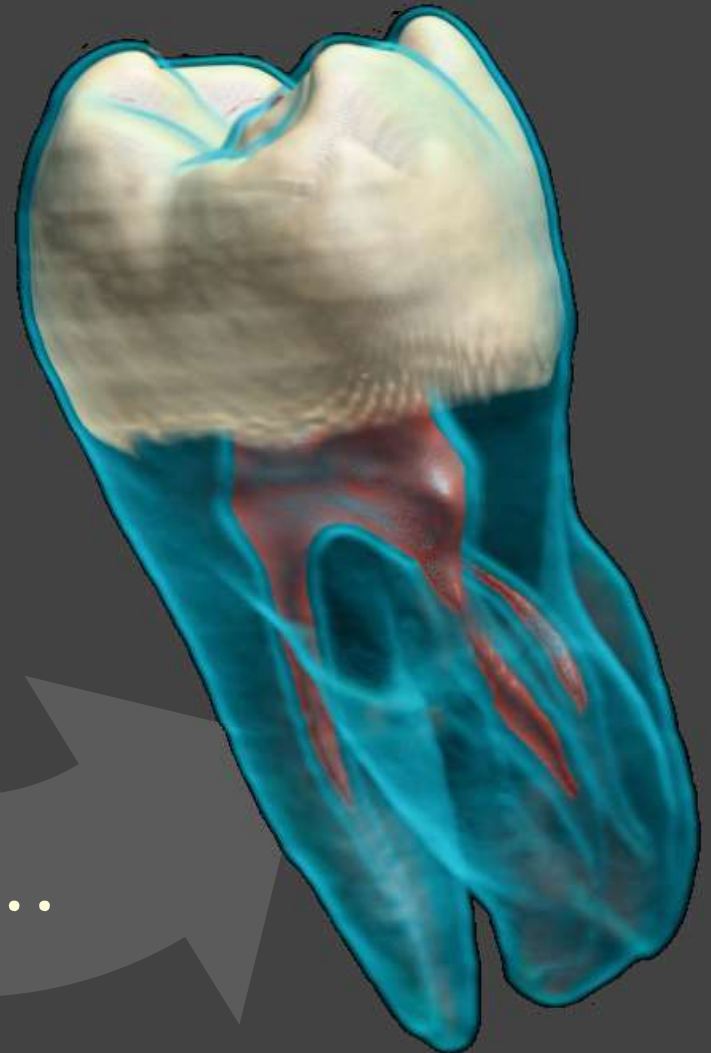
Simple (usual) case: Map data value f to color and opacity



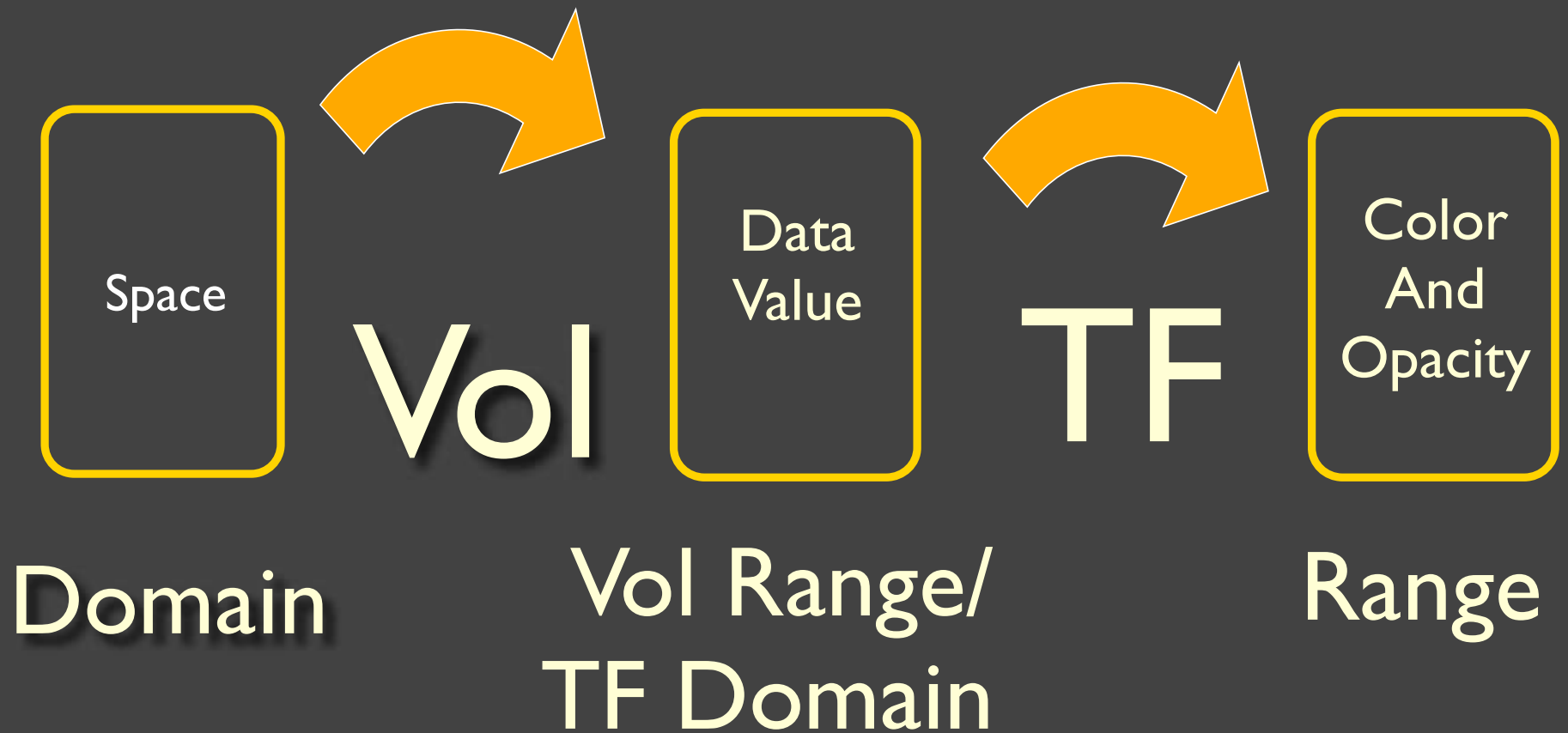
Human Tooth CT



Shading,
Compositing...



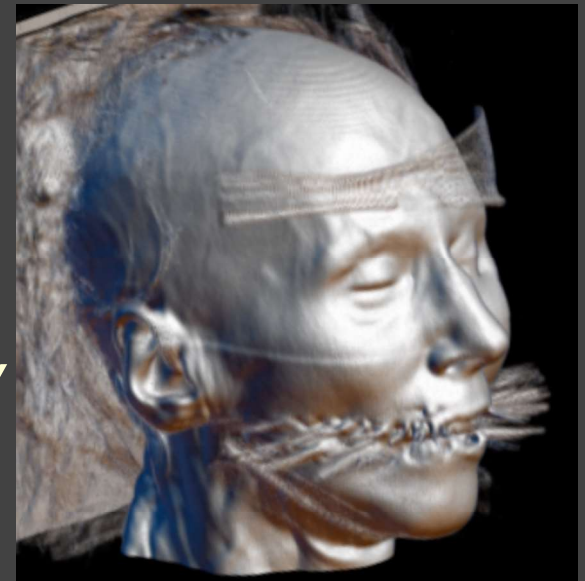
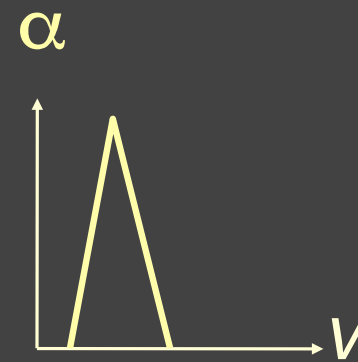
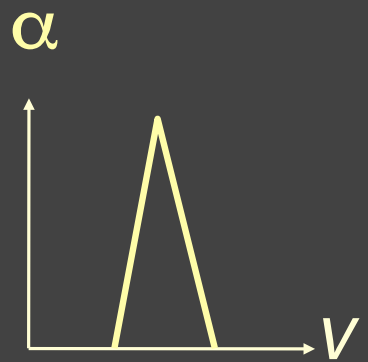
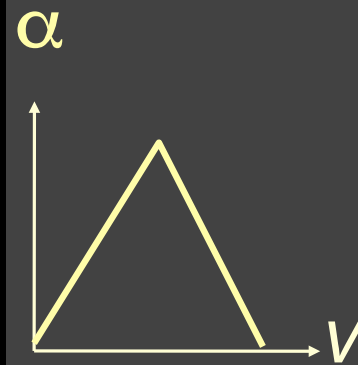
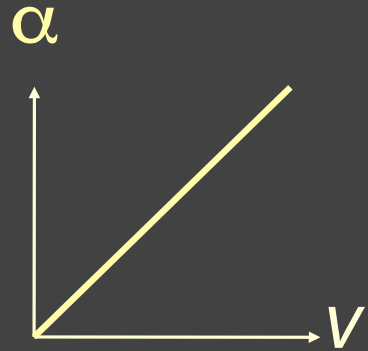
Basic Transfer Functions:



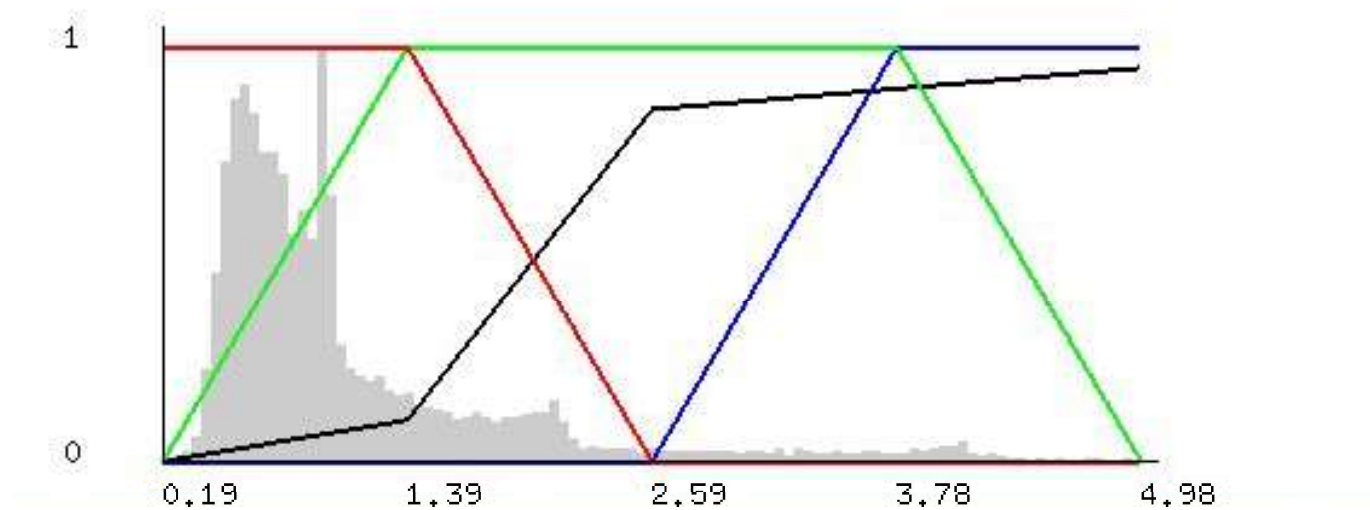
What Can Be Controlled by the Transfer Function?

- Optical Properties: Anything that can be composited with a standard graphics operator (“over”)
 - Opacity: “opacity functions”
 - Color: Can help distinguish features
 - Phong parameters (k_a , k_d , k_s)
 - Index of refraction

Setting Transfer Function: Hard

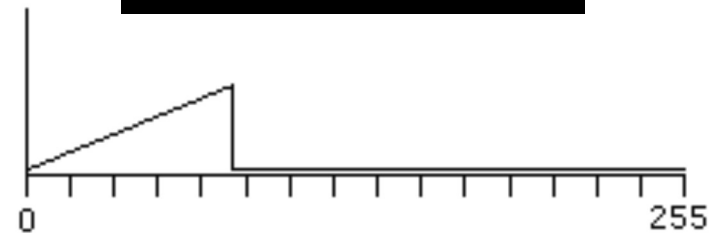
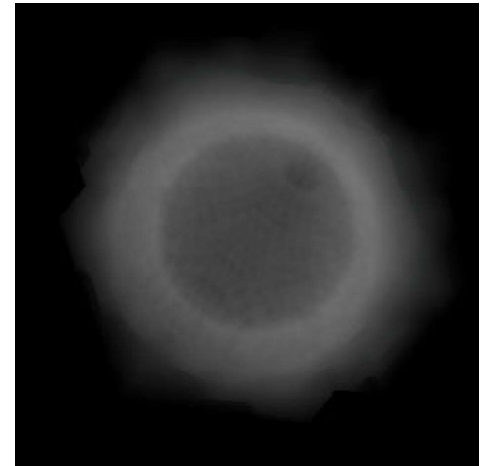
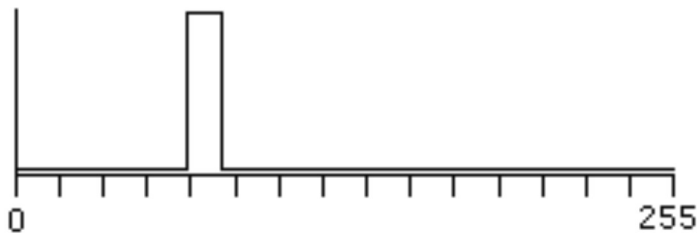
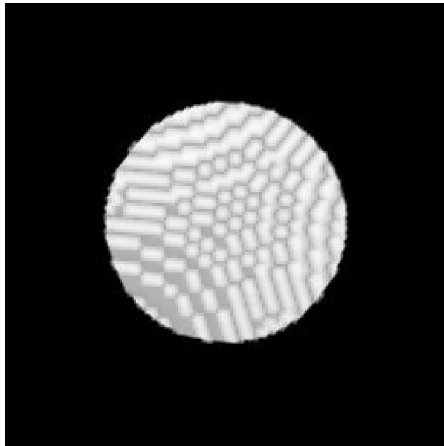


Transfer Function

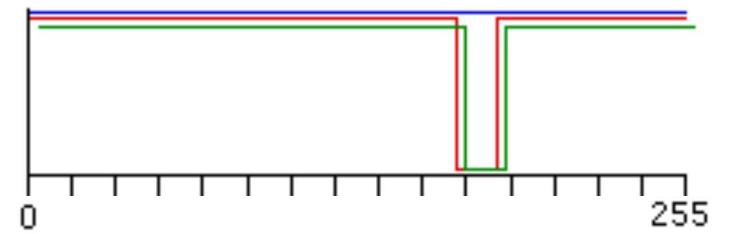
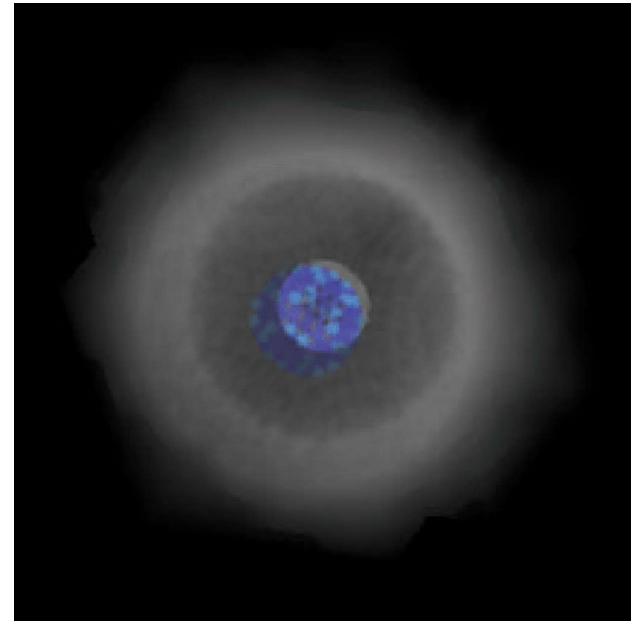
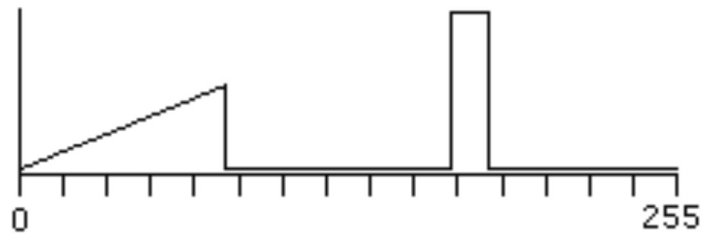
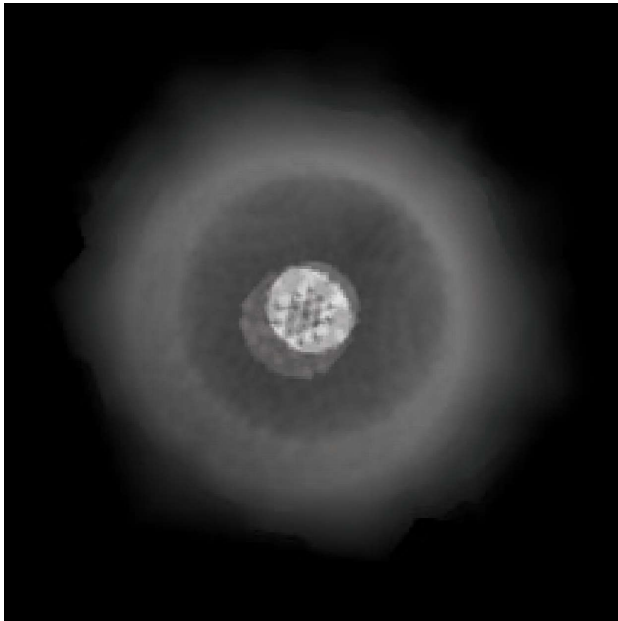


- » RGB components
- » Opacity

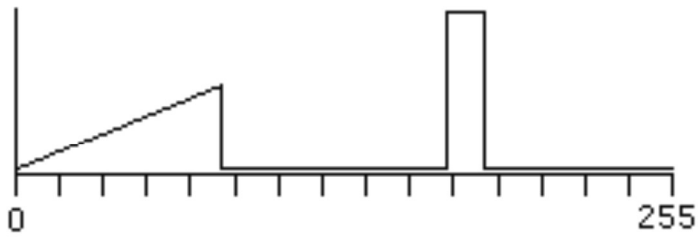
Transfer Function



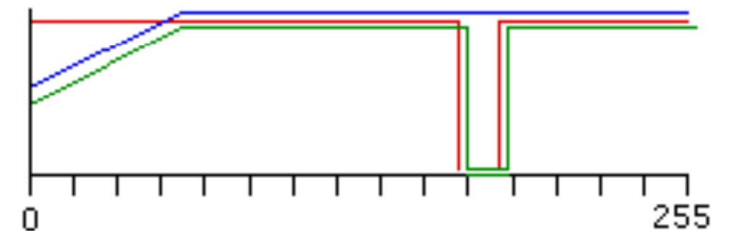
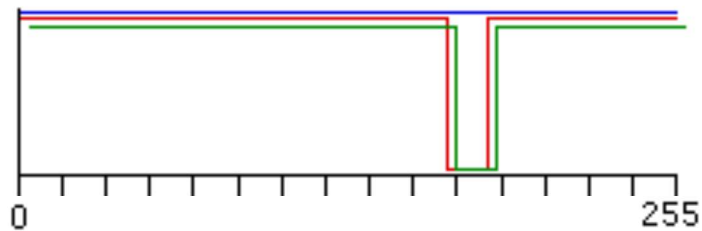
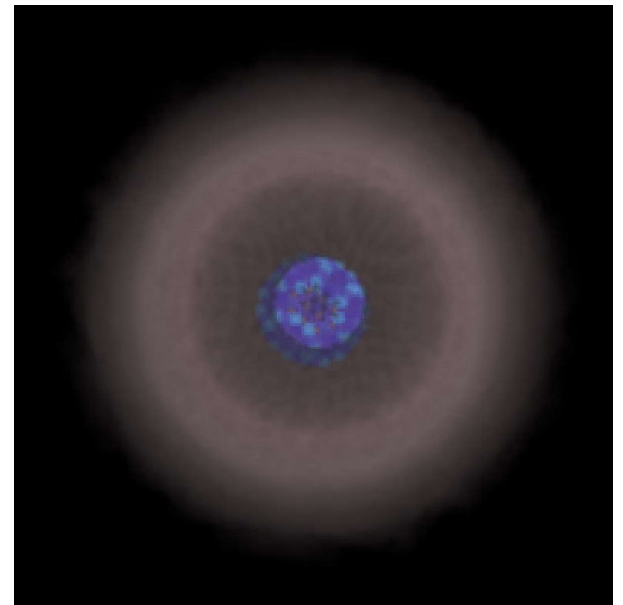
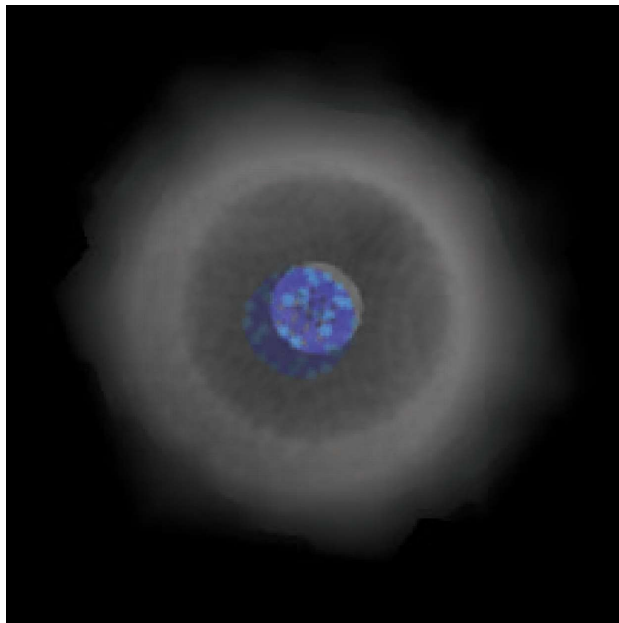
Transfer Function



Transfer Function

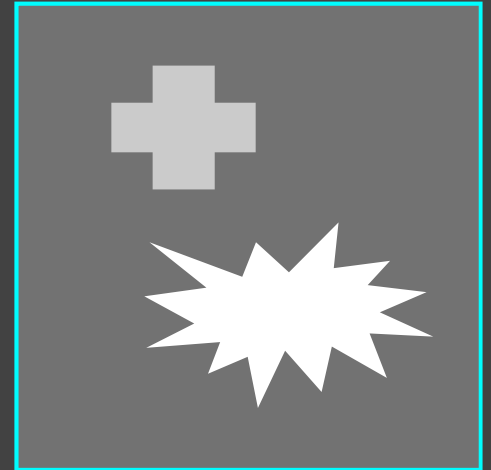
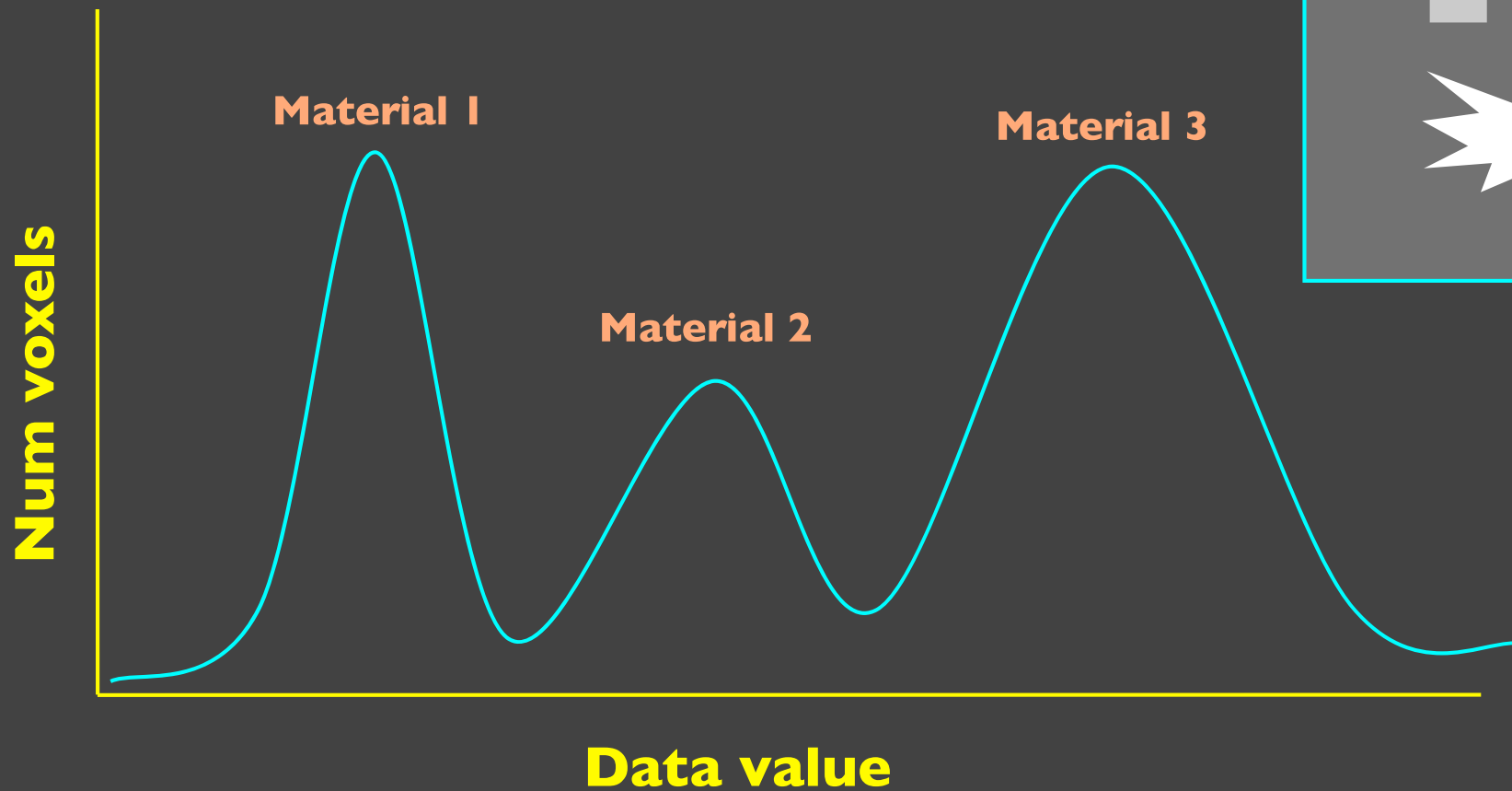


Different colors, same opacity



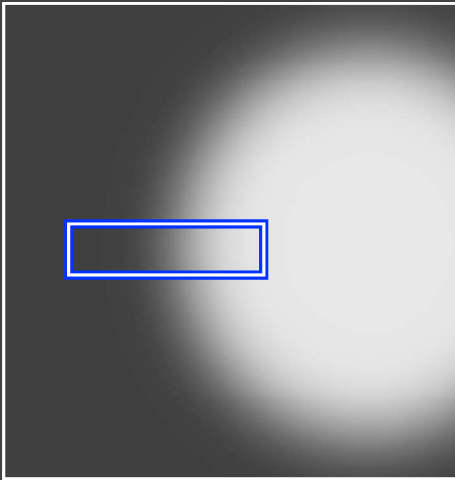
Volumes as Consisting of Materials

Grey-Level Histogram

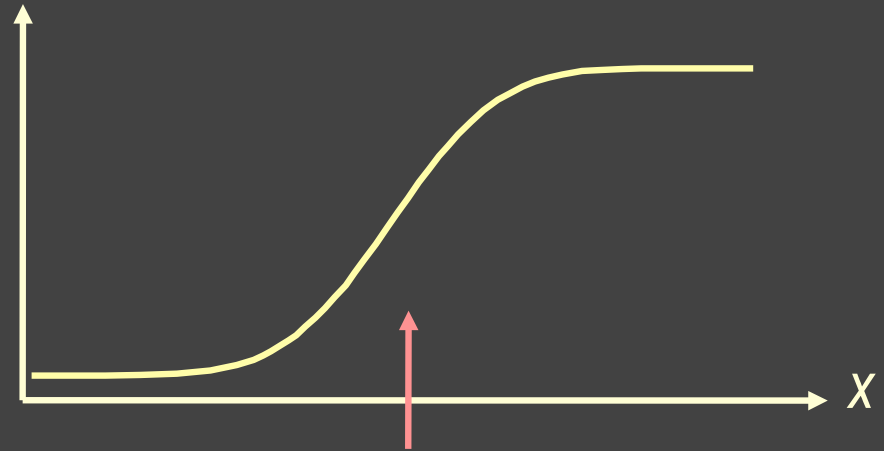


Finding edges: easy

“Where’s the edge?”



$$v = f(x)$$

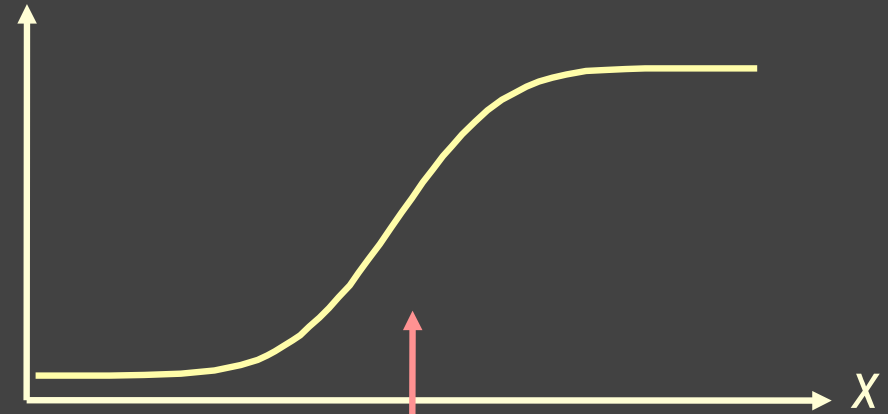
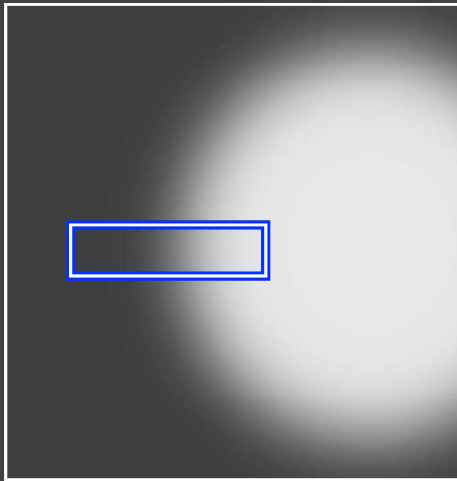


“ here’s the edge ”

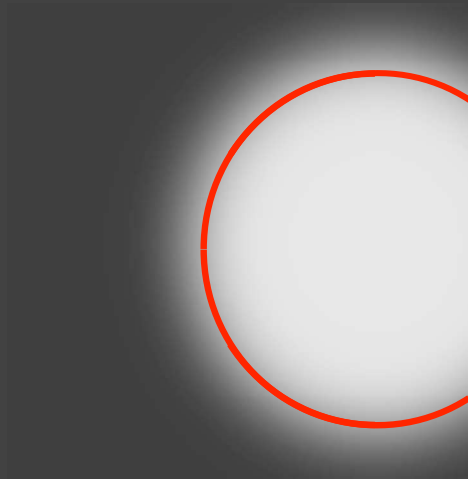
Finding edges: easy

“Where’s the edge?”

$$v = f(x)$$



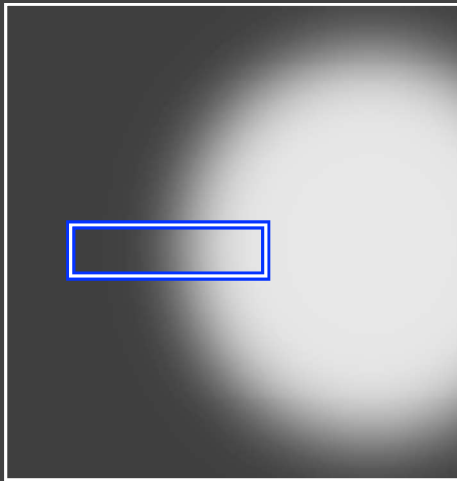
“here’s the edge”



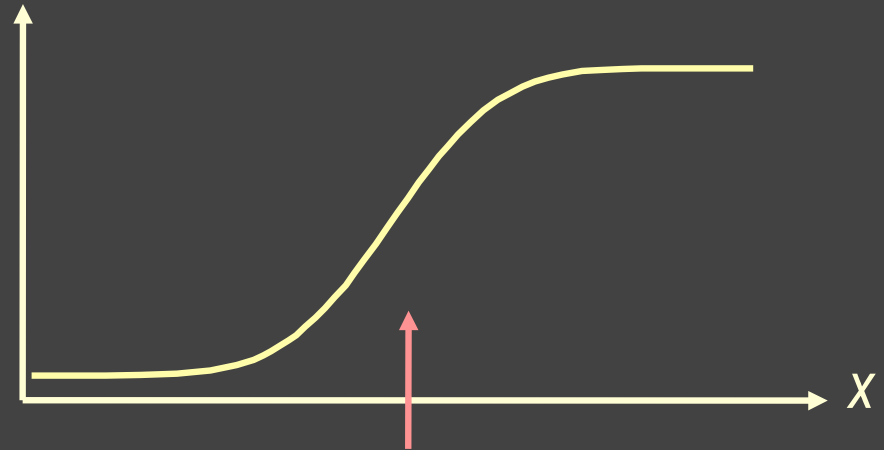
Result: edge pixels

Finding edges: easy

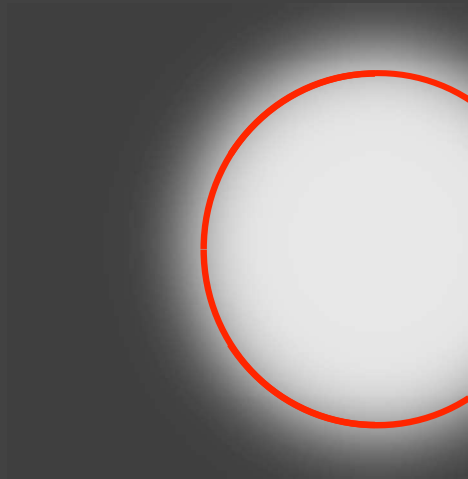
“Where’s the edge?”



$$v = f(x)$$



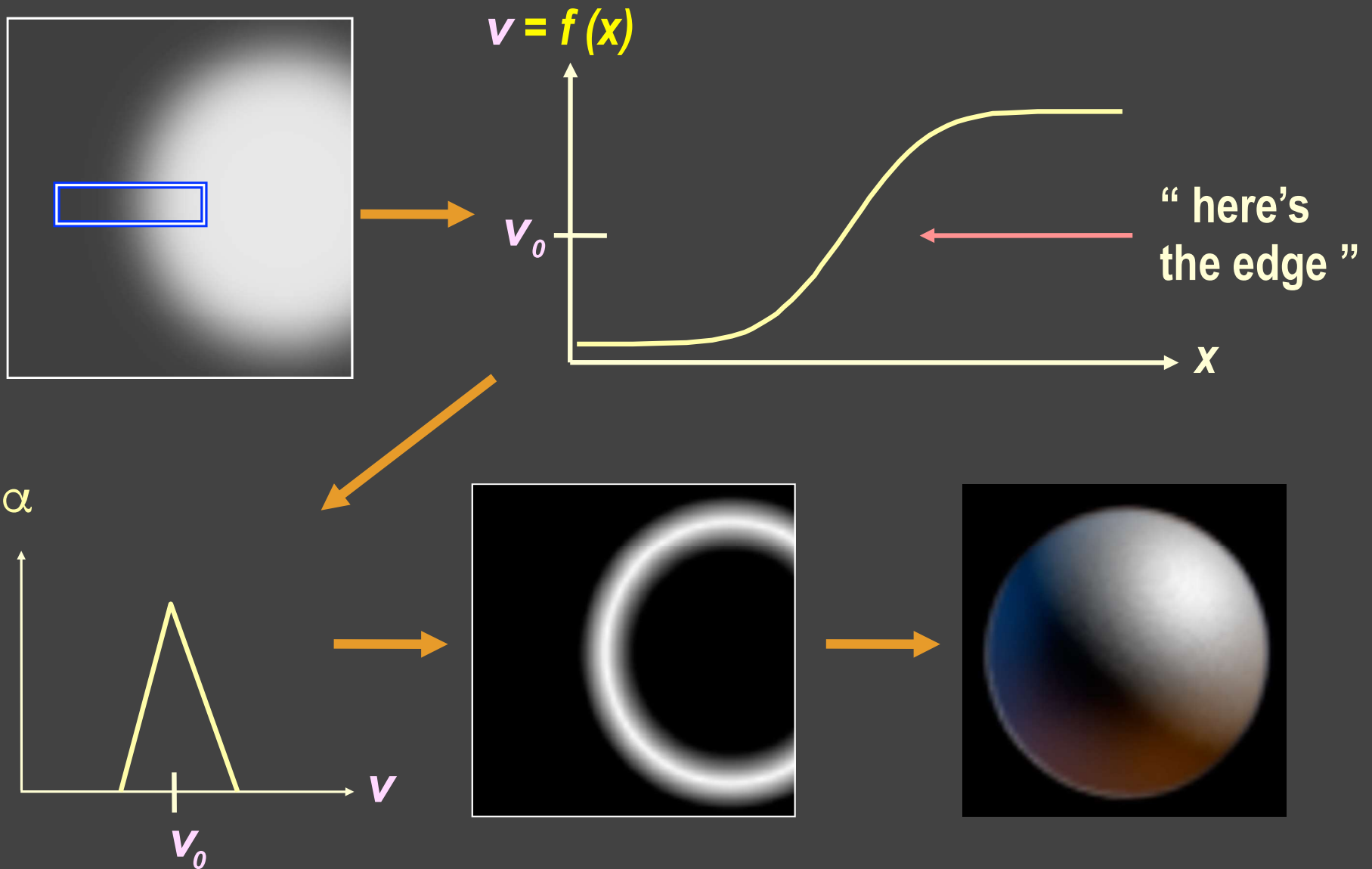
“here’s the edge”



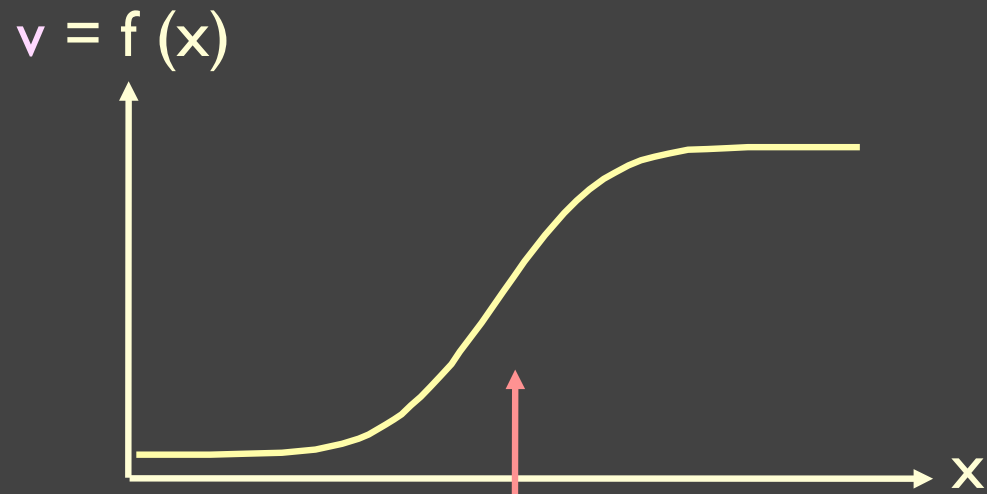
Result: edge pixels



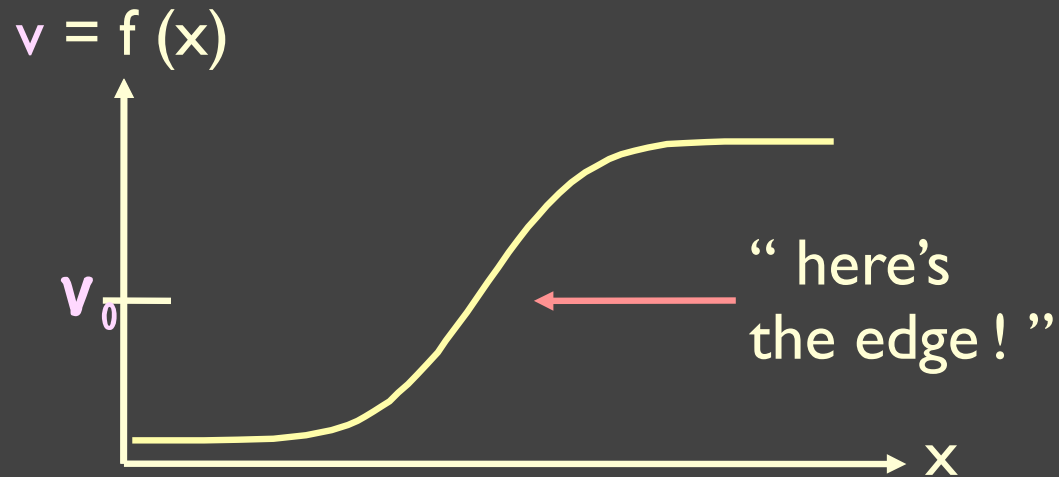
Transfer function Unintuitive



TFs as feature detection

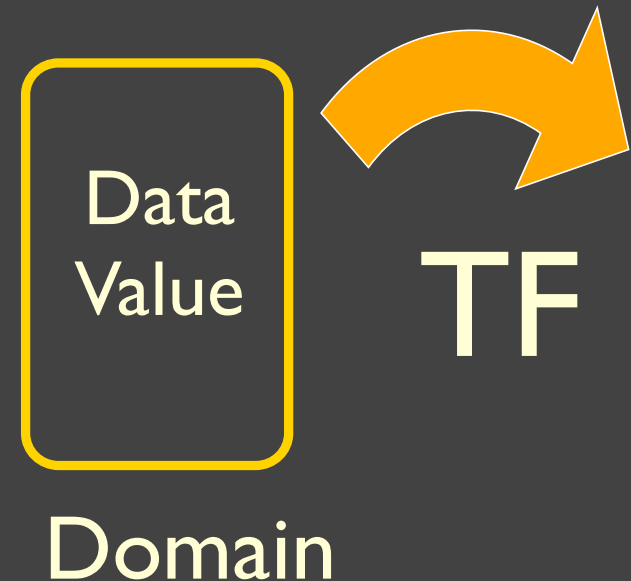


“ here’s the edge ! ”



“ here’s the edge ! ”

Domain of the transfer function does not include position



What Makes Designing TF's Challenging?

1. Non-spatial: spatial isolation doesn't imply data value isolation
2. Many degrees of freedom
3. No constraints or guidance
4. Material uniformity assumption

Goals for TF Design

- Make good renderings easier to come by
- Make space of TFs less confusing
- Remove excess “flexibility”
- Provide one or more of:
 - Information
 - Guidance
 - Semi-automation / Automation

TF Techniques/Tools

1. **Trial and Error**
2. Image-Centric Approach (not discussed)
3. Data-Centric Approach

I. Trial and Error

1. Manually edit graph of transfer function
2. Enforces learning by experience
3. Get better with practice
4. Can make terrific images



William Schroeder, Lisa Sobierajski Avila, and Ken Martin; Transfer Function Bake-off Vis '00





Break for Questions

TF Techniques/Tools

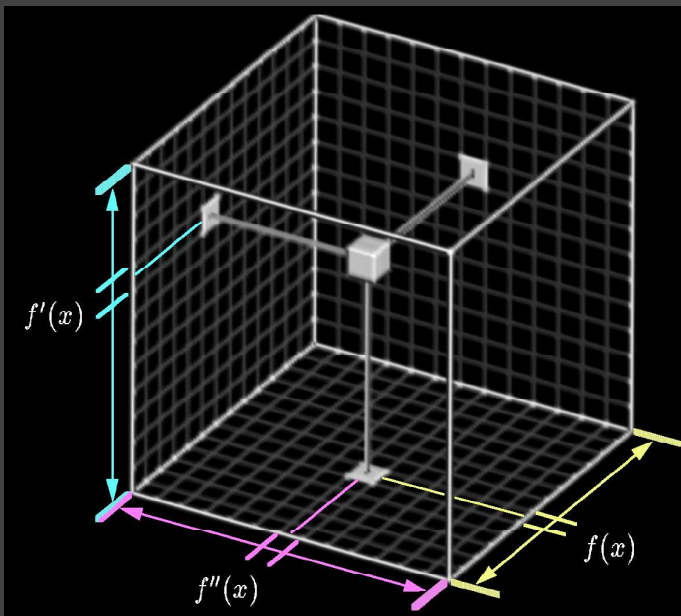
1. Trial and Error
2. Image-Centric Approach (not discussed)
3. **Data-Centric Approach**

Use derivatives

Reasoning:

- TFs are volume-position invariant
- Histograms “project out” position
- Interested in boundaries between materials
- Boundaries characterized by derivatives

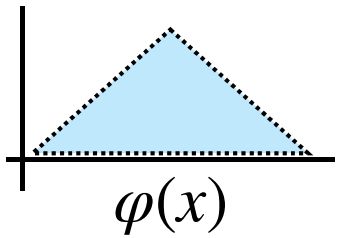
Make 3D histograms of value, 1st, 2nd deriv.



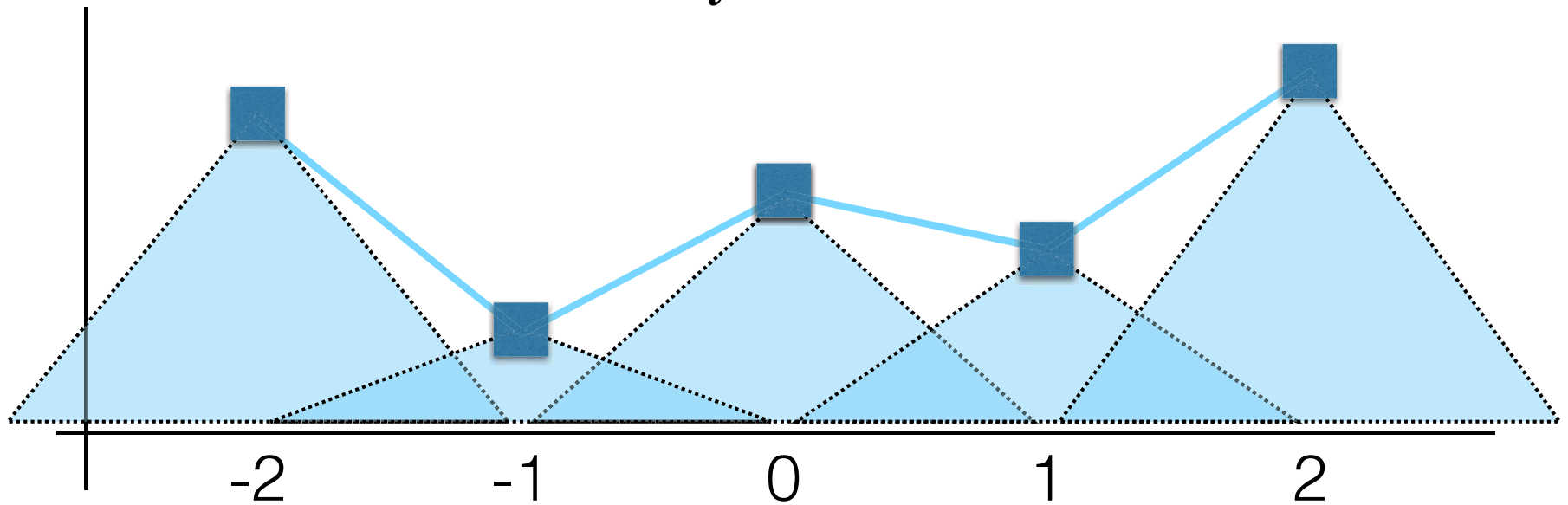
By (1) **inspecting** and
(2) algorithmically **analyzing**
histogram volume, we can
create transfer functions

Some Background: Gradients

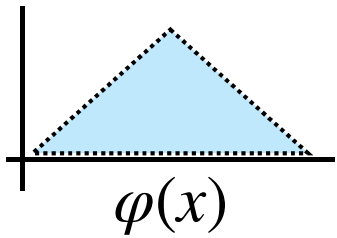
Recall: Finite-Dimensional Function Spaces



$$f(x) = \sum_i \varphi(x - i)c_i$$



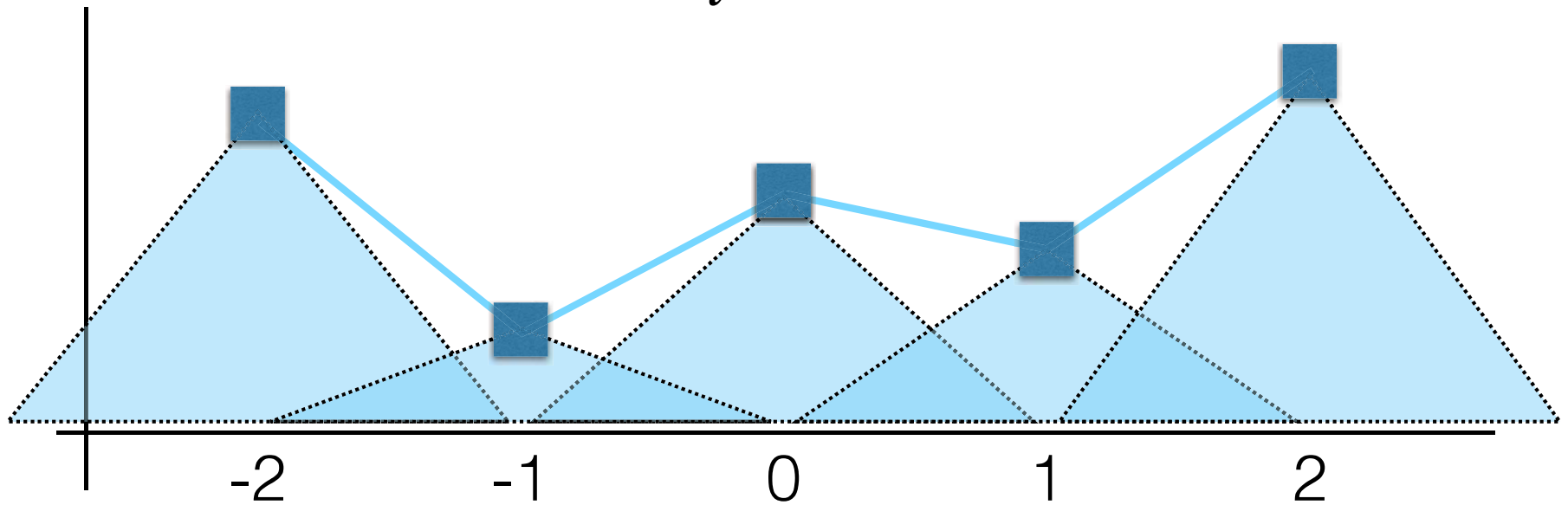
Recall: Finite-Dimensional Function Spaces



$$f(x) = \sum_i \varphi(x - i) c_i$$

Annotations for the equation:

- sums**: points to the summation symbol $\sum_i$.
- simple functions**: points to the basis function $\varphi(x - i)$.
- shifts**: points to the shift term $x - i$.
- scales**: points to the coefficient c_i .



Derivatives of Finite-Dimensional Function Spaces

$$f(x) = \sum_i c_i \varphi(x - i)$$

$$\frac{df}{dx}(x) = \frac{d}{dx} \sum_i c_i \varphi(x - i)$$

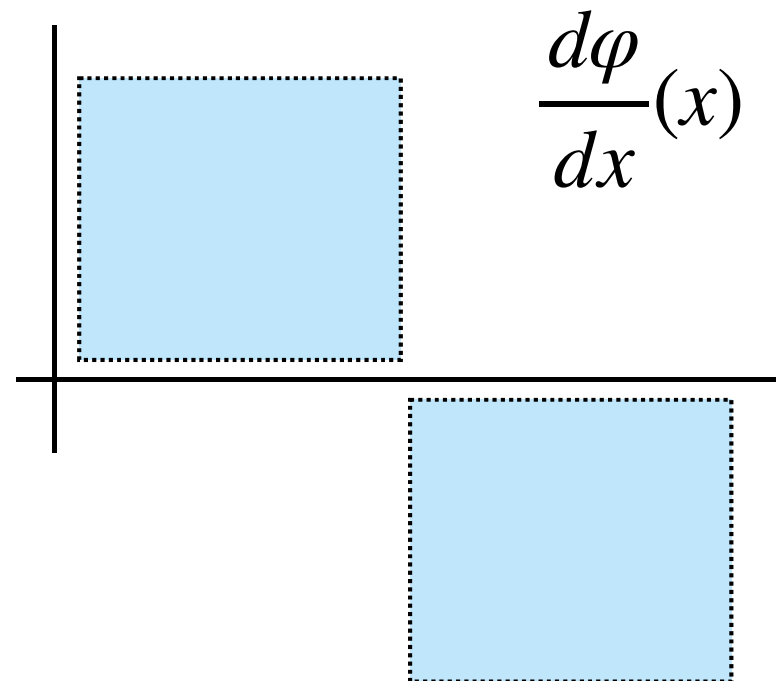
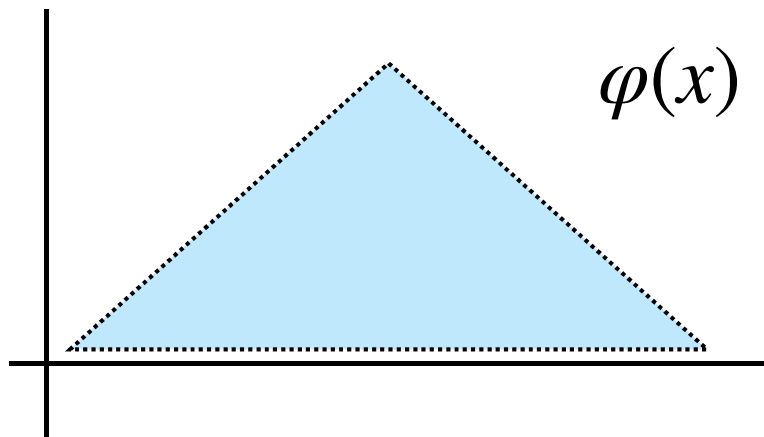
Derivatives of Finite-Dimensional Function Spaces

$$f(x) = \sum_i c_i \varphi(x - i)$$

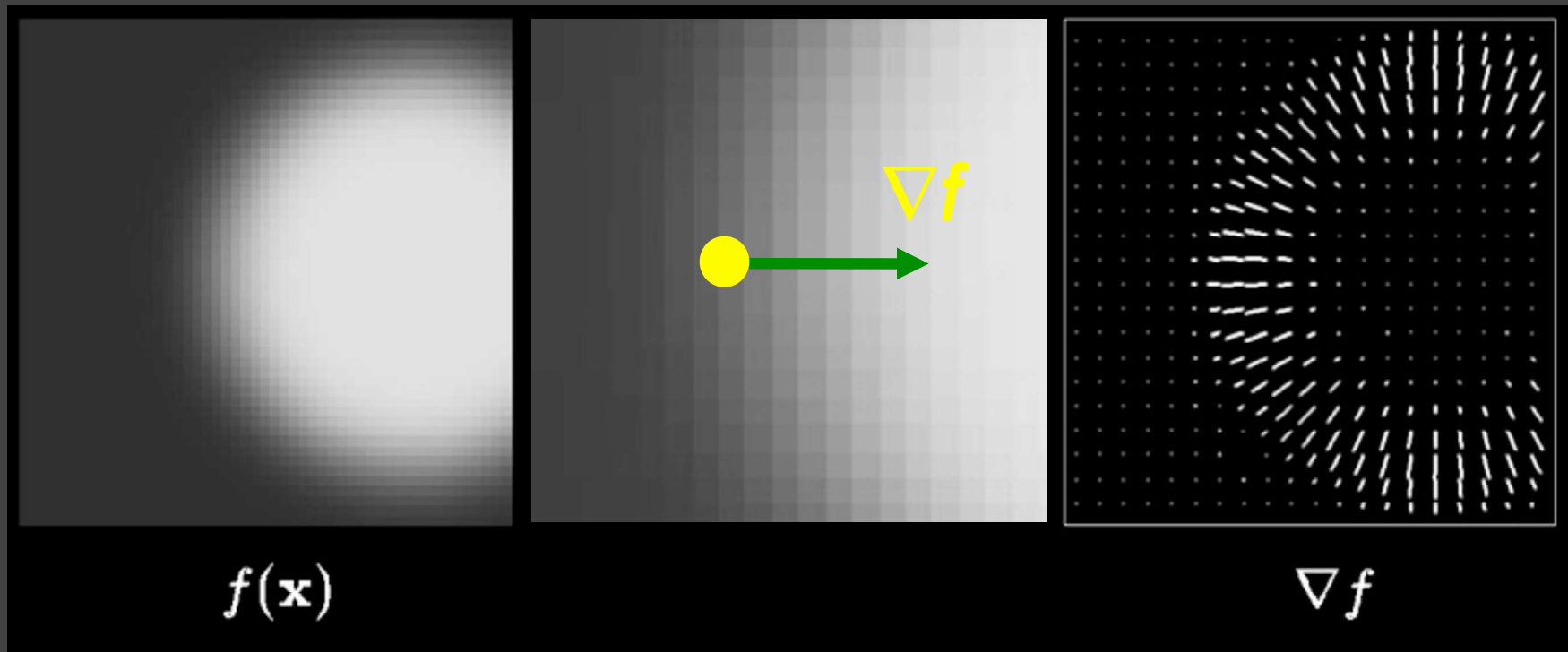
$$\frac{df}{dx}(x) = \sum_i c_i \frac{d\varphi}{dx}(x - i)$$

- Derivatives are just another type of function space where we change the “simple function”

Derivatives of Finite-Dimensional Function Spaces

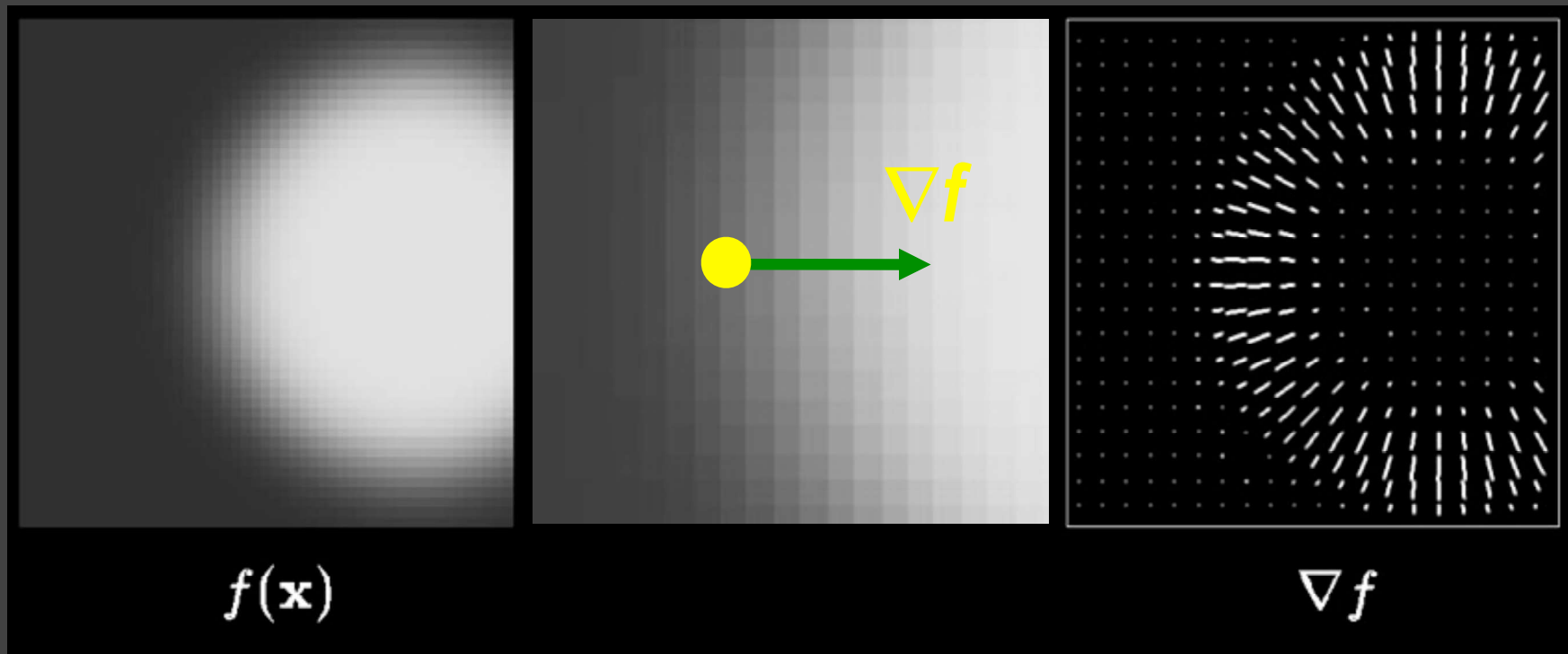


Gradient



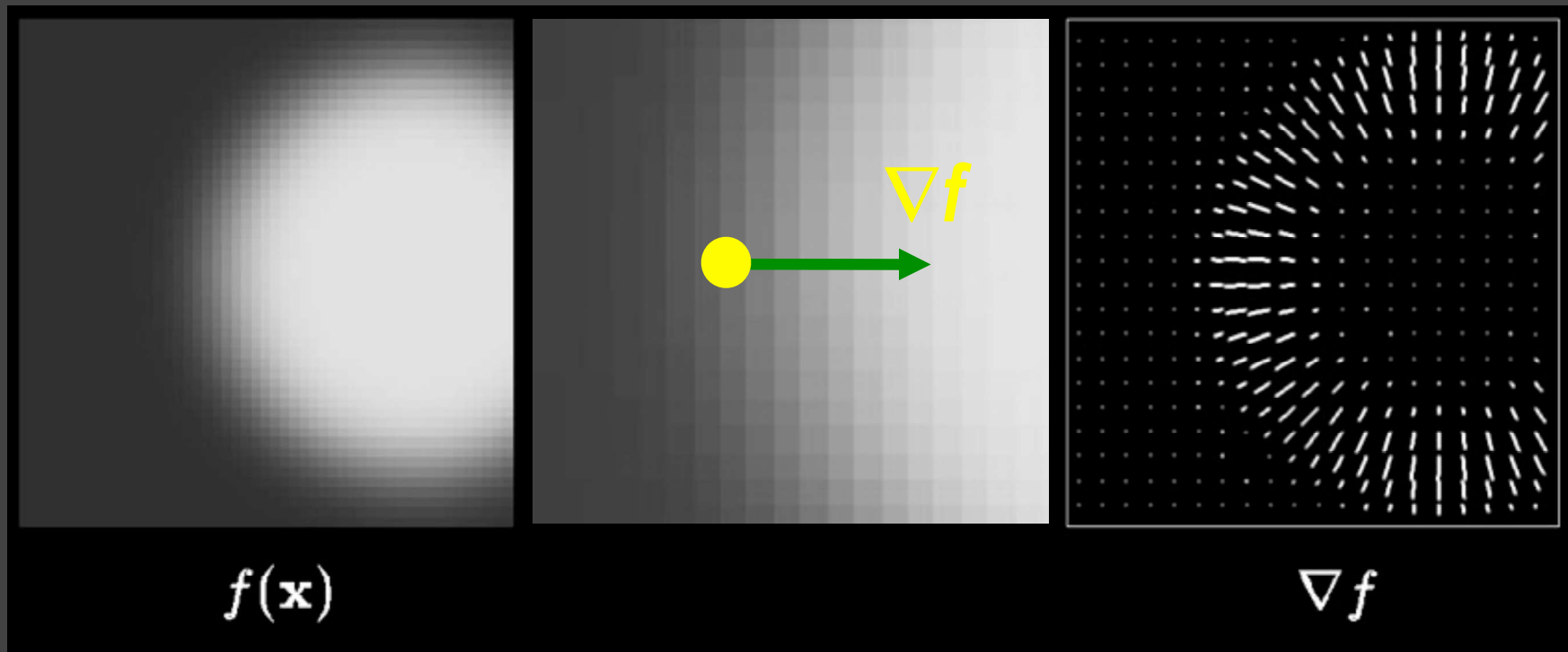
Gradient

$$\nabla f = (dx, dy, dz)$$



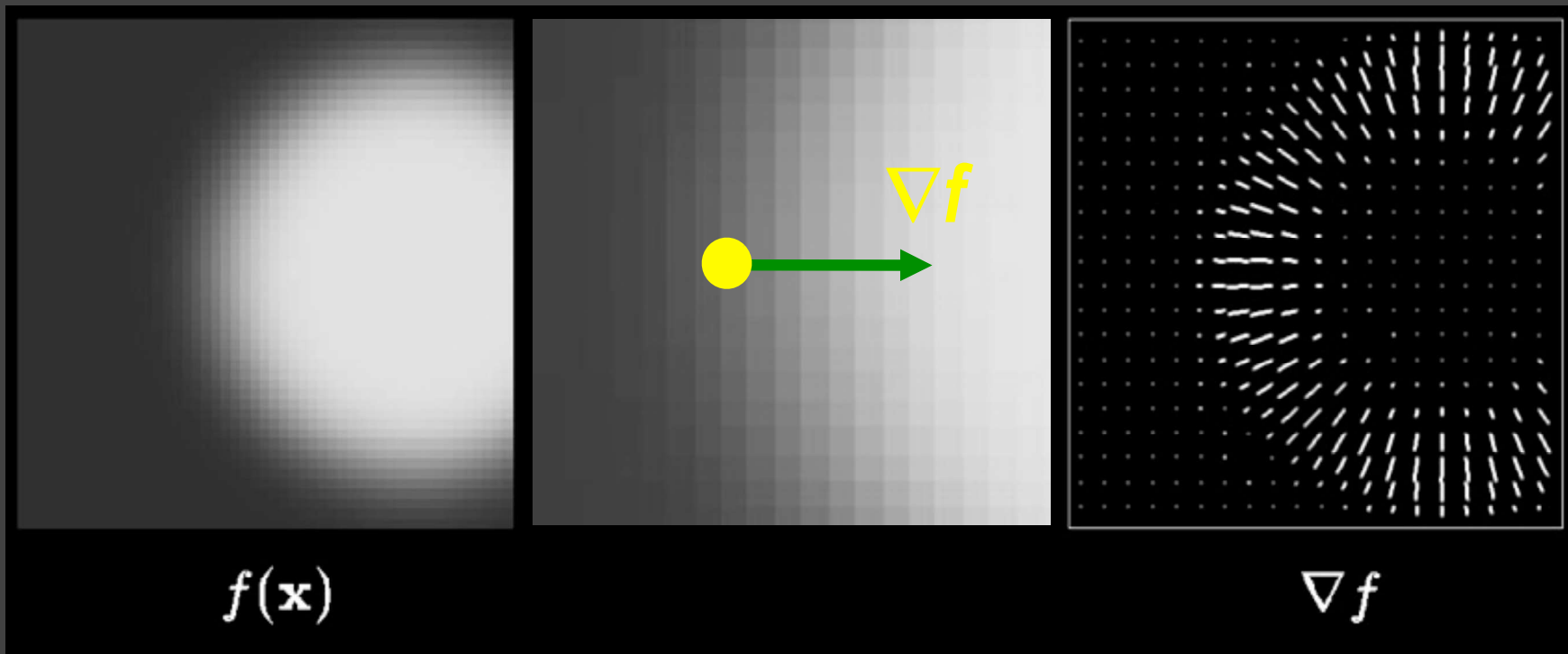
Gradient

$$\begin{aligned}\nabla f &= (dx, dy, dz) \\ &= ((f(1,0,0) - f(-1,0,0))/2,\end{aligned}$$



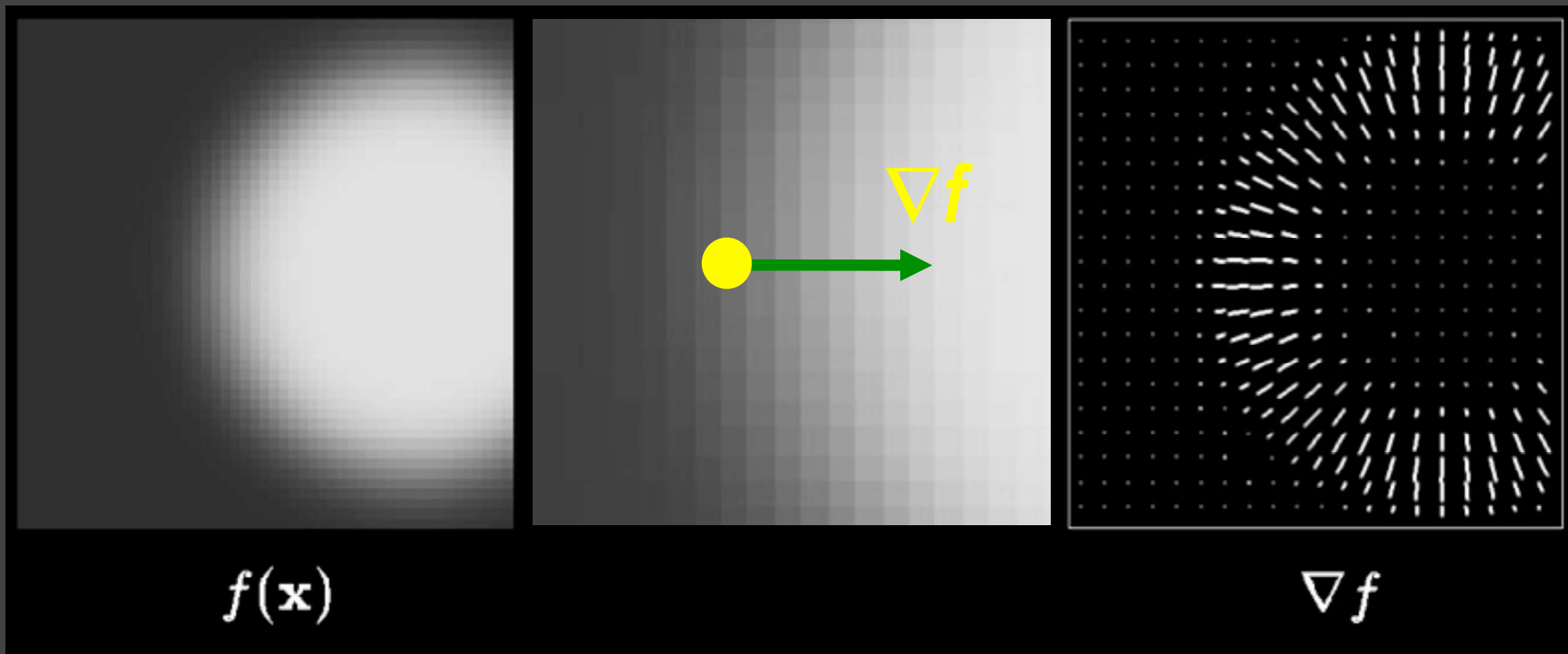
Gradient

$$\begin{aligned}\nabla f &= (dx, dy, dz) \\ &= ((f(1,0,0) - f(-1,0,0))/2, \\ &\quad (f(0,1,0) - f(0,-1,0))/2, \\ &\quad \dots)\end{aligned}$$



Gradient

$$\begin{aligned}\nabla f &= (dx, dy, dz) \\ &= ((f(1,0,0) - f(-1,0,0))/2, \\ &\quad (f(0,1,0) - f(0,-1,0))/2, \\ &\quad (f(0,0,1) - f(0,0,-1))/2)\end{aligned}$$

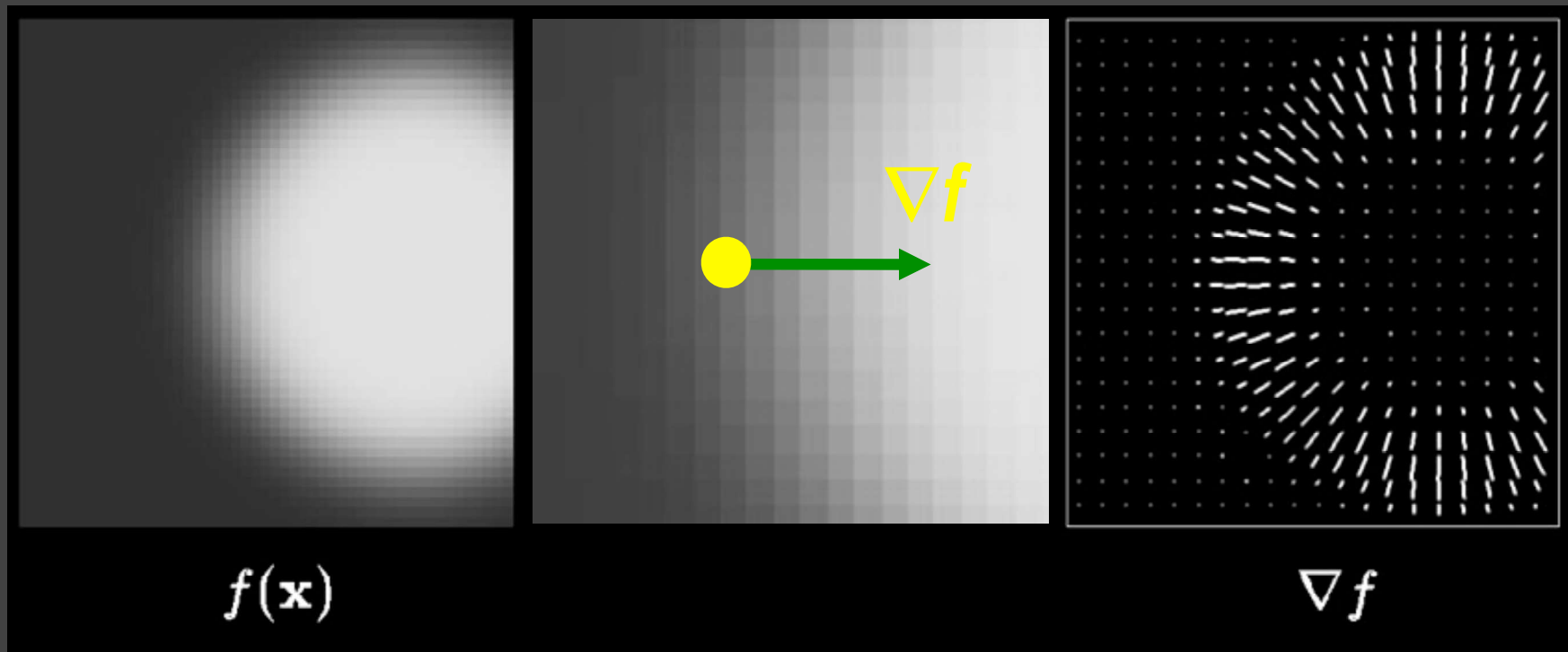


Gradient

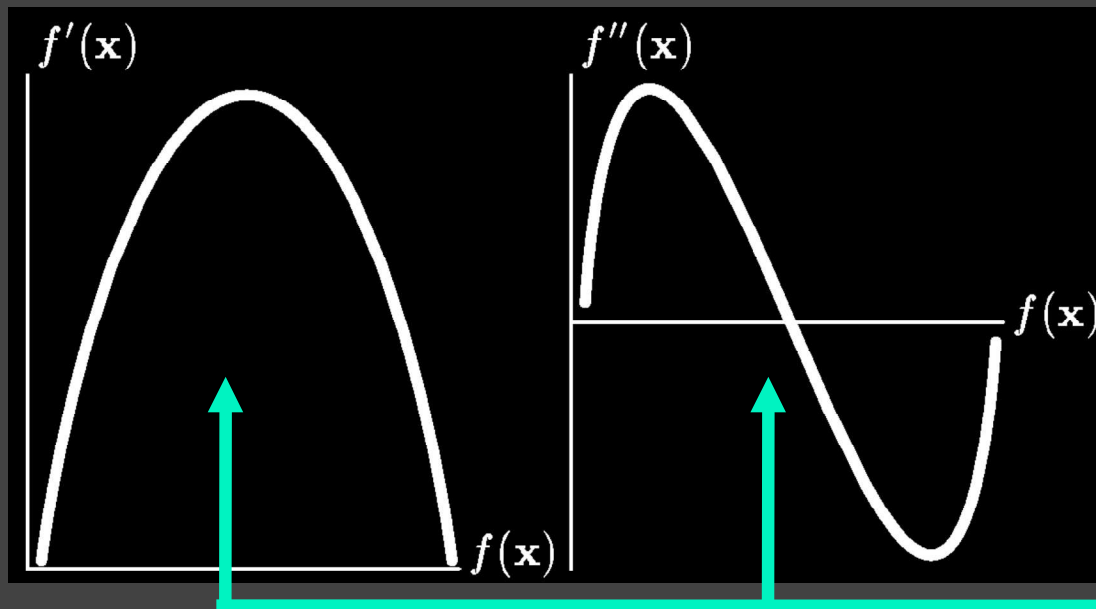
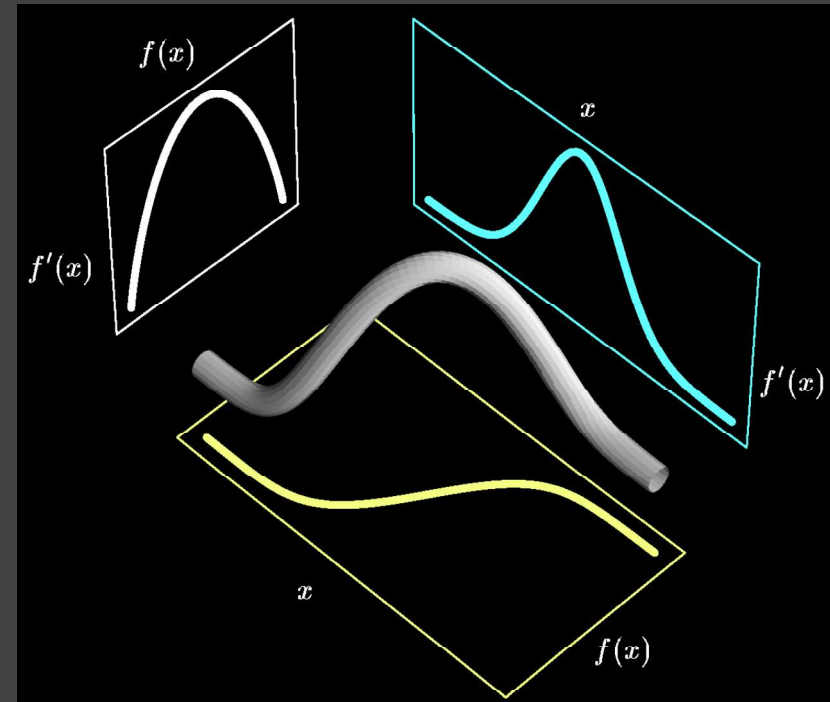
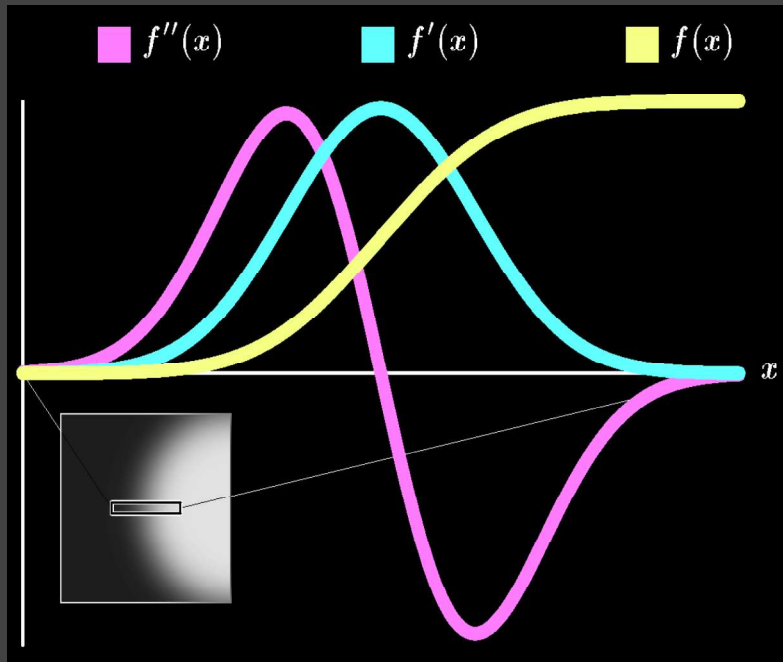
$$\nabla f = (dx, dy, dz)$$

$$= ((f(1,0,0) - f(-1,0,0))/2, \\ (f(0,1,0) - f(0,-1,0))/2, \\ (f(0,0,1) - f(0,0,-1))/2)$$

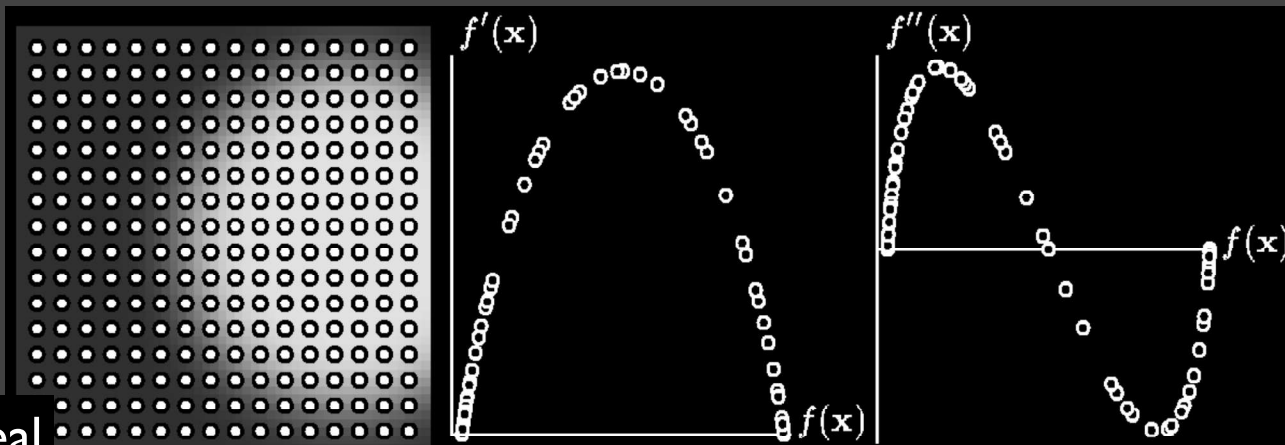
- Approximates "surface normal" (of isosurface)



Derivative relationships



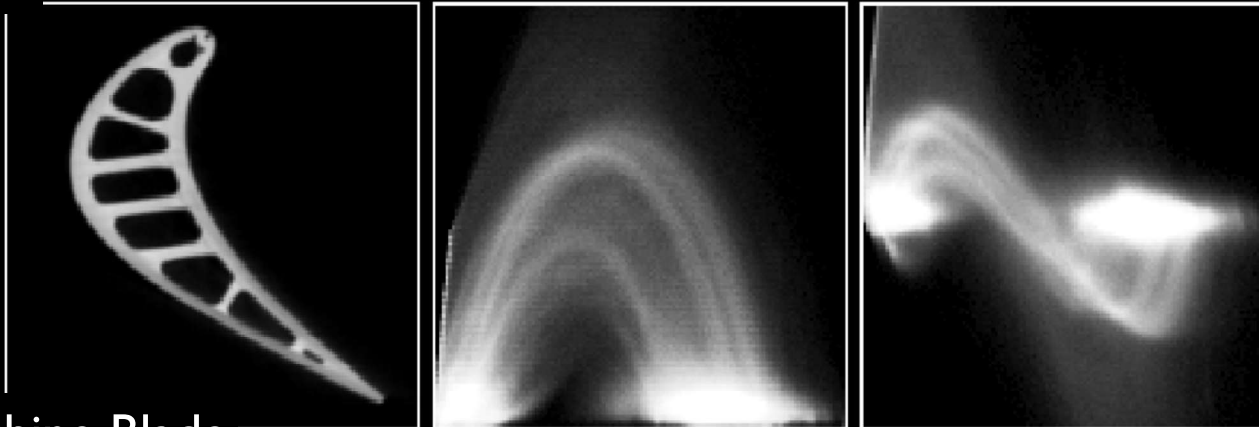
Edges at maximum
of 1st derivative or
zero-crossing of 2nd



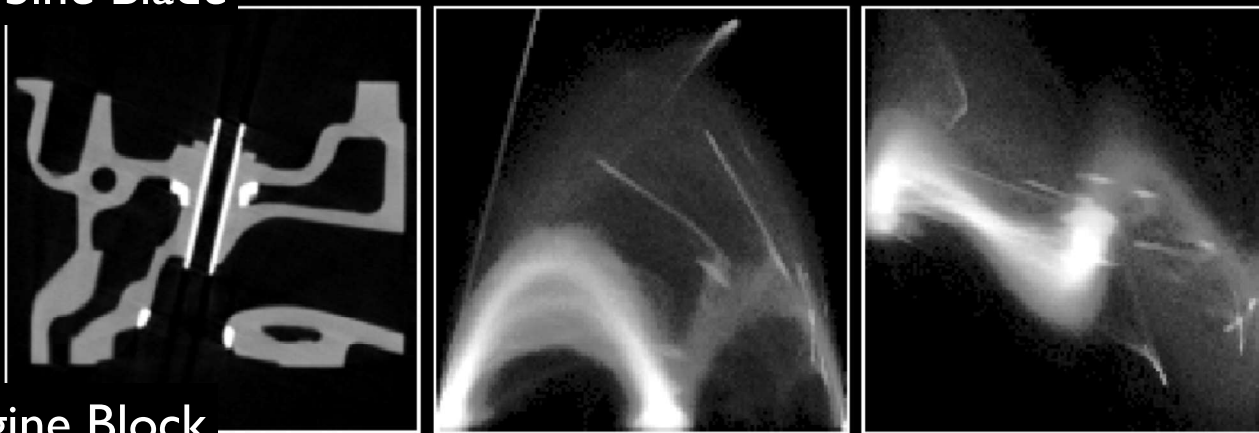
Ideal

Project histogram
volume to 2D
scatterplots

- Visual summary
- Interpreted for TF guidance
- No reliance on boundary model at this stage



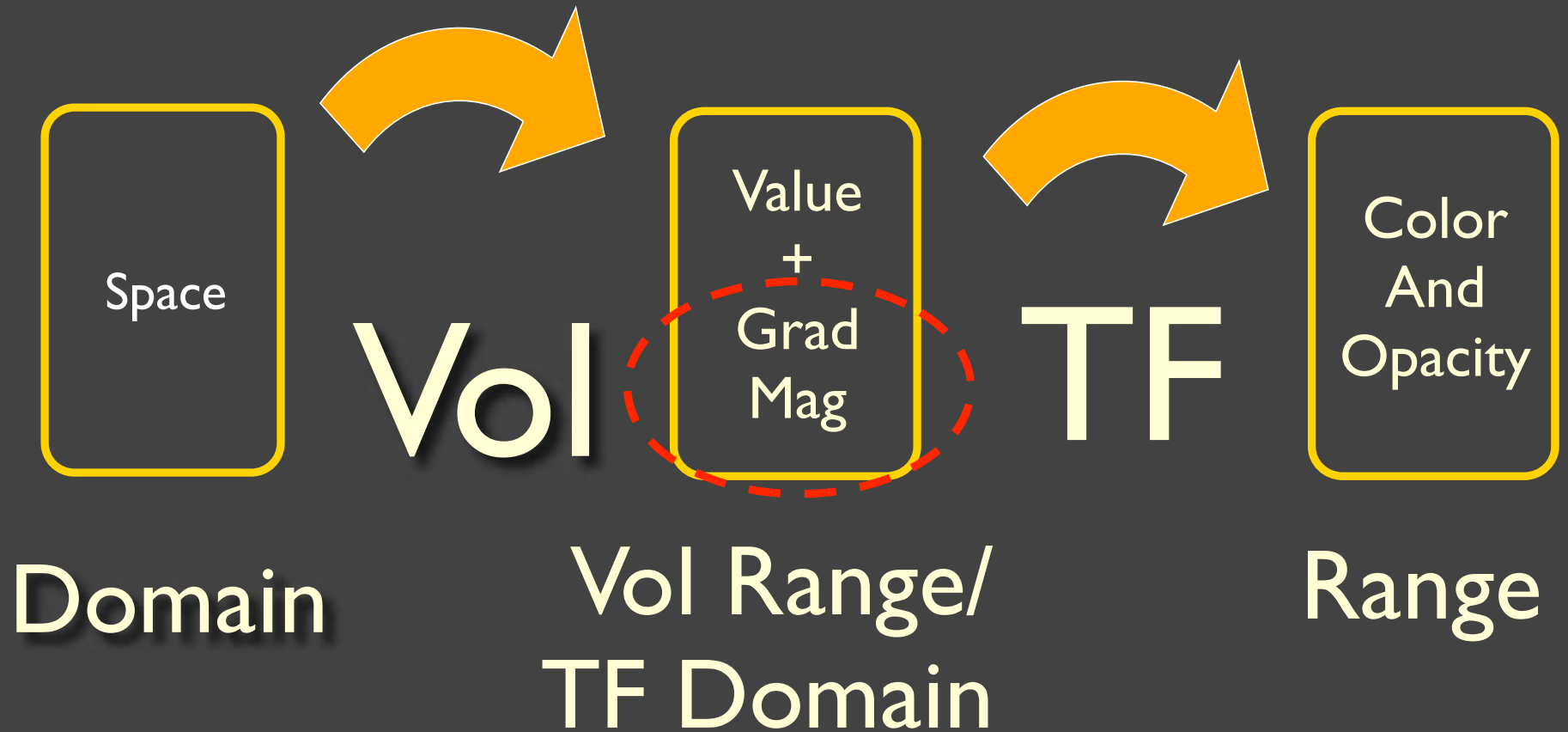
Turbine Blade



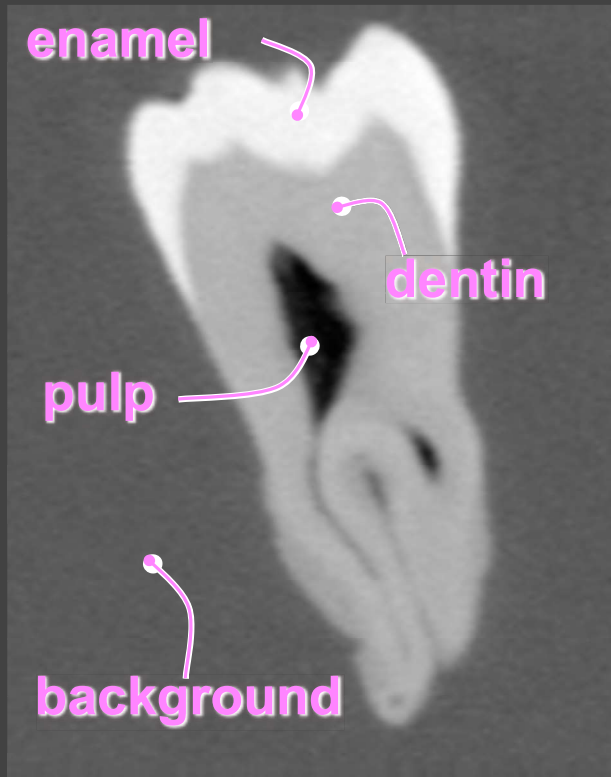
Engine Block

Higher-Dimensional Transfer Functions

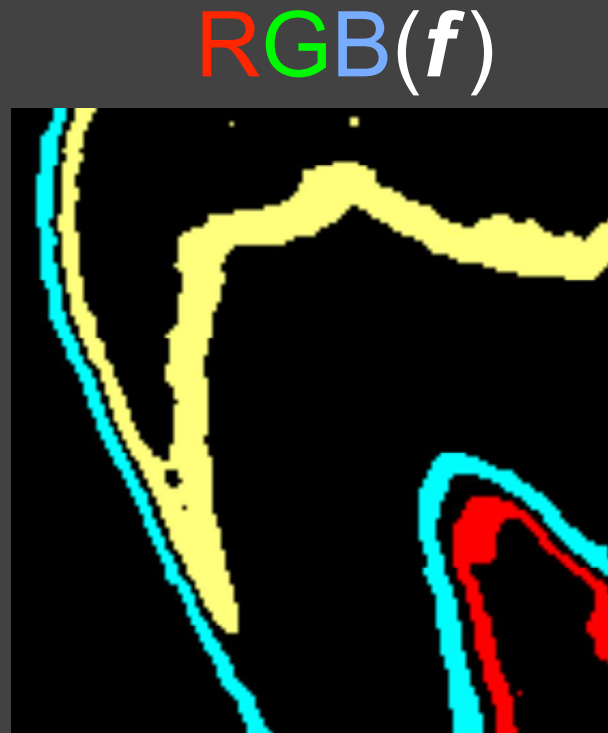
Basic Transfer Functions:



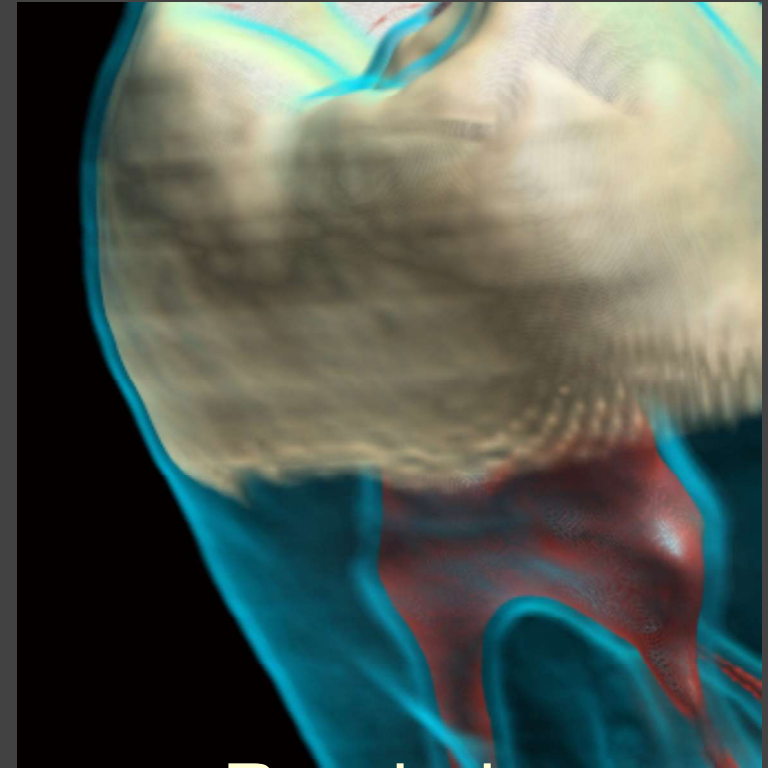
1D TFs: limitation



Slice



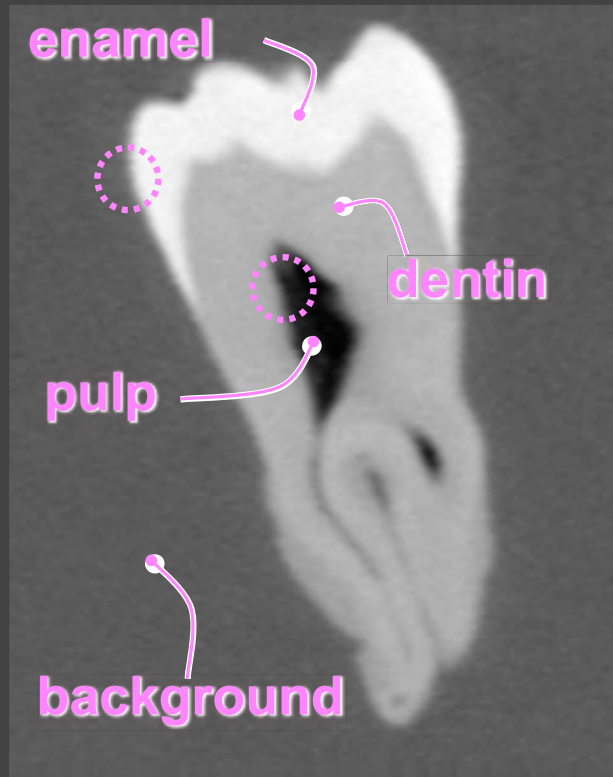
1D TF output



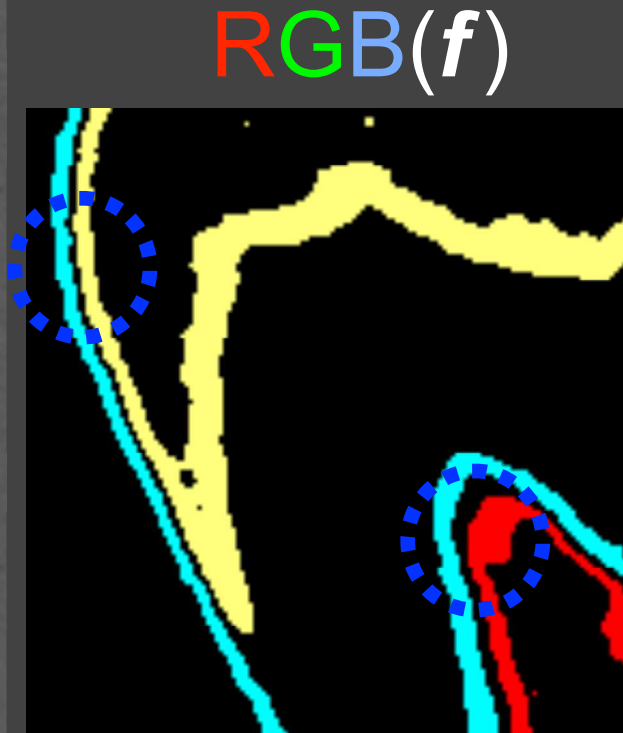
Rendering

1D transfer functions can not accurately capture all material boundaries

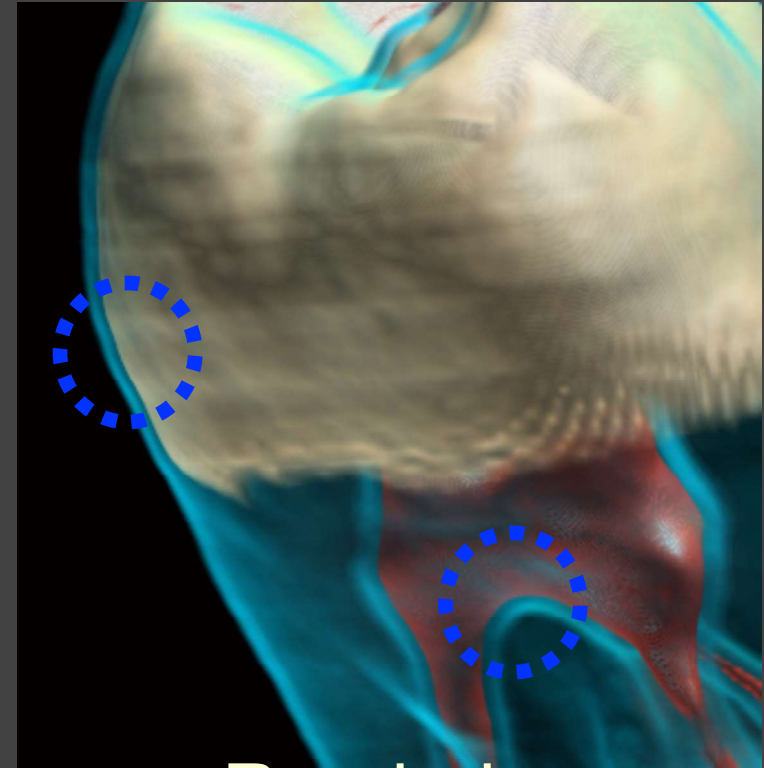
1D TFs: limitation



Slice



1D TF output



Rendering

1D transfer functions can not accurately capture all material boundaries

1D $\rightarrow$ 2D Transfer Function



$\text{RGB}(f)$
 $\alpha(f)$ } Generalize...



1D $\rightarrow$ 2D Transfer Function



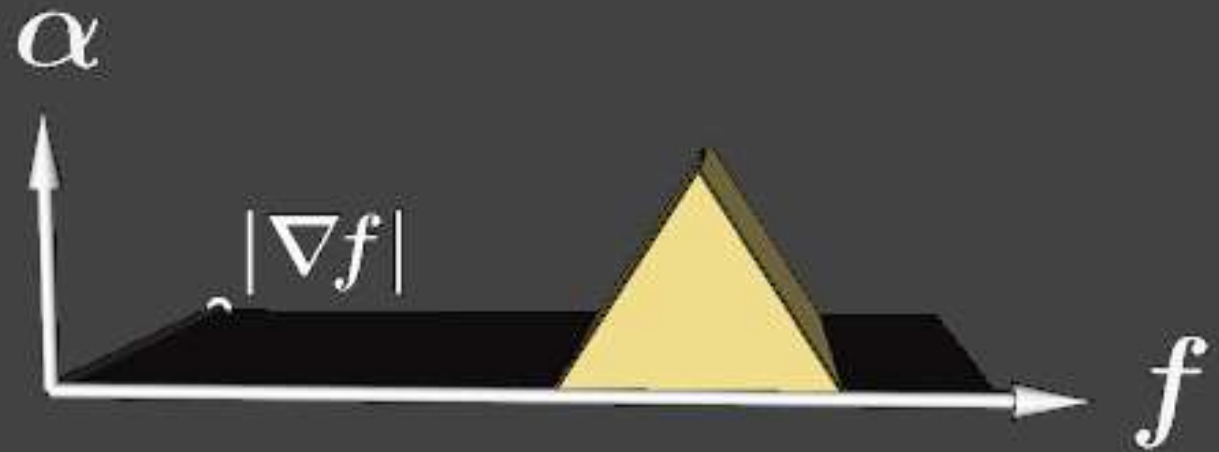
$\text{RGB}(f)$
 $\alpha(f)$ } Generalize...



1D $\rightarrow$ 2D Transfer Function



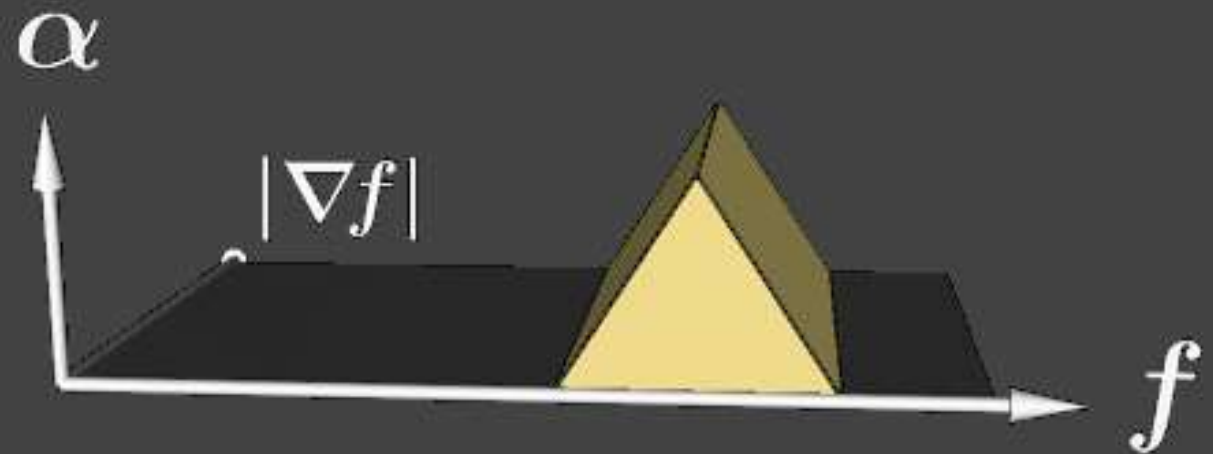
$$\left. \begin{array}{l} \text{RGB}(f) \\ \alpha(f) \end{array} \right\} \text{Generalize...}$$



1D $\rightarrow$ 2D Transfer Function



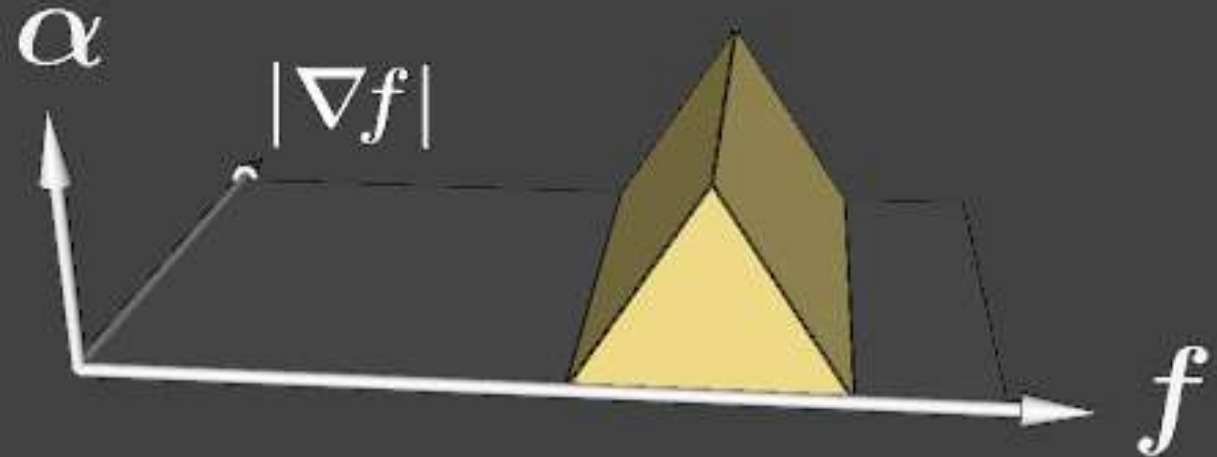
$\left. \begin{matrix} \text{R} \text{G} \text{B}(f) \\ \alpha(f) \end{matrix} \right\} \text{Generalize...}$



1D $\rightarrow$ 2D Transfer Function



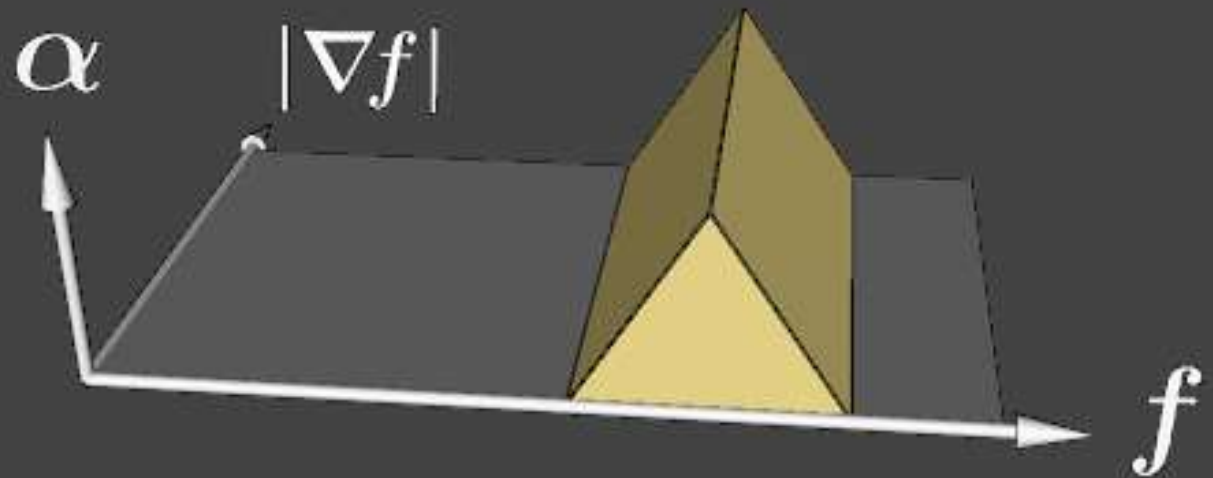
$\text{RGB}(f)$
 $\alpha(f)$ } Generalize...



1D $\rightarrow$ 2D Transfer Function



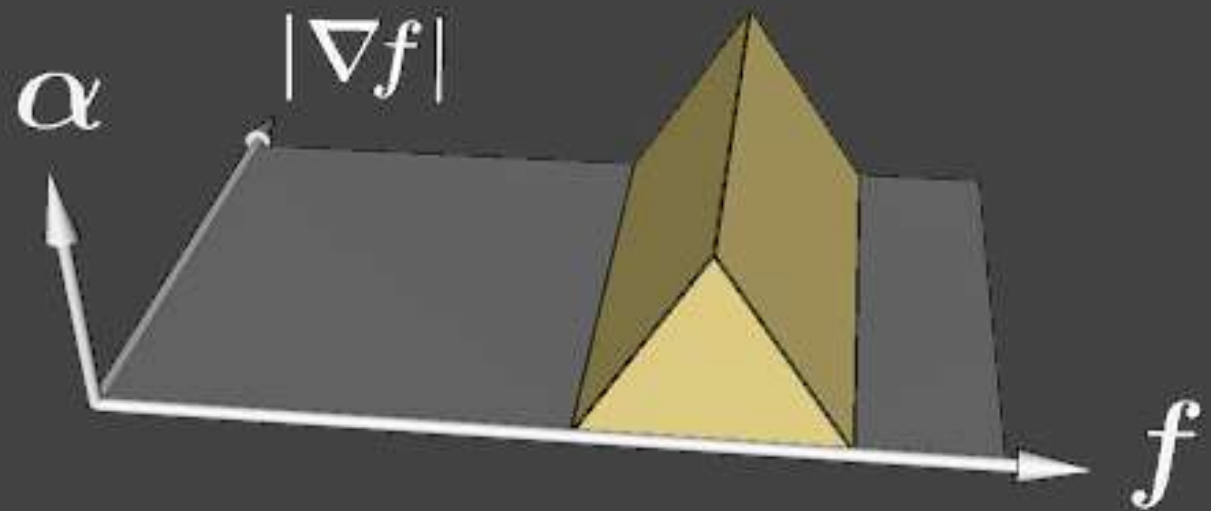
$\text{RGB}(f)$
 $\alpha(f)$ } Generalize...



2D Transfer Function



$$\left. \begin{array}{l} \text{RGB}(f) \\ \alpha(f) \end{array} \right\} \text{Generalize...}$$

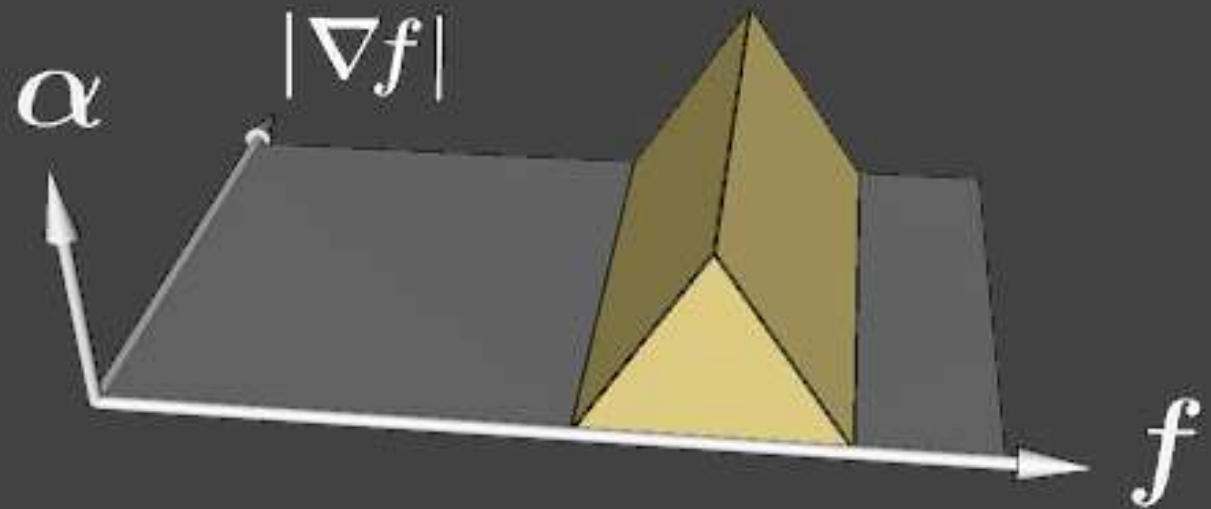


$$\rightarrow \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array}$$

2D Transfer Function



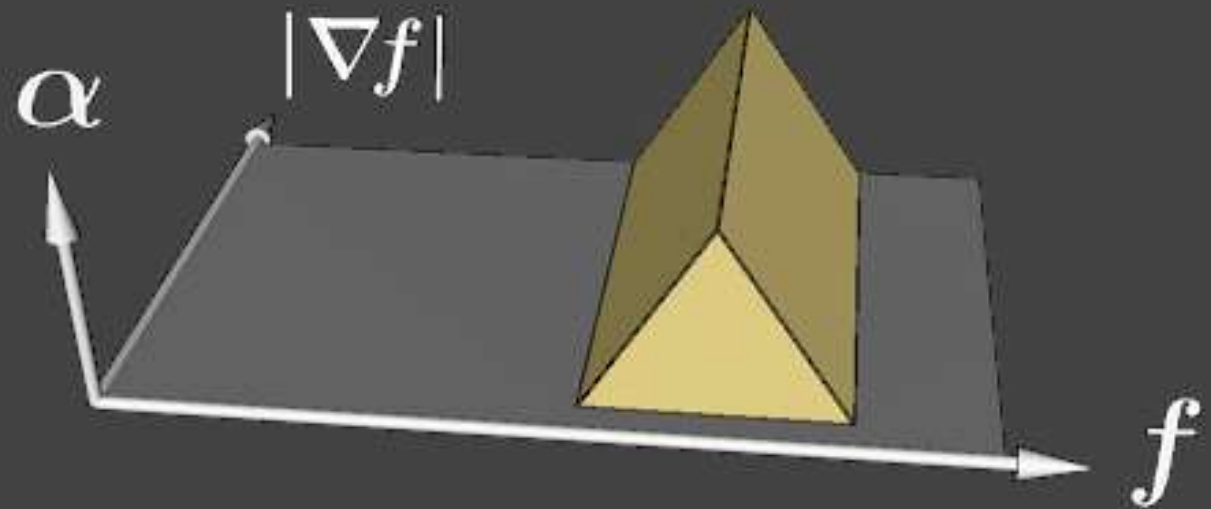
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



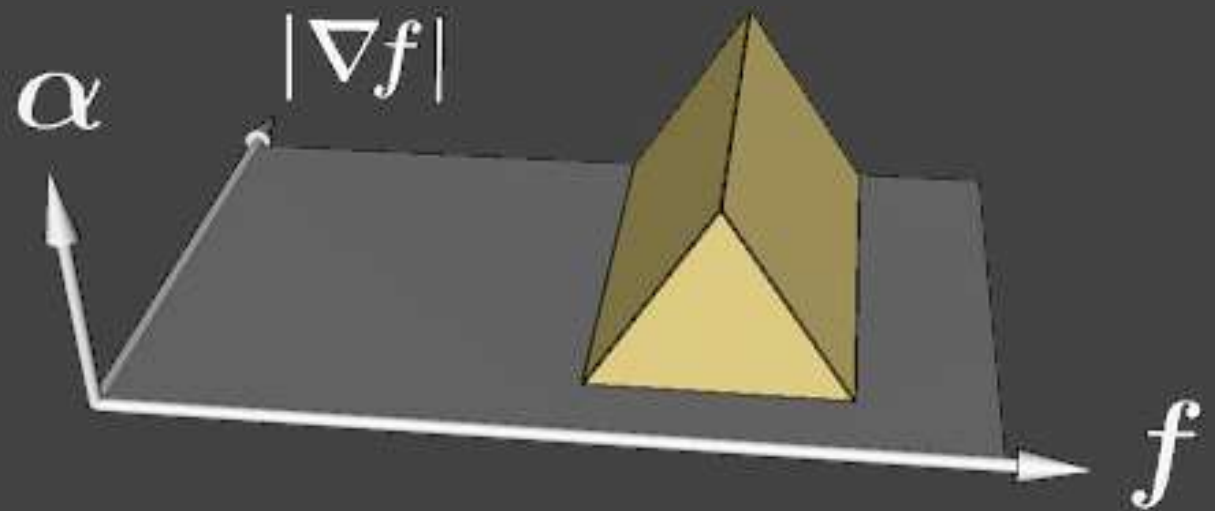
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



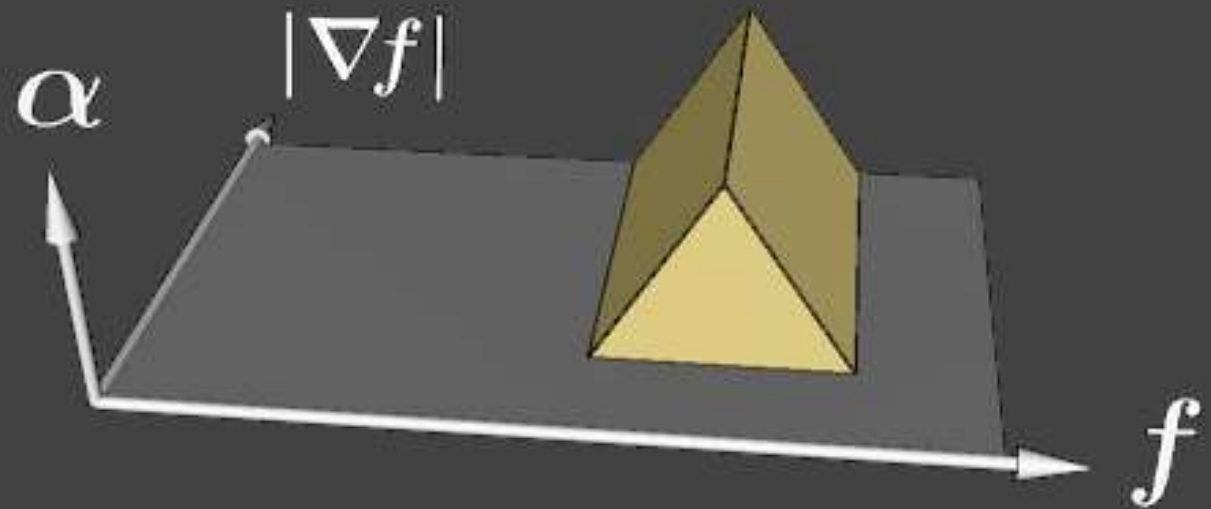
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



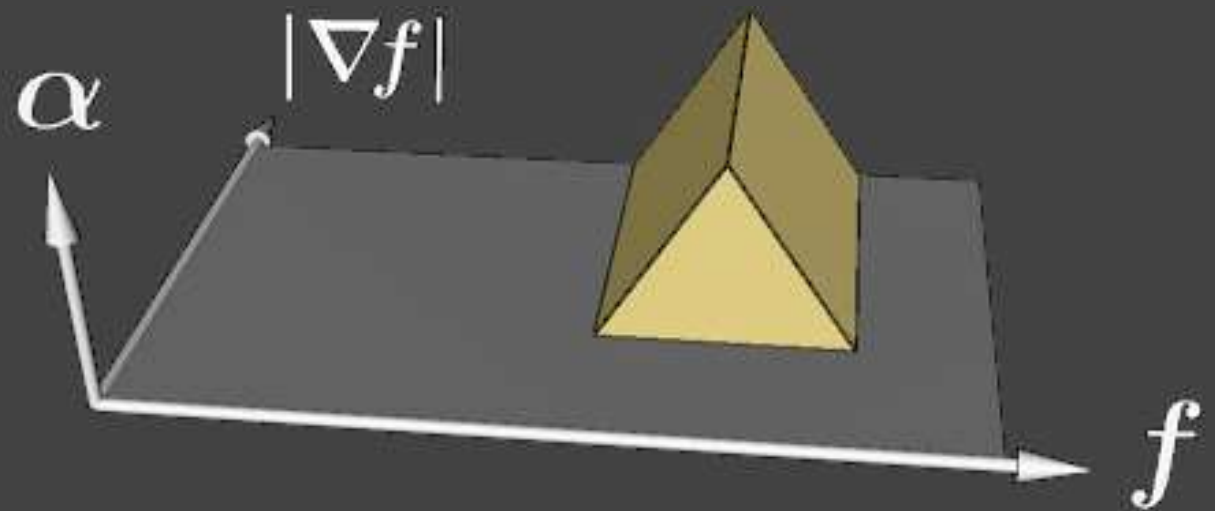
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



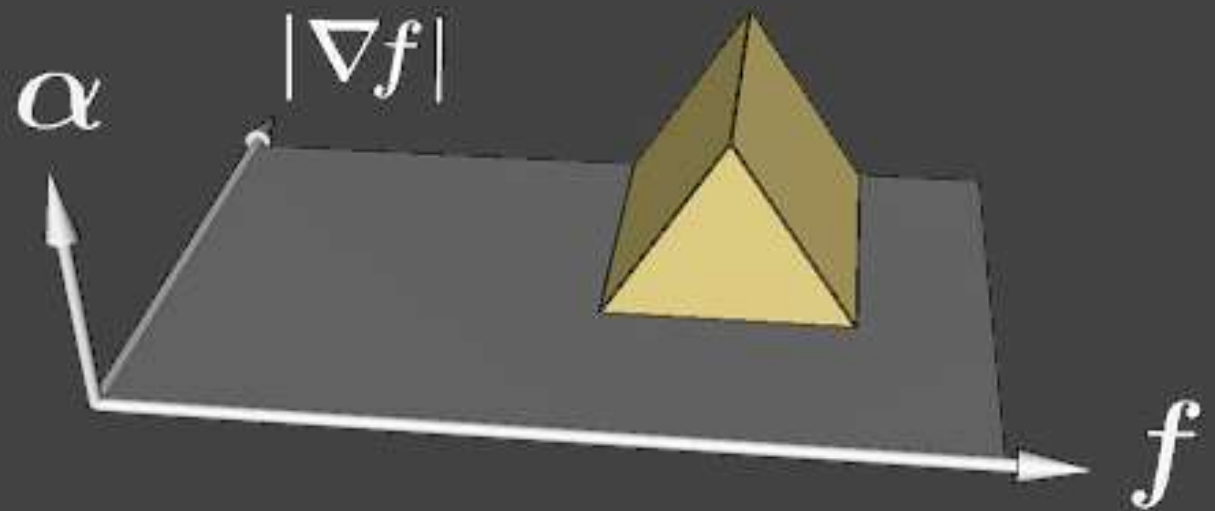
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



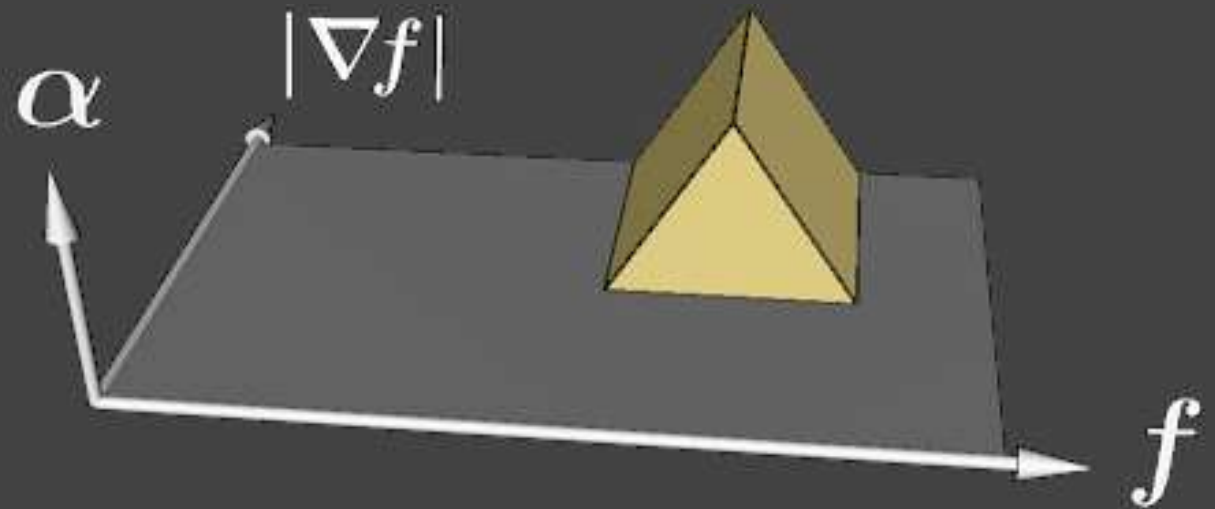
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



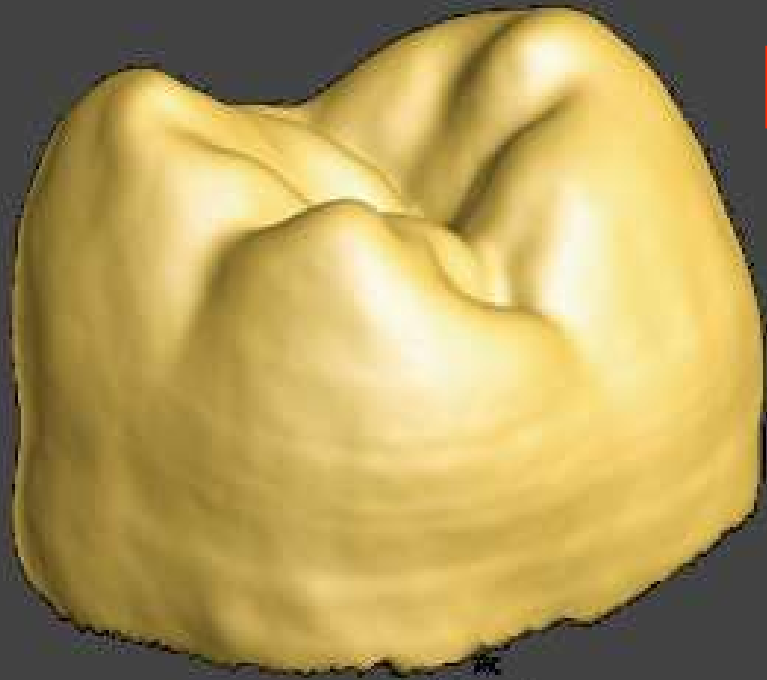
2D Transfer Function



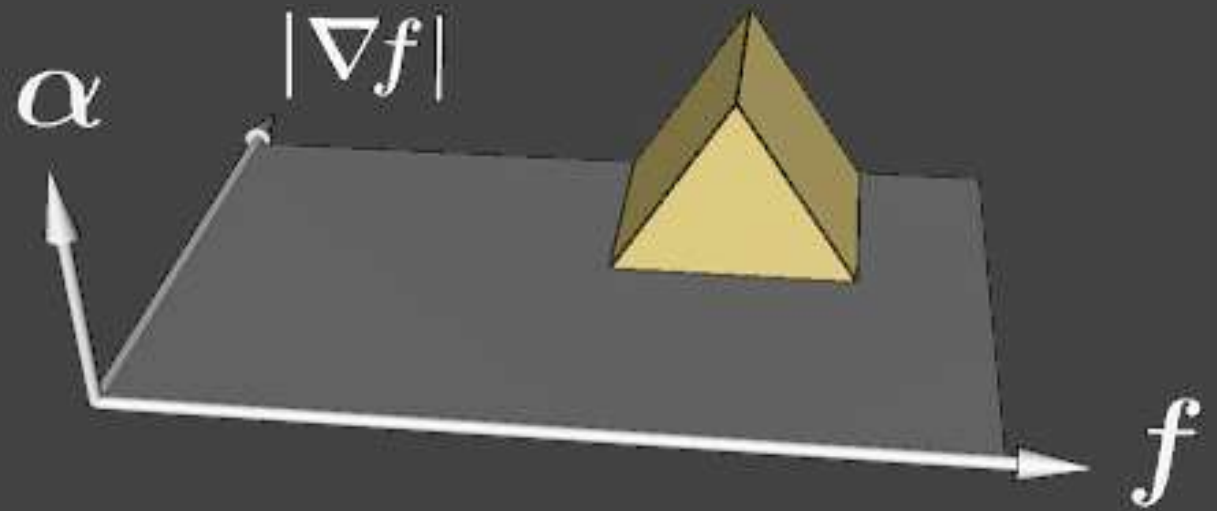
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



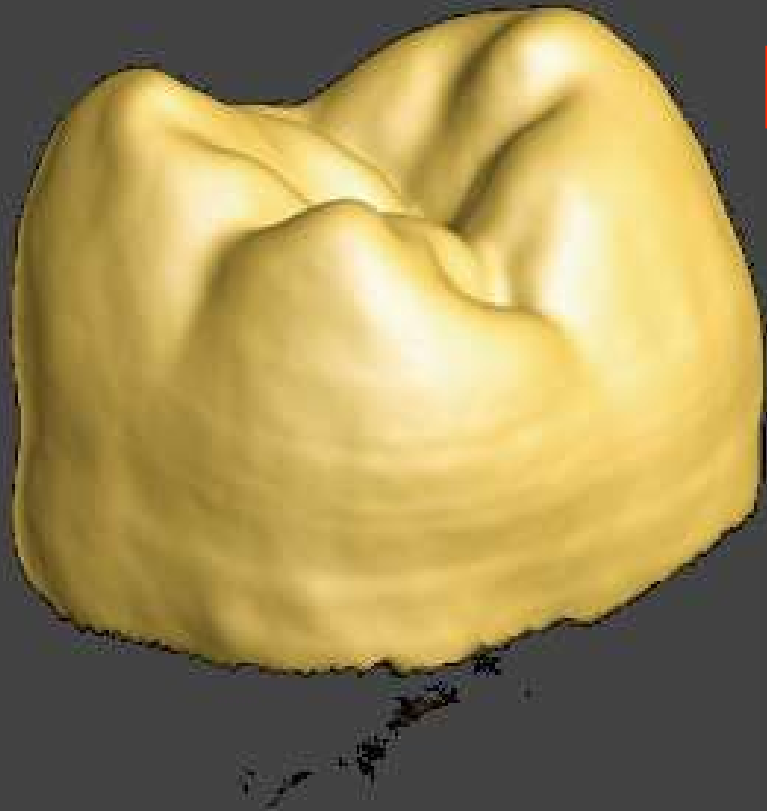
2D Transfer Function



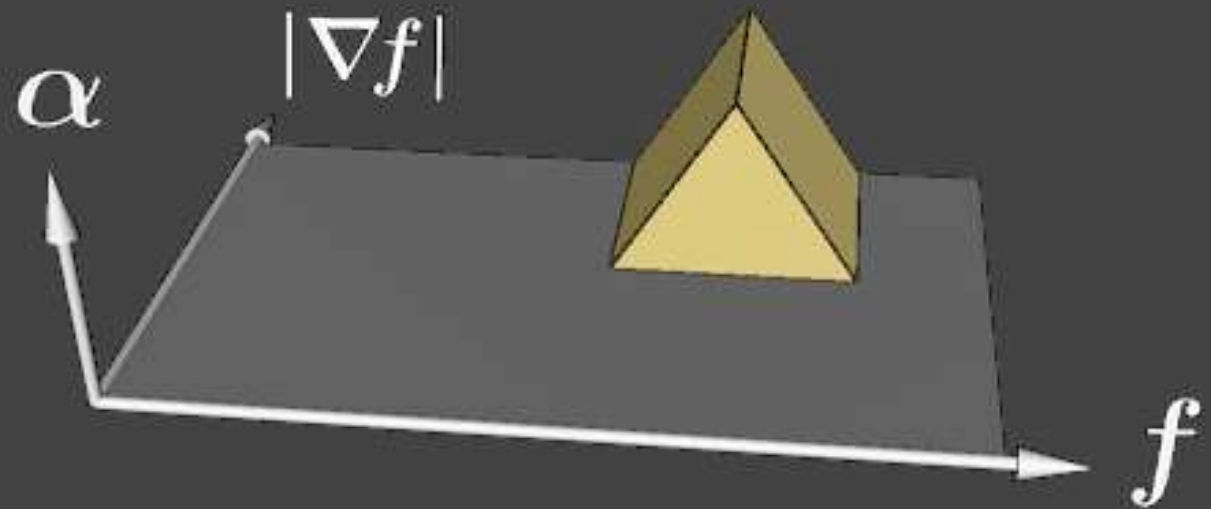
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



2D Transfer Function



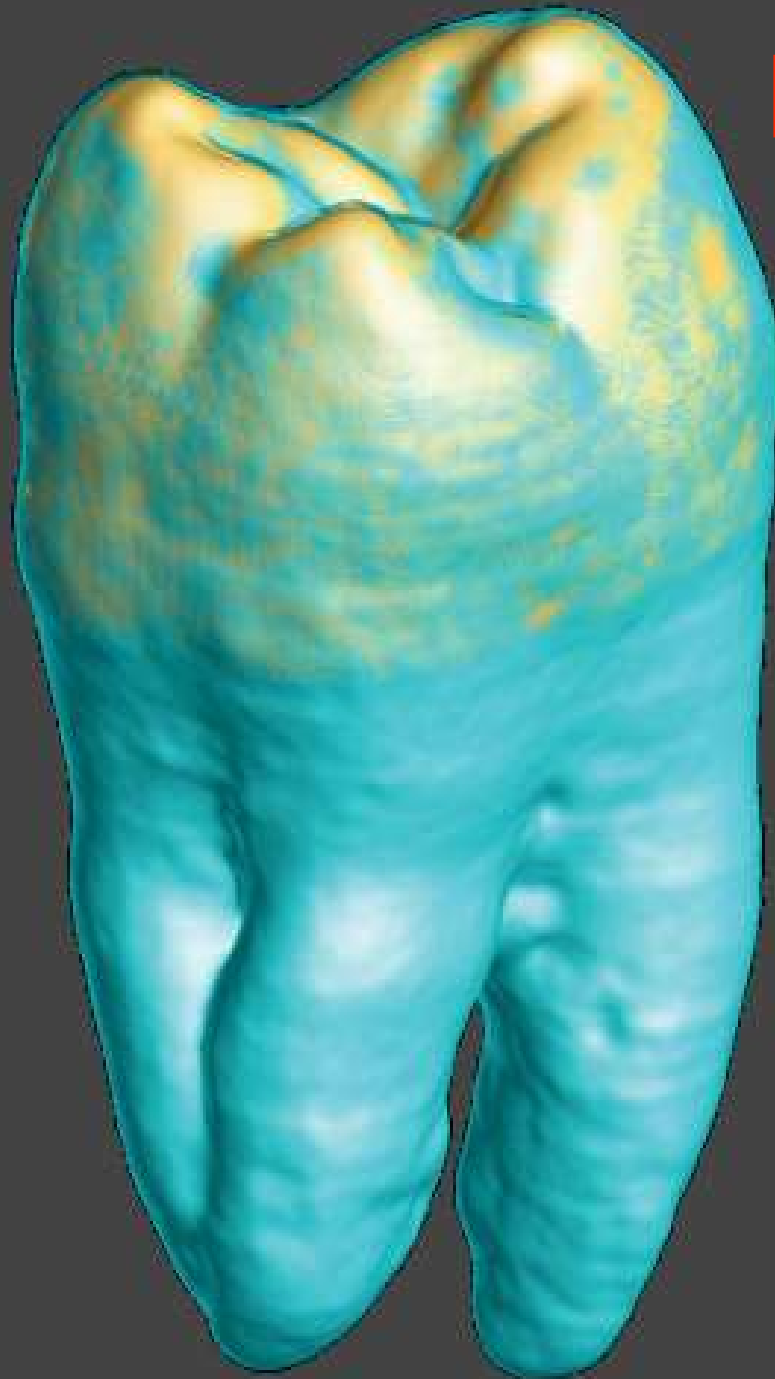
$$\left. \begin{array}{l} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{array} \right\} \text{Modify...}$$



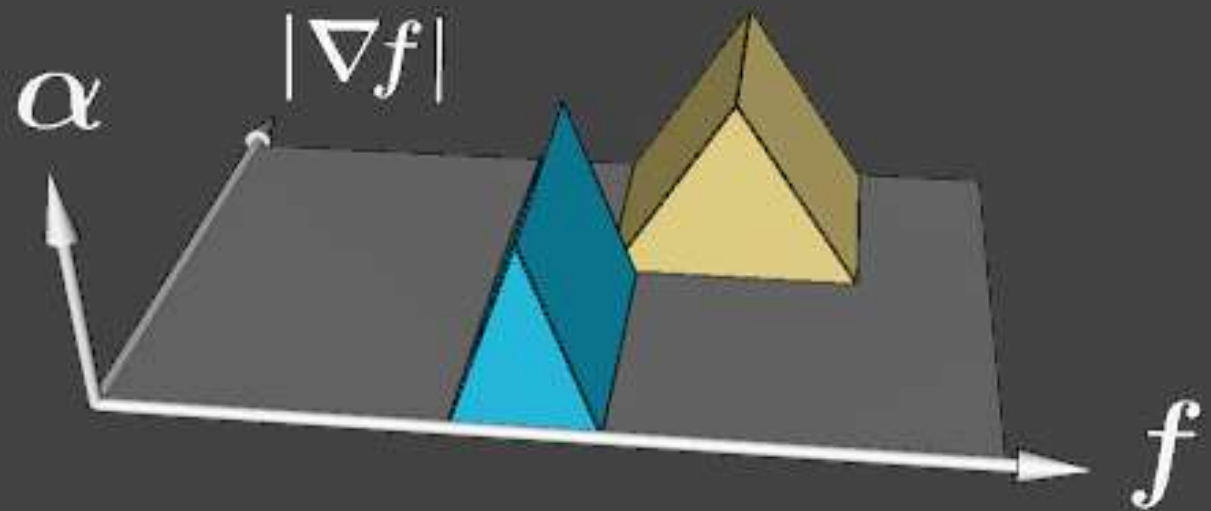
2D transfer functions give greater **flexibility** in boundary visualization

Display of Surfaces from Volume Data, Levoy 1988

2D Transfer Function



$$\left. \begin{matrix} \text{RGB}(f, |\nabla f|) \\ \alpha(f, |\nabla f|) \end{matrix} \right\} \text{Modify...}$$



Trying to reintroduce
dentin / background
boundary ...





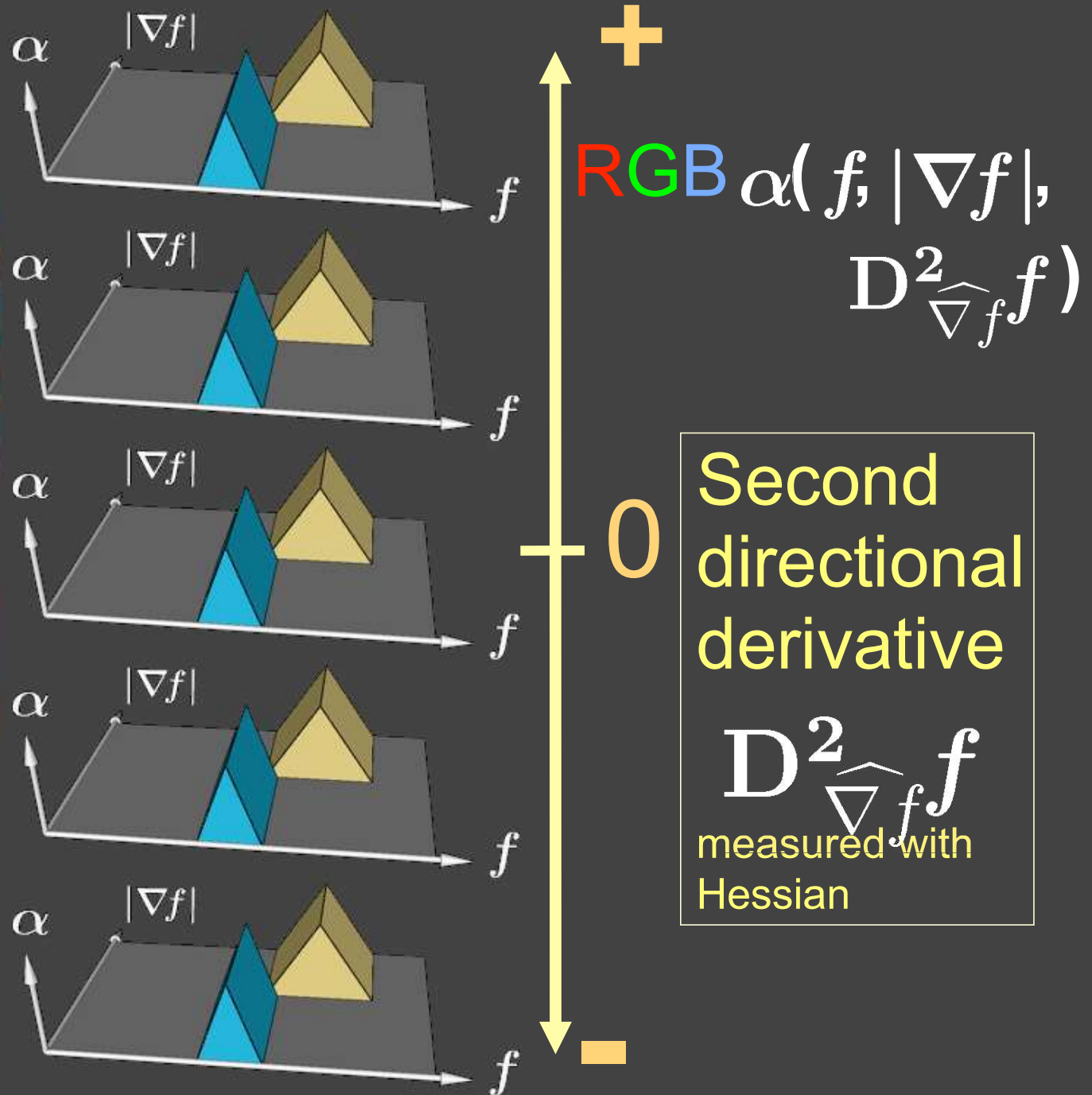
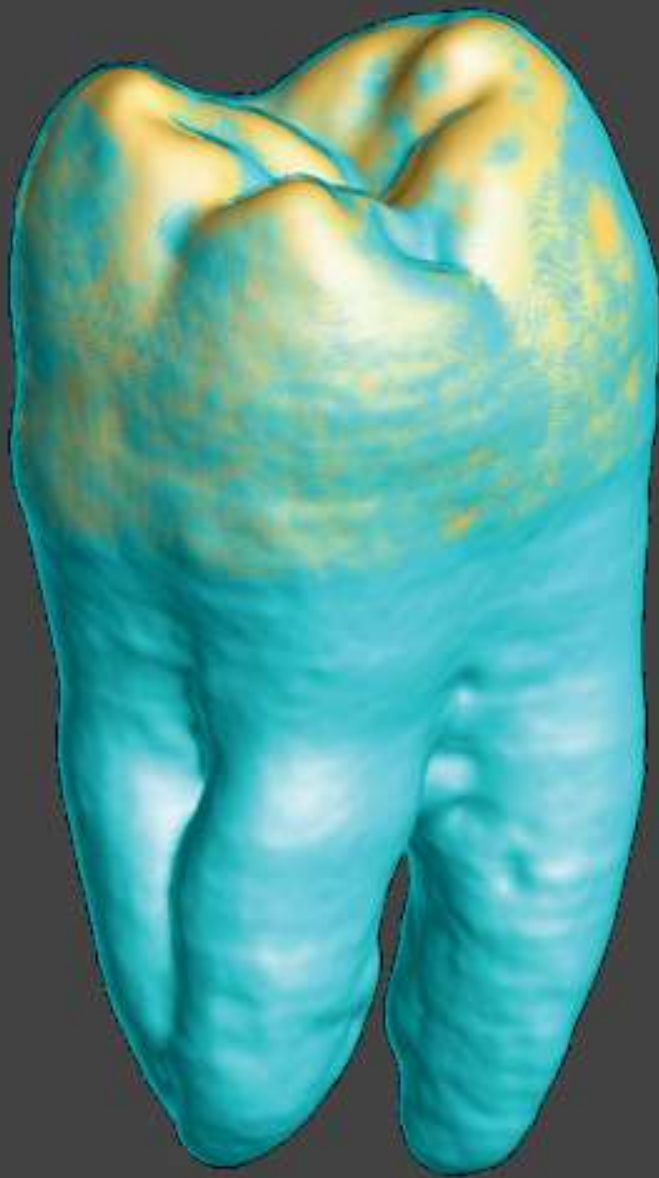
Break for Questions

2-Dimensional TFs

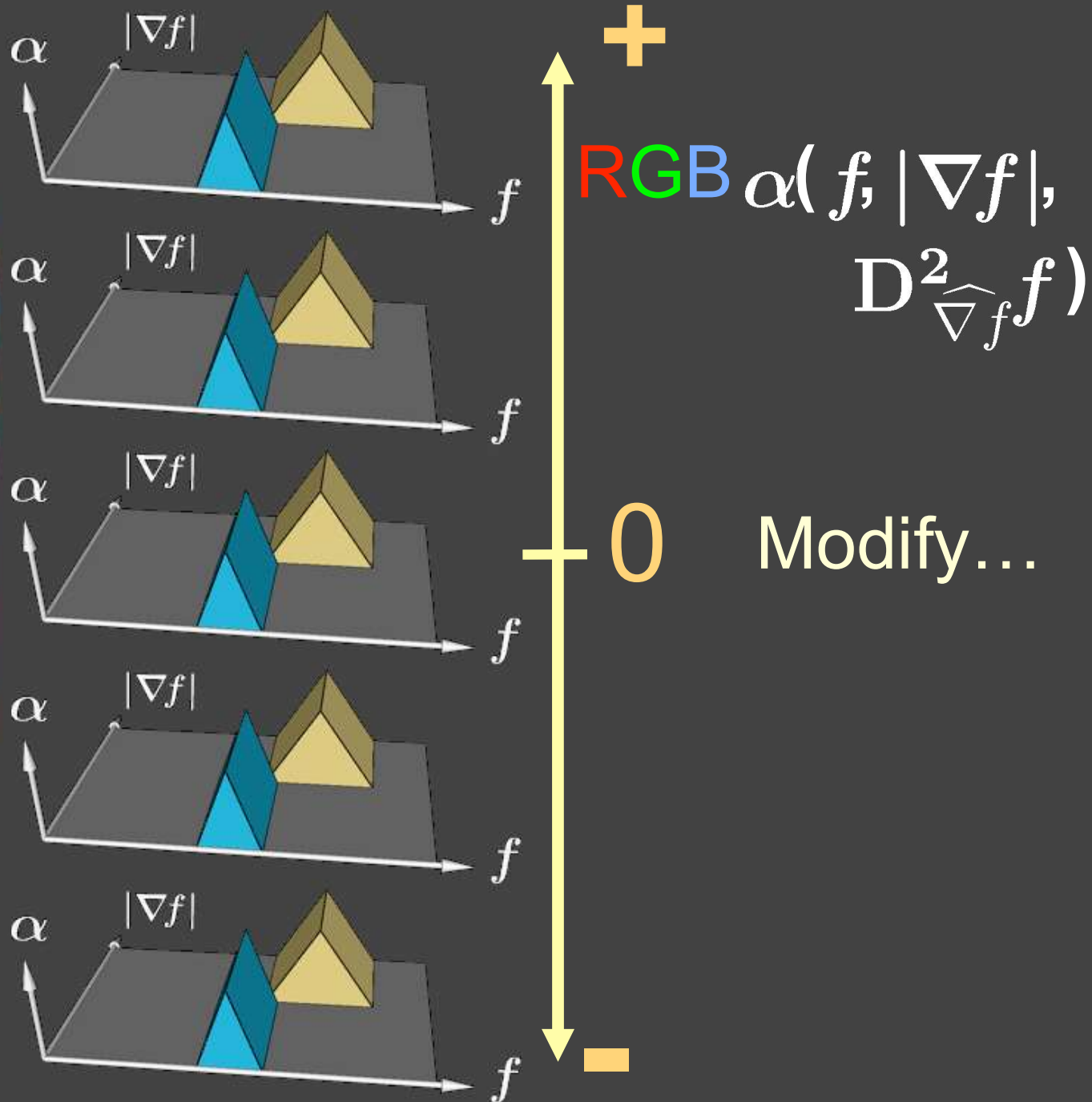
- Strengths:
 - Better flexibility, specificity
 - Higher quality visualizations
- Weaknesses:
 - Can be even harder to specify
 - Unintuitive relationship with boundaries
 - Greater demands on user interface

Other TF Methods

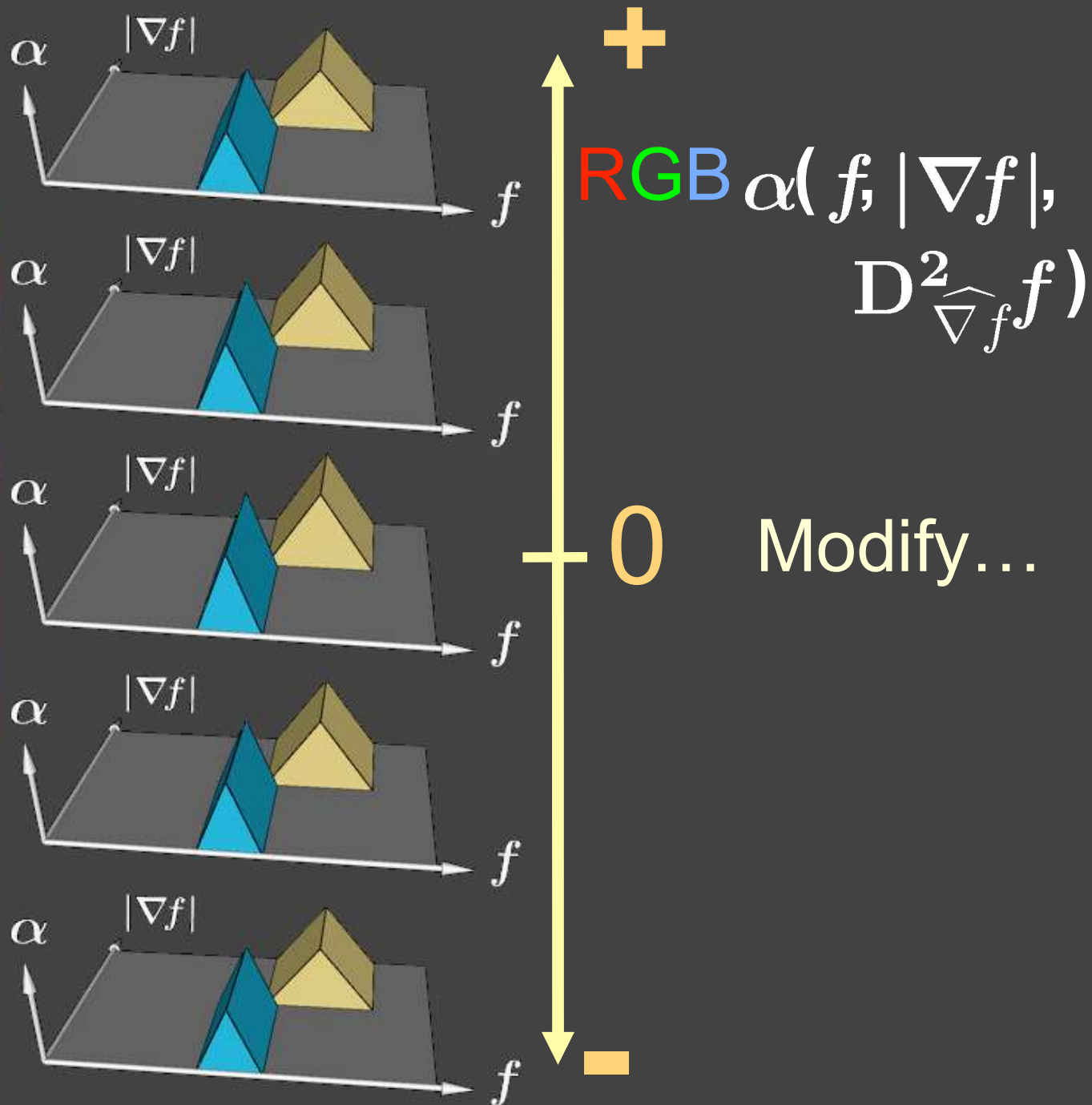
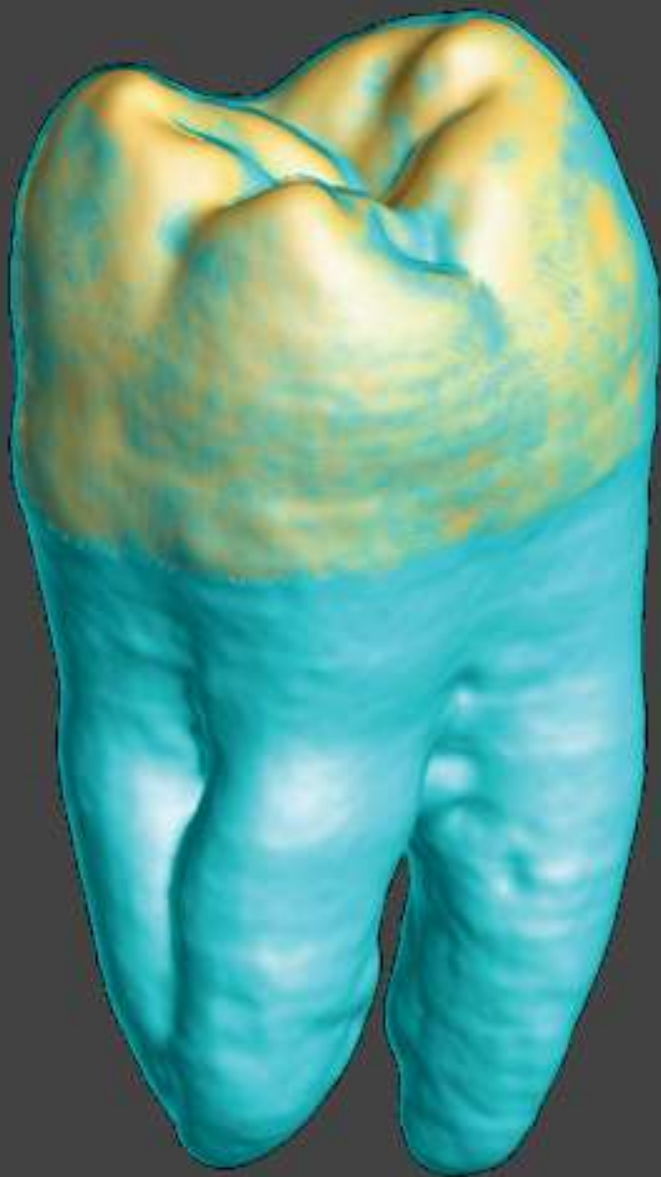
2D → 3D Transfer Function



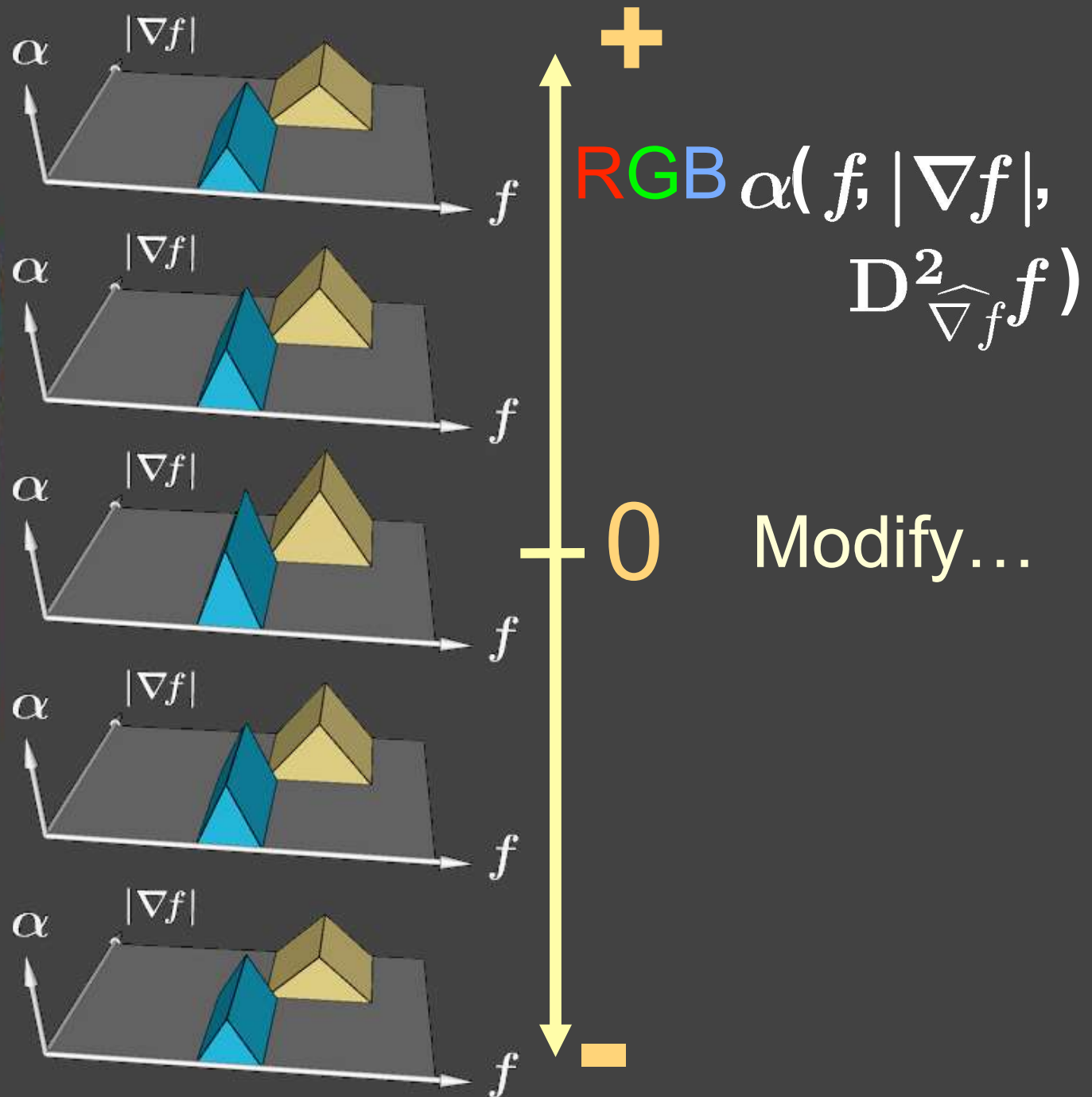
3D Transfer Function



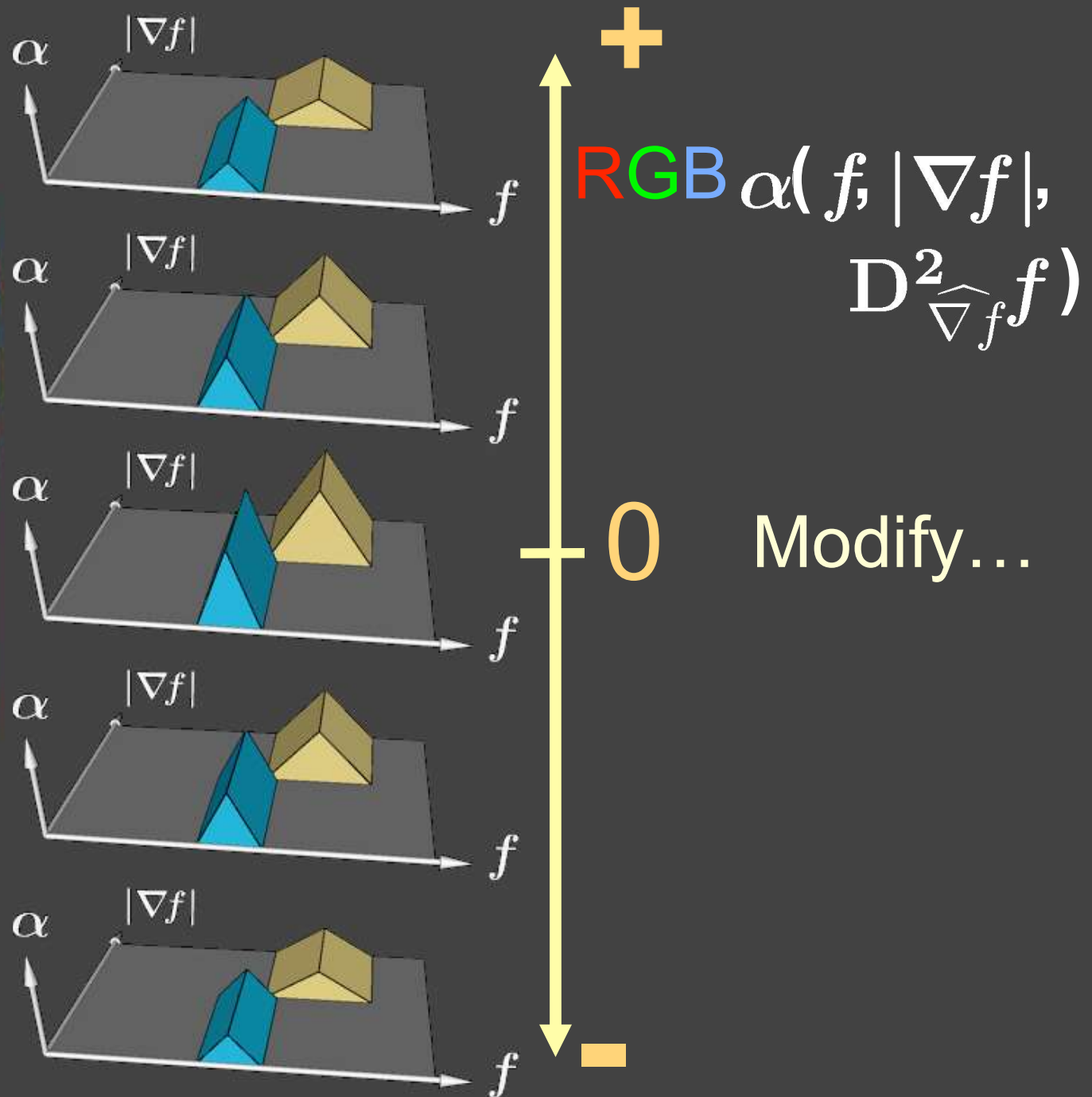
3D Transfer Function



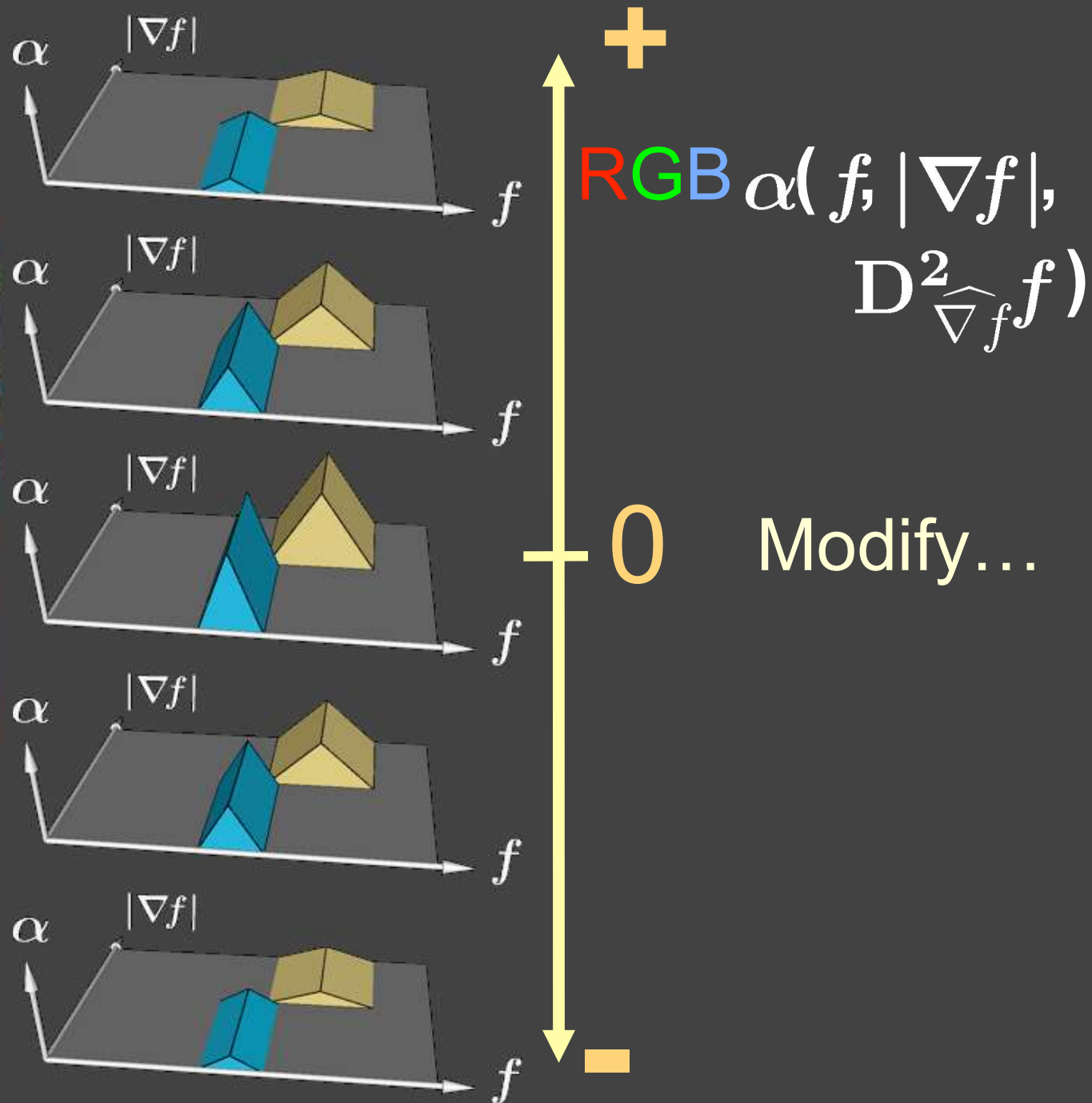
3D Transfer Function



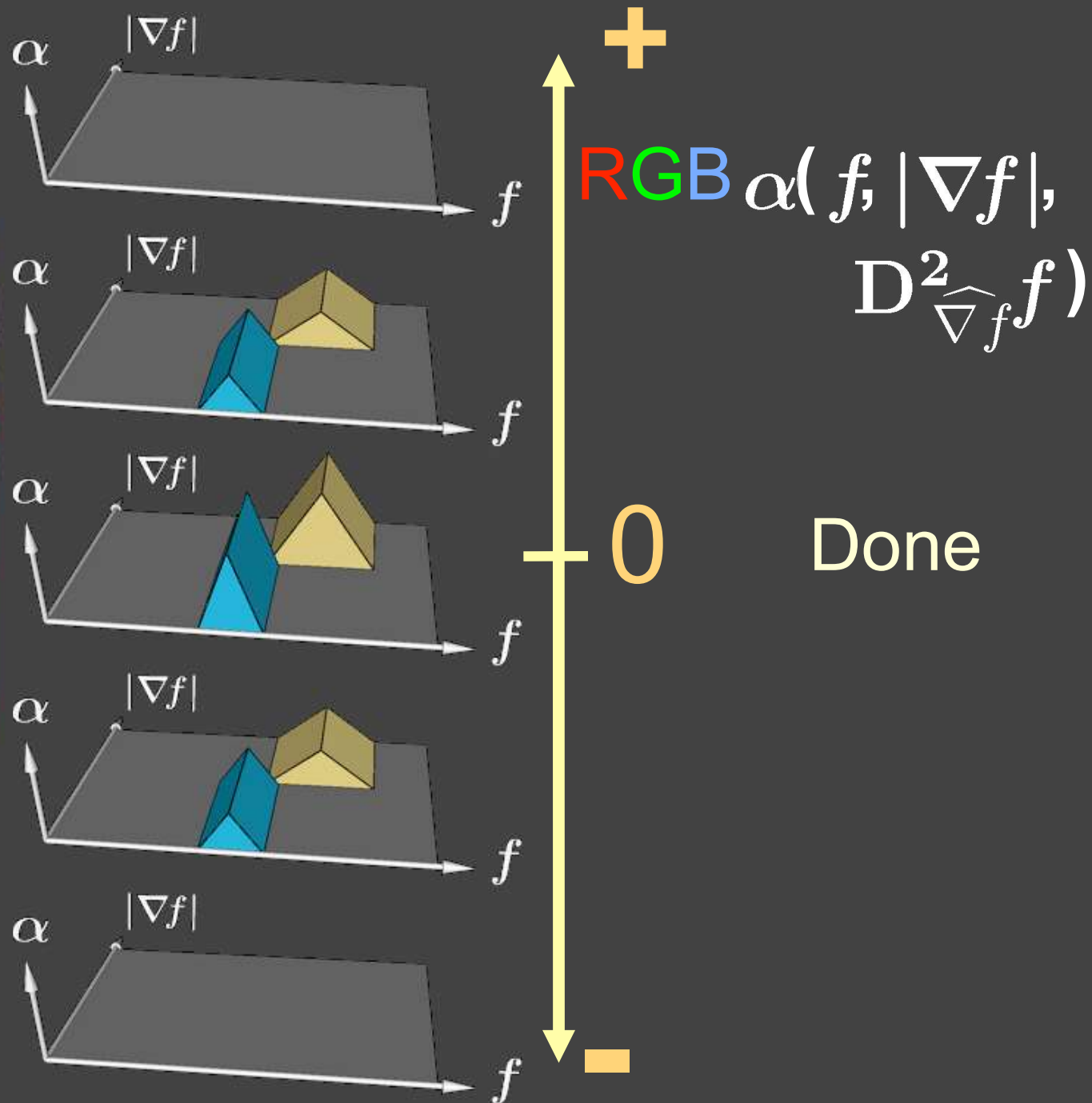
3D Transfer Function



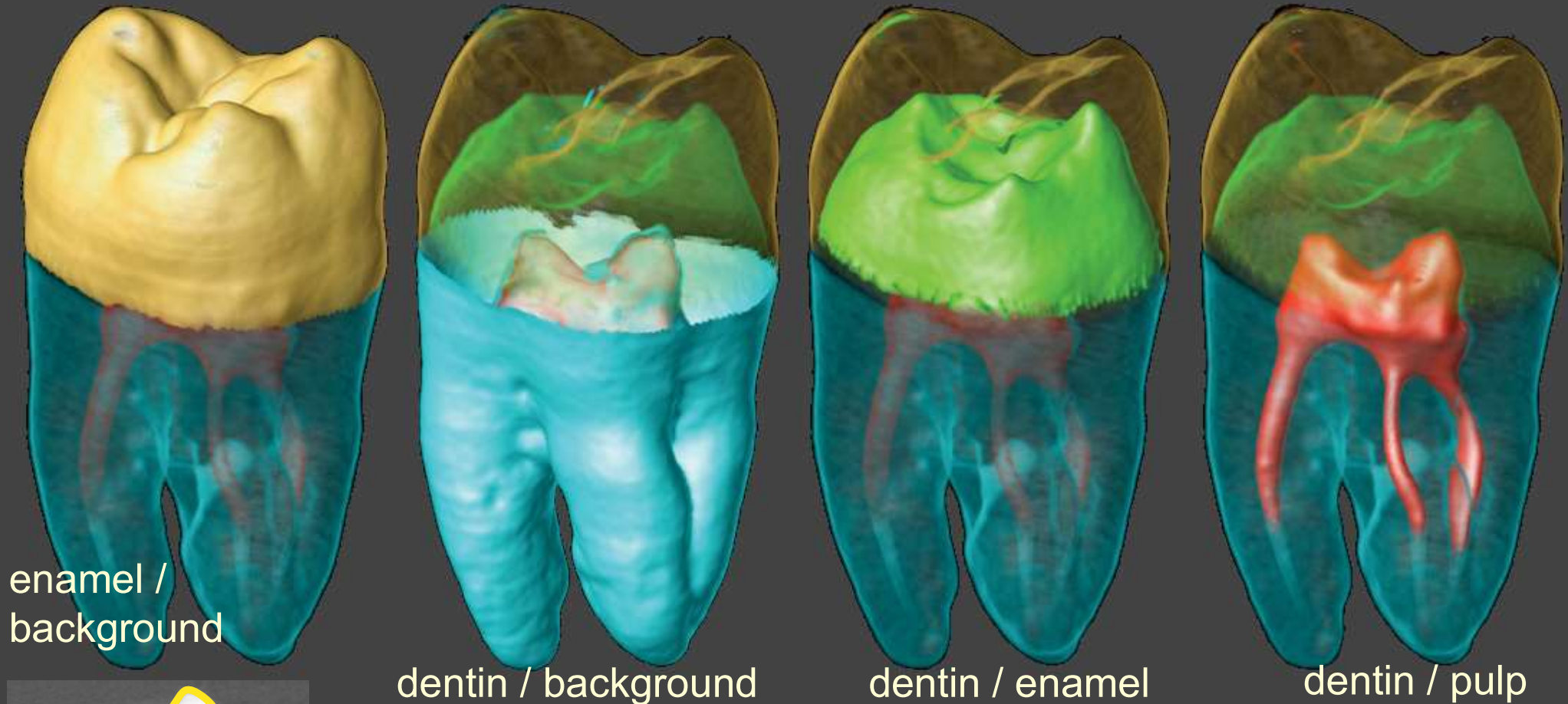
3D Transfer Function



3D Transfer Function



3D Transfer Function



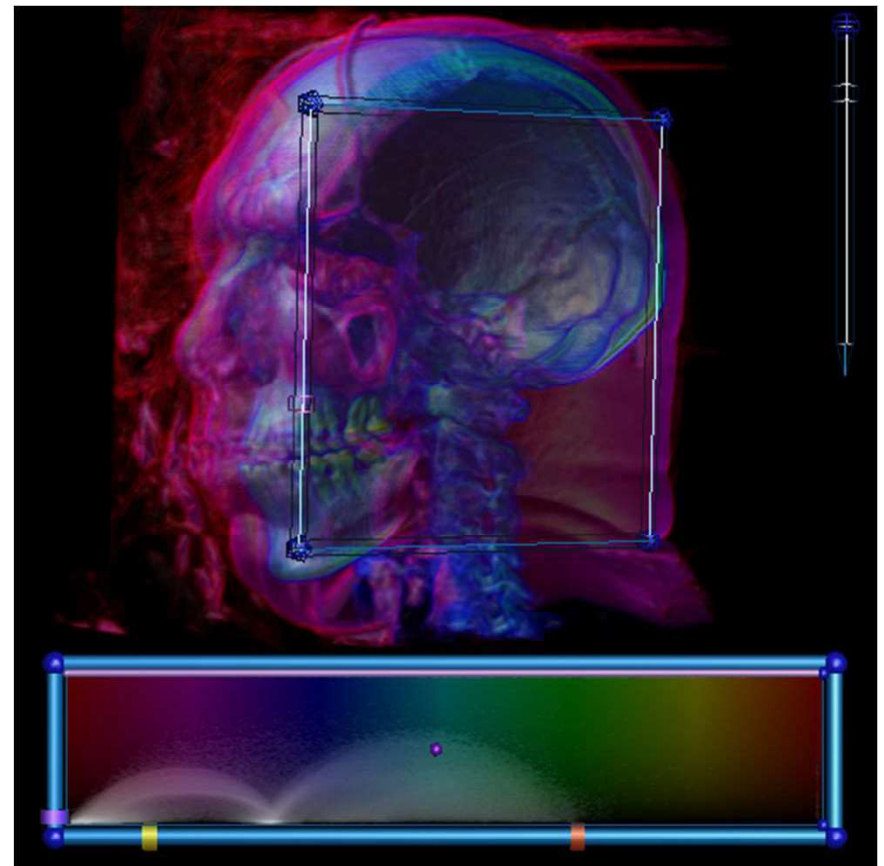
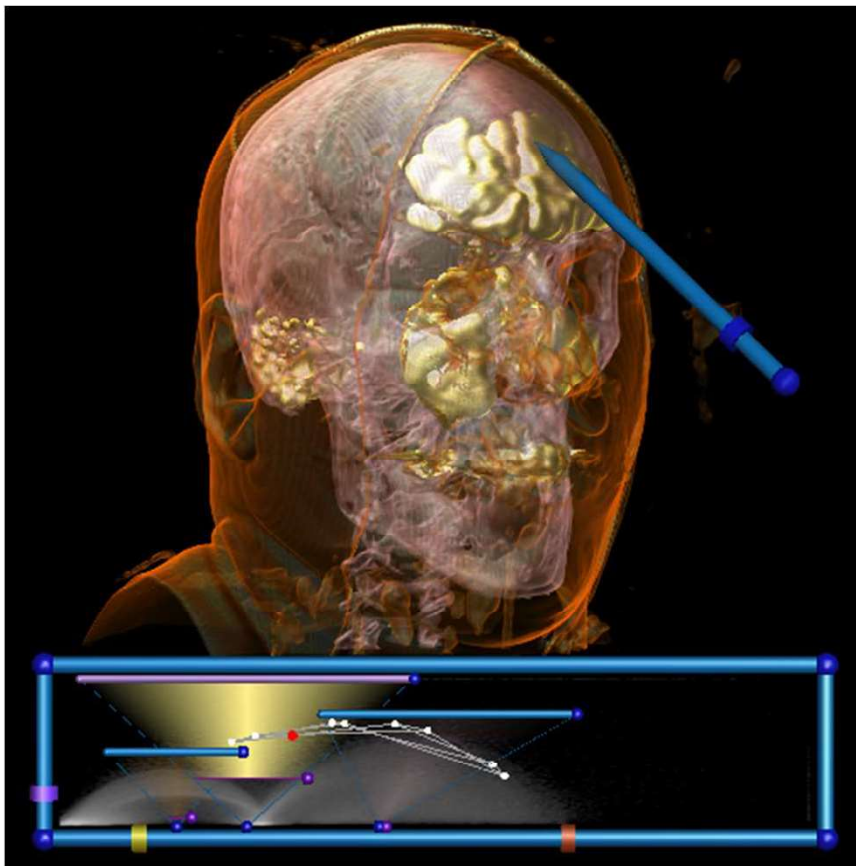
1D: not possible

2D: **specificity** not as good

Different Interaction

“Interactive Volume Rendering Using Multi-Dimensional Transfer Functions and Direct Manipulation Widgets” Kniss, Kindlmann, Hansen: Vis '01

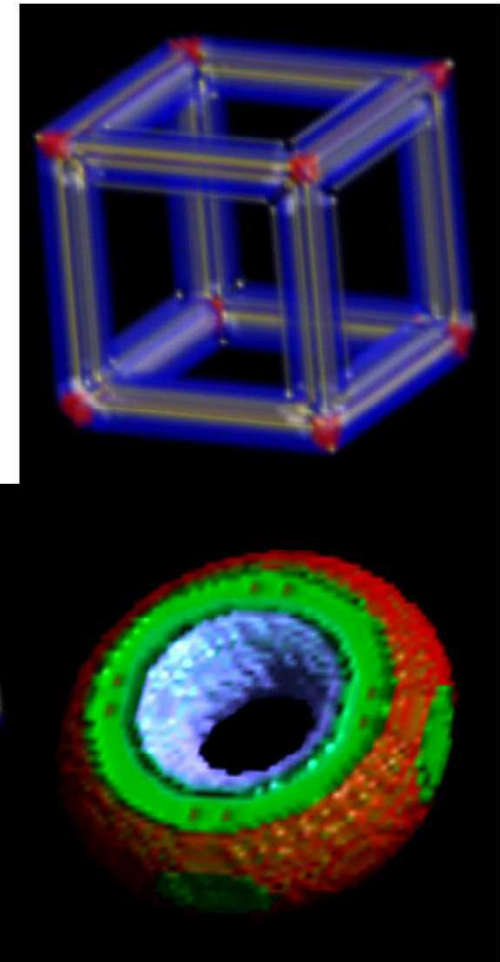
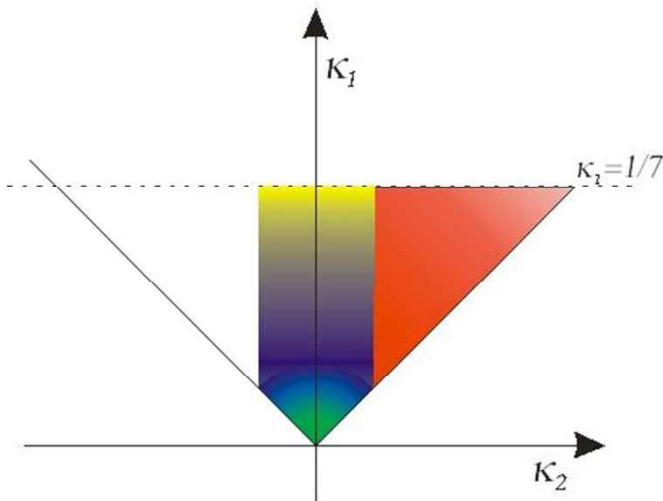
- Make things opaque by pointing at them
- Uses **3D** transfer functions (value, 1st, 2nd derivative)
- “Paint” into the transfer function domain



Curvature

“Curvature-Based Transfer Functions for Direct Volume Rendering”, Hladuvka, König, Gröller: SCCG '00

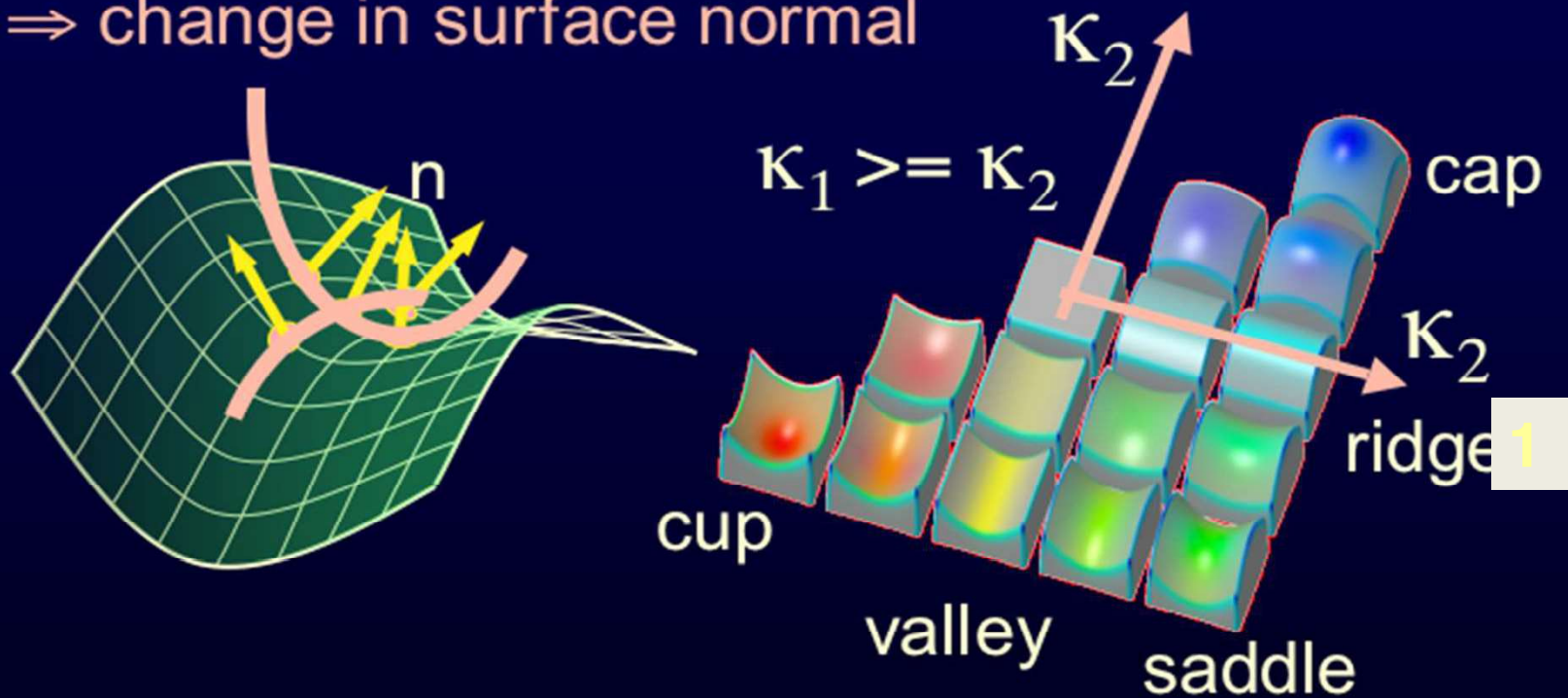
- Uses 2D space of K_1 and K_2 : principal curvatures of isosurface at a given point
- Graphically indicates aspects of local shape
- Specification is simple



Curvature

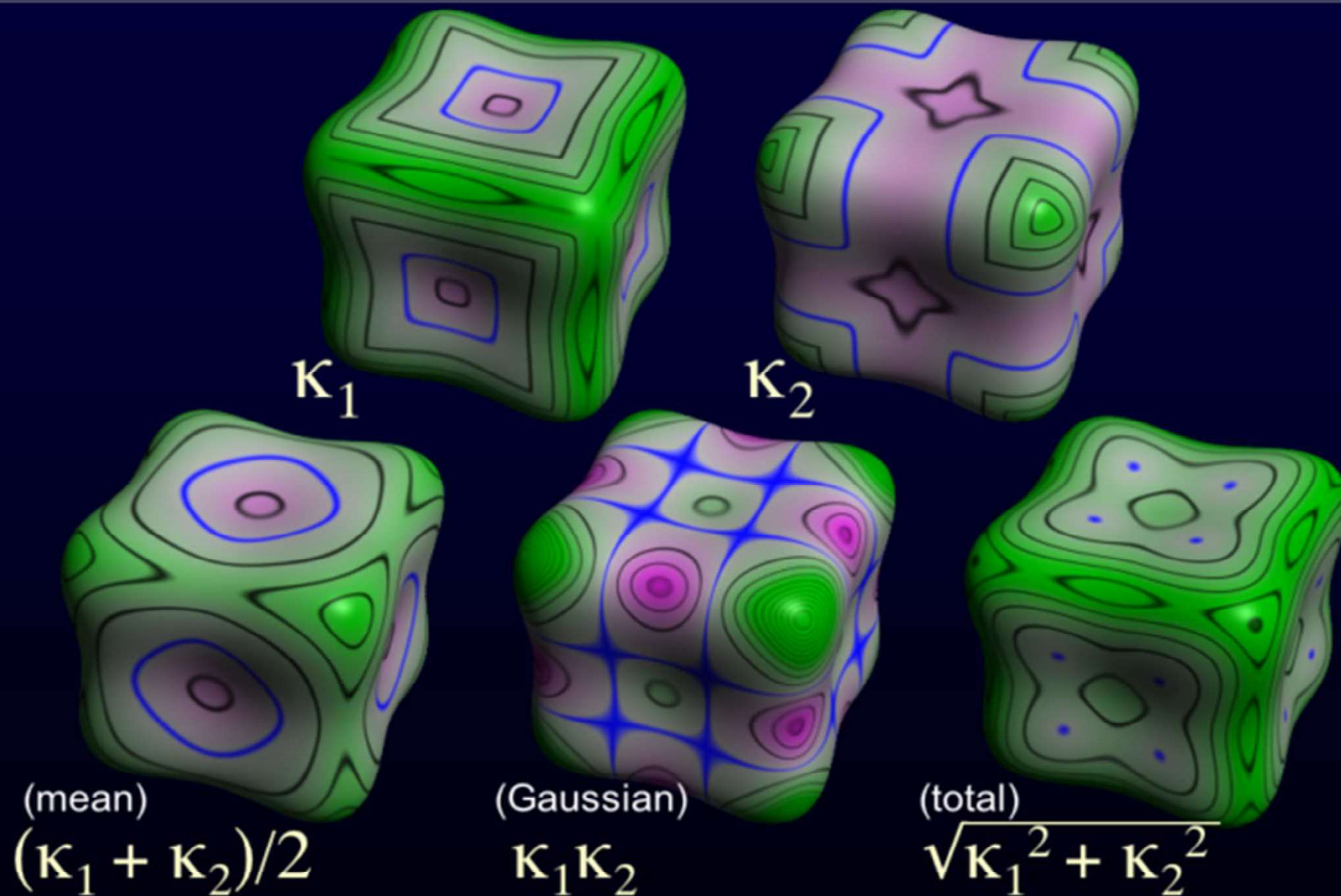
What is curvature

Small movements along the surface
⇒ change in surface normal



Principal curvature magnitudes
Principal curvature directions

Curvature measures



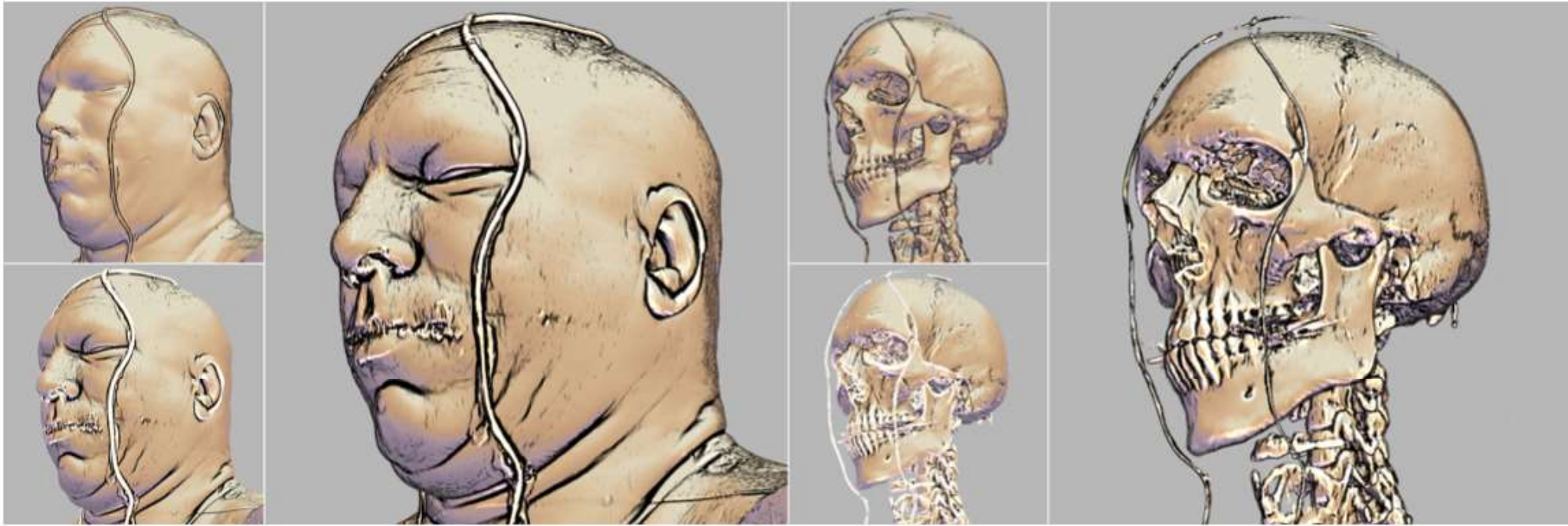


Figure 9: Curvature-based non-photorealistic volume rendering. The two small images depict the contribution of thickness-controlled contours (top) and ridge and valley emphasis (bottom). The large images show the combination of the two effects.

Curvature-Based Transfer Functions for Direct Volume Rendering: Methods and Applications

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Ross Whitaker¹

Tolga Tasdizen¹

Torsten Möller²